Admin Panel Guide

Introduction

- This document provides detailed information about the DNS Admin Panel functionalities, modules, and workflows. It explains the behavior of different modules such as PowerDNS, Cloudflare Secondary Zone, Cloudflare Native Zone, Forwarding, General Settings, Logs, and WebJobs.
- This document is prepared based on the current QA/Staging environment and can be used to understand system functionality, testing flows, validations, and module behavior.
- It also contains Production and QA/Staging environment details, along with important notes related to access permissions, configuration handling, and feature implementation.

What is DNS:

- DNS stands for **D**omain **N**ame **S**ystem.
- Mainly Used is convert human friendly domain name (www.google .com) into IP address.
- DNS is a directory service that provides a mapping between the name of a host on the network and its numerical address.
- User can know in depth about DNS by using below link: https://www.javatpoint.com/computer-network-dns

How DNS Works:

- DNS servers convert URLs and domain names into IP addresses that computers can understand and use. They translate what a user types into a browser into something the machine can use to find a webpage. This process of translation and lookup is called *DNS resolution*. [Text Wrapping Break]
- The process of DNS resolution involves converting a hostname (such as www.example.com) into a computer-friendly IP address (such as 192.168.1.1). An IP address is given to each device on the Internet, and that address is necessary to find the appropriate Internet device.
- User can know in depth about how to DNS work by using below link : [Text Wrapping

Break]https://www.techtarget.com/searchnetworking/definition/domain-name-system

Use Of Admin Panel:

- **The main use of the admin panel is for web forwarding and email forwarding.** For example: If the user wants to forward a website to another website, then the user can use web forwarding and if the user wants to forward an email to another email ID. So he/she can use email forwarding.
- To use web and email forwarding, the user has to register a domain, create records like A, AAAA, MX records and **all this process will be done using the admin panel.**

Use Of API's:

- Here, the user API is used to create, update and delete the domain, creating records within the domain.
- Users can run all these APIs through swagger.

Who will use the API's ? :

- **As Admin :-** Admin will be able to **use all API's by going through swagger.**
- **As API User :-** API user will not be able to **use all API's by going through swagger.**
- **As Normal Agent :-** Here a normal agent cannot use **direct** swagger from the admin panel to run API.
  - **A normal agent can use Swagger by opening Swagger's URL directly in a new tab to run the API.**[Text Wrapping Break]

Who will use the admin panel? :-[Text Wrapping Break]

- **As Admin :-** Admin will be able to access the **entire admin panel.**
- **As API User :-** API user will not be able to access the **entire admin panel.**
- **As a Normal Agent :-** Here, a normal agent will not be able to access the entire admin panel, **It has access to only 6 modules..**
  1. Domain Search
  2. API Search Log
  3. Error Log

4. API Search Chart
5. Domain Count
6. Domain Count Chart

What Is Cloudflare:

- **Cloudflare** is a content delivery network (CDN) and cloud security platform that provides website optimization, security, and performance services. It acts as a mediator between a website's server and its visitors, improving the speed and reliability of the website while also protecting it from online threats.
- User can know in depth about DNS by using below link: https://www.geeksforgeeks.org/what-is-cloudflare

How Cloudflare Works:

- To use Cloudflare, website owners need to sign up for an account and add their website to the Cloudflare dashboard. Cloudflare will scan the website for DNS records and configure its CDN and security features automatically. Once configured, website owners can monitor and adjust Cloudflare's settings from the dashboard.
- User can know in depth about DNS by using below link: https://www.geeksforgeeks.org/what-is-cloudflare

QA Site for accessing Macaria DNS application : https://admin-dns-api-qa-core.dns.nexus/login

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DNS Admin Panel – User Management

1. User Roles

The system shall support the following user types:

- Super User
- Admin
- Normal Agent
- API User

Each user type shall have different permissions based on their assigned role.

-When a user itself change the role it will logout and asked to login again means I am a super user and I changed my role to admin the next screen is the login screen.

2. User Listing

The system shall display users according to the logged-in user's permissions. The user list shall always be sorted by the most recently updated user in descending order.

Super User

The Super User shall be able to:

- View all users.
- View both Active and non-Active users.
- Manage every user type.

User Enable / Disable functionality

Only the Super User shall have access to Enable/Disable functionality.

The Admin shall not:

- View the toggle
- Enable users
- Disable users

The Super User shall not disable their own account.

Super User Protection

The system shall always maintain at least one Active Super User.
If only one Active Super User exists:

- The role shall not be changed.
- The role shall not be downgraded.
- The system shall display:

"At least one active SuperUser must be present."
The existing role shall remain unchanged.

Admin

The Admin shall only be able to:

- View Admin users.
- View Normal Agent users.
- View API users.
- View only Active users.

3. Role-Based Access

Super User

The Super User shall have unrestricted access to:

- DNS Admin Panel
- Swagger API
- PowerDNS
- Cloudflare Native Zone

- Cloudflare Secondary Zone

Admin

The Admin shall have full access to:

- DNS Admin Panel
- Swagger API

The Admin shall not have Super User administrative privileges to view other super user or disabled user.

Normal Agent

The Normal Agent shall have access only to:

- PowerDNS module
- PowerDNS API

The Normal Agent shall not access administrative modules and the Swagger platform as well.

API User

The API User shall only be permitted to access the PowerDNS API using a valid API token.
The API User shall not access the Admin Panel.

4. User Creation

Super User

The Super User shall be allowed to create:

- Super User
- Admin
- Normal Agent
- API User

Admin

The Admin shall be allowed to create:

- Admin
- Normal Agent
- API User

The system shall prevent creation of duplicate usernames.
If a duplicate username is entered, the system shall display:
"User already exists."

5. User Update

The system shall allow editing user information according to role permissions.

Super User

The Super User may update:

- First Name
- Last Name
- Username
- User Type
- Password
- Comments

For API Users, password modification shall not be available.

Admin

The Admin may update:

- First Name
- Last Name
- Username
- Comments

The Admin shall not modify:

- User Type
- Password

The system shall validate username uniqueness during updates.
If another user already owns the username, the system shall display:
"UserName you try to update is already exist."

6. Token Management

Only the Super User shall have permission to:

- View generated tokens
- Generate new tokens
- Regenerate existing tokens

Admin users shall only be able to view the default token assigned during user creation.
Admin users shall not regenerate tokens.

7.Disabled User Behavior

Disabled users shall not authenticate successfully.

Normal Agent

A disabled Normal Agent:

- Cannot log in.
- Cannot access PowerDNS.
- Cannot use PowerDNS API.
- Shall receive HTTP 401 Unauthorized for API requests.

API User

A disabled API User:

- Cannot access PowerDNS API.
- Shall receive HTTP 401 Unauthorized.

Admin

A disabled Admin:

- Cannot log in.
- Cannot access APIs.
- Shall receive HTTP 401 Unauthorized.

Super User

A disabled Super User:

- Cannot log in.
- Cannot access APIs.
- Shall receive HTTP 401 Unauthorized.

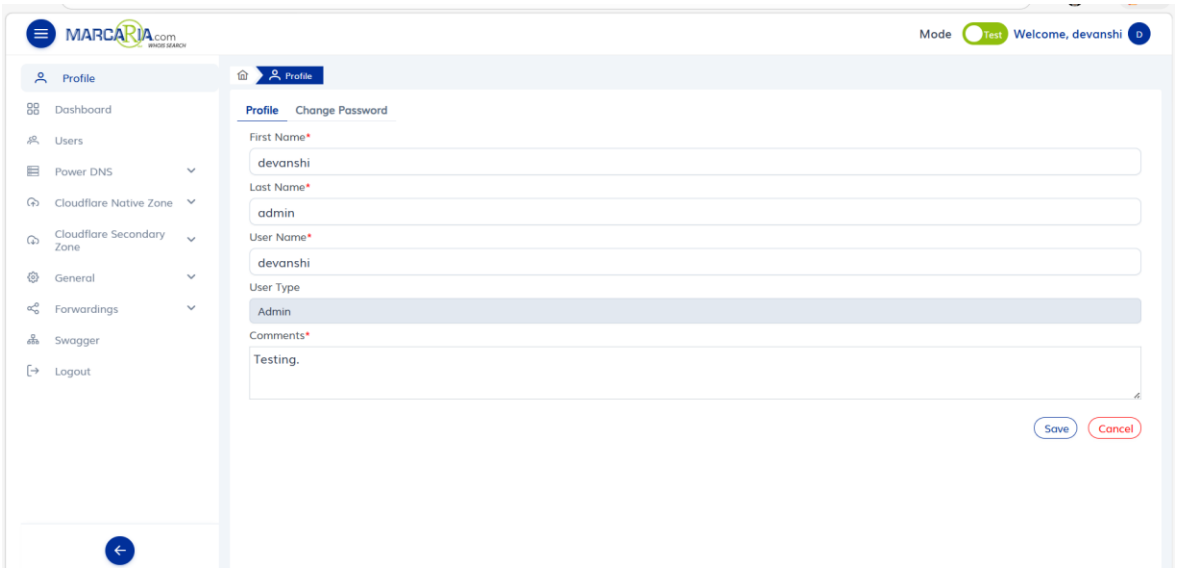
PROFILE

- Click on the Profile option from the side menu bar.
- The Profile module allows logged-in users to view and update their profile details.
- This module contains the following two sections:
    - Profile
    - Change Password

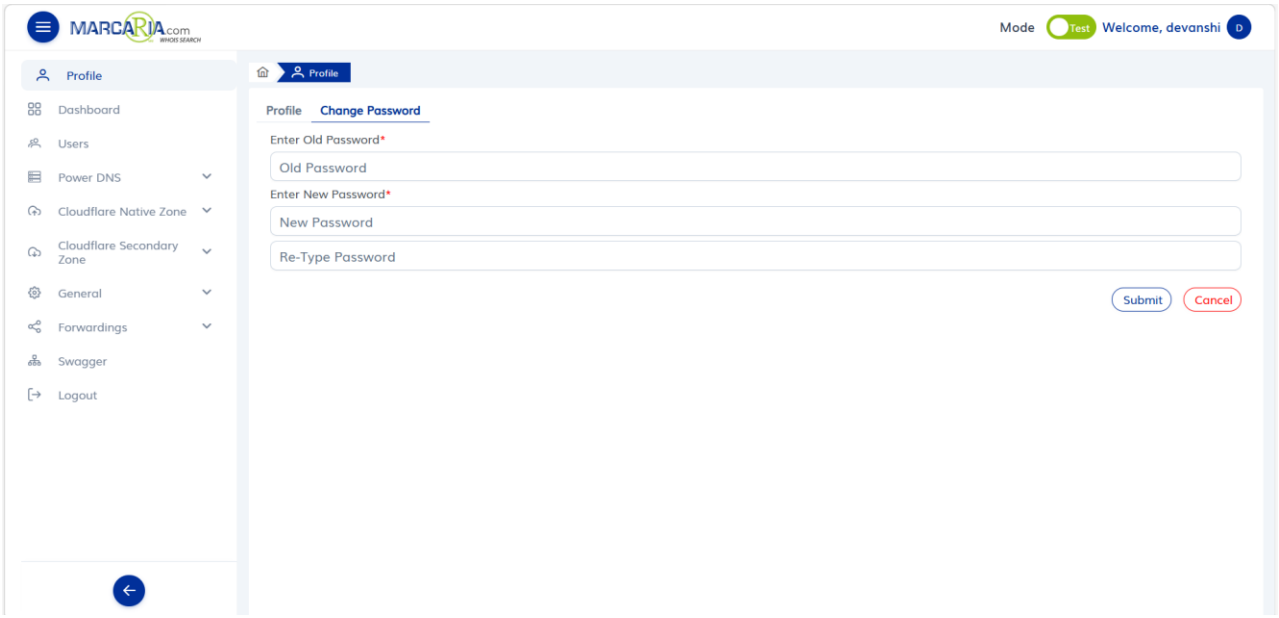
● Profile Section

- The following fields are available in the Profile section:
    - **First Name:** Used to enter or update the user's first name.
    - **Last Name :** Used to enter or update the user's last name.
    - **User Name** : Displays the login username of the current user.
    - **User Type** : User Type is only displayed and cannot be changed.
    - **Comments** : Add Comment.

- **Save button :** Click on Save button to update and store profile details.
- **Cancel button :** Click on Cancel button to discard unsaved profile changes.



- **<u>Change Password Section</u>**
  - Click on the Change Password tab to update the password of the logged-in user.
  - The Change Password section allows users to update their login password.
    - **Old Password:** Used to enter the current password of the logged-in user.
    - **New Password** : Used to enter a new password for the account.
    - **Re-Type Password** : Used to confirm the new password by entering it again.
    - **Submit button :** Click on Submit button to update the account password.
    - **Cancel button :** Click on Cancel button to cancel the password update process.

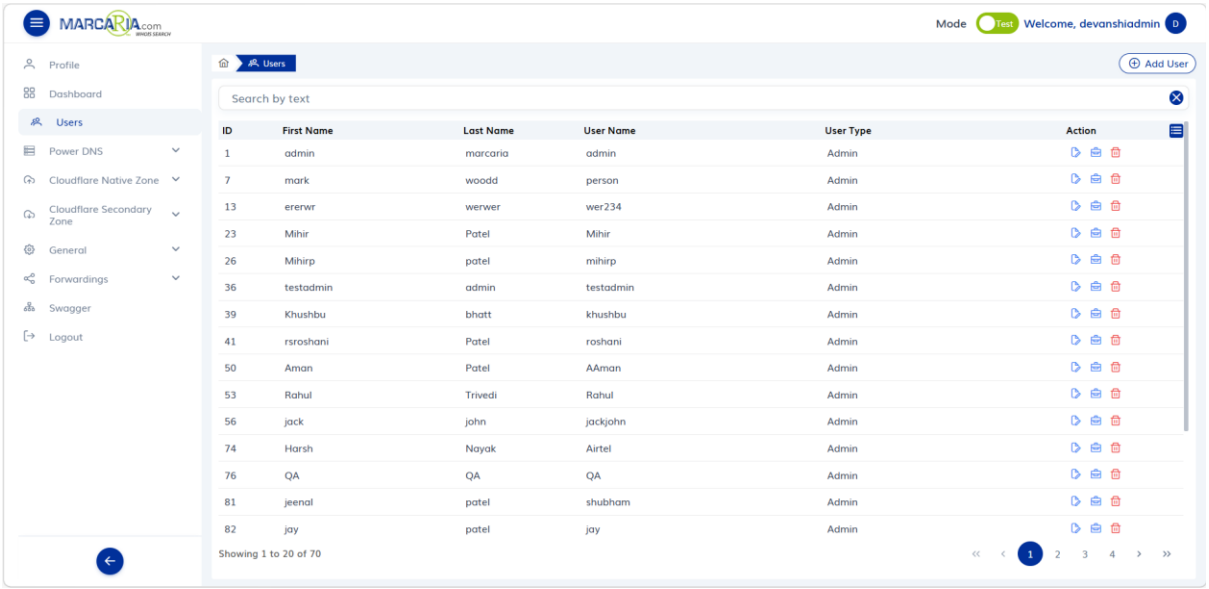


DASHBOARD.

- The Dashboard module provides an overview of DNS Admin Panel activities and summary data.
- Users can access the Dashboard module from the left-side menu.
- Dashboard data is displayed based on the selected module from the dropdown available in the top-right section.
- The following modules are available in the dropdown:
    - Power DNS
    - Cloudflare Native
    - Cloudflare Secondary
- The Dashboard helps users monitor overall system activity and summary information from a single page.
- Dashboard information is displayed in graphical and table formats for better visibility and monitoring.
- The API Error log or Search log are show always be sorted by the most recently updated in descending order
- Only authenticated users can access the Dashboard module.
- The Dashboard page displays API search count, added vs deleted domain count, domain list, recent API error logs, and recent API search logs based on the selected module.

USERS

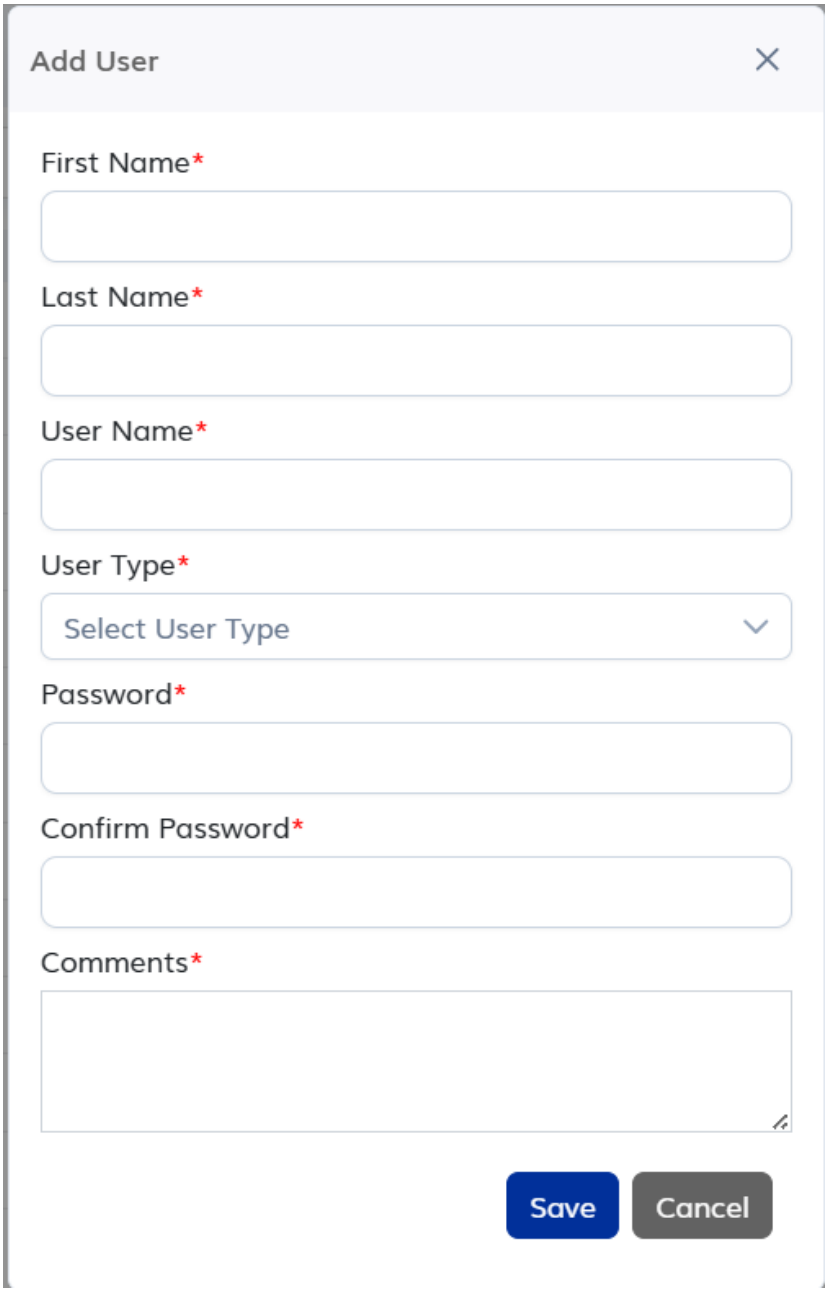
- Refer to User management tab to understand the different roles and it permission to perform actions.
- Click on the Users option from the side menu bar.
- Users are able to view a list of API user tables with fields like ID, First Name, Last Name, User Name, User Type, Action, Search Box, Add button, sorting and pagination functionality.
- Users can search any user using the search box.



❖ Add User:

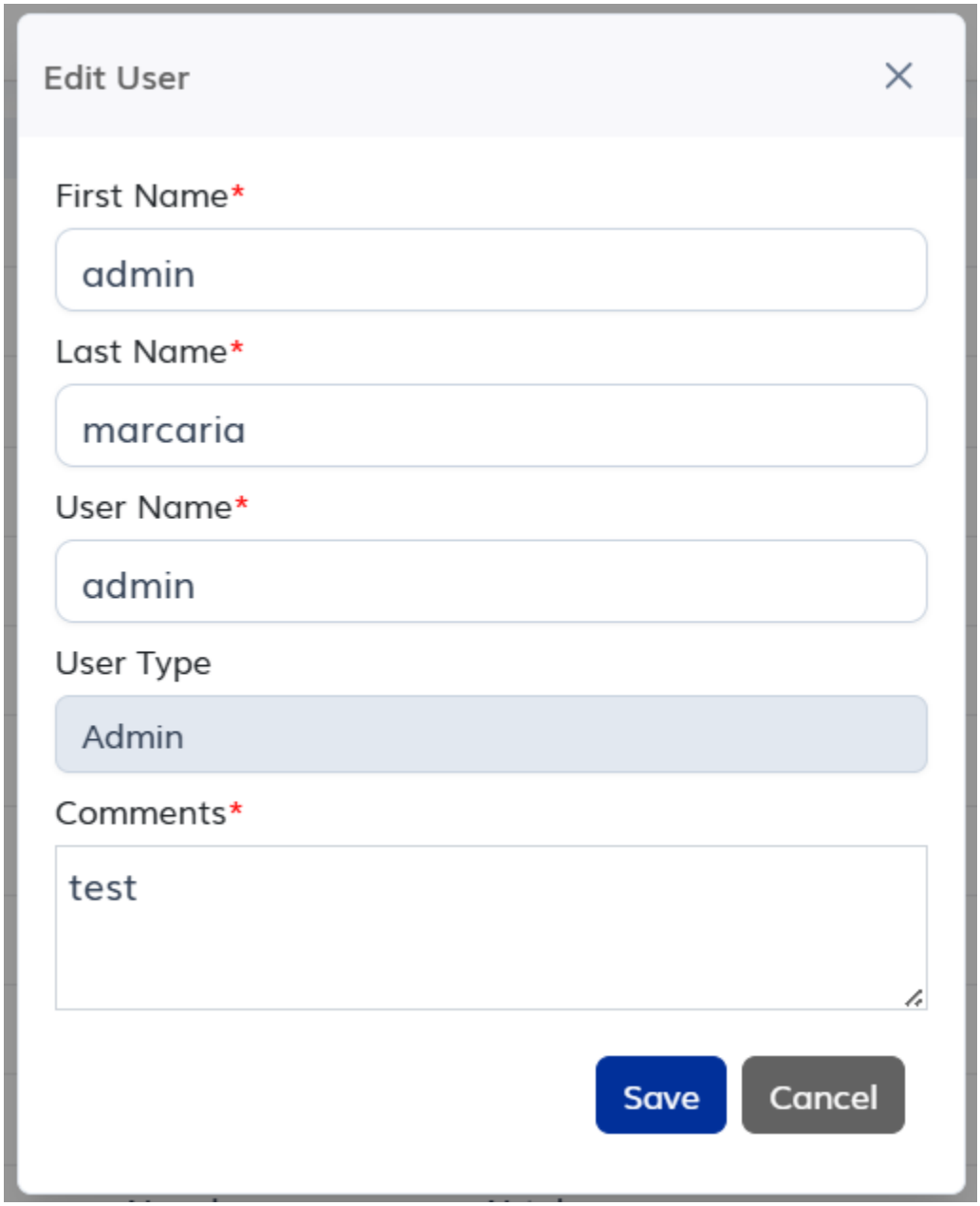
- Click on Add button it will open the window for Add User.
- It has the following options..
  - **First Name:** Add First Name of the User
  - **Last Name:** Add Last Name of the User
  - **User Name:** Add User name which you need to set as a User Name.
  - **User Type:** Select User Type from dropdown i.e. API User, Normal Agent, Admin
  - **Password :** Used to set the user's login password.

- **Confirm Password** : Enter the same password again to verify it.
- **Comments:** Add comment
- **Submit button:** Click on the submit button it will generate the profile with the given details.
- **Cancel button:** It will cancel the process and add user pop will be closed.



❖ Edit Icon:

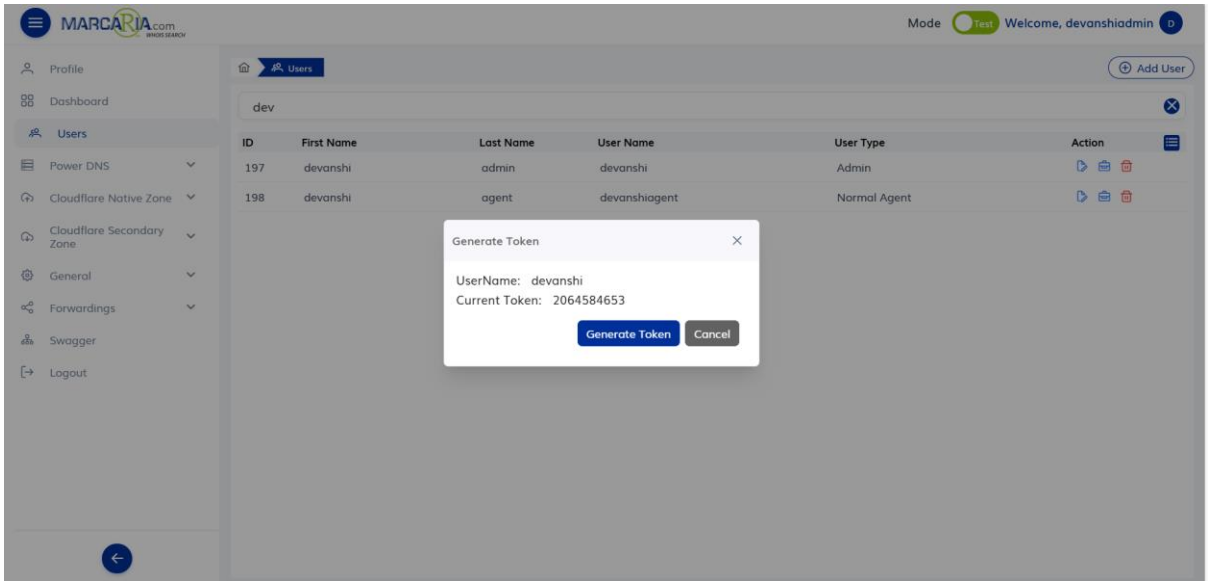
- Click on the Edit icon and it will open the Edit User popup.
- It allows us to edit First Name, Last Name, User Name, User Type, Comments and Save & Cancel button.
- Here, there are three type of the user type
    - Admin.
    - Users.
    - Normal Agent.
- **User Type**: User Type is displayed in read-only mode and cannot be changed.
    - **API User** - Authorized third-party users can access the API when it is specifically intended for use by clients' approved external users. It can be directly integrated into their applications.
    - **NormalAgent -** It has access only to the PowerDNS module in the Admin Panel, but it also has access to all APIs in Swagger, including PowerDNS, Cloudflare Native Zone, and Cloudflare Secondary Zone, etc.
    - **Admin -** Admin user can access and manage all modules of the DNS Admin Panel.
- **Save Button**: Click on the Save button to update the user details.
- **Cancel Button**: Click on the Cancel button to close the popup without saving changes.



- After successfully updation user will get the notification message like - “Record Successfully Updated”

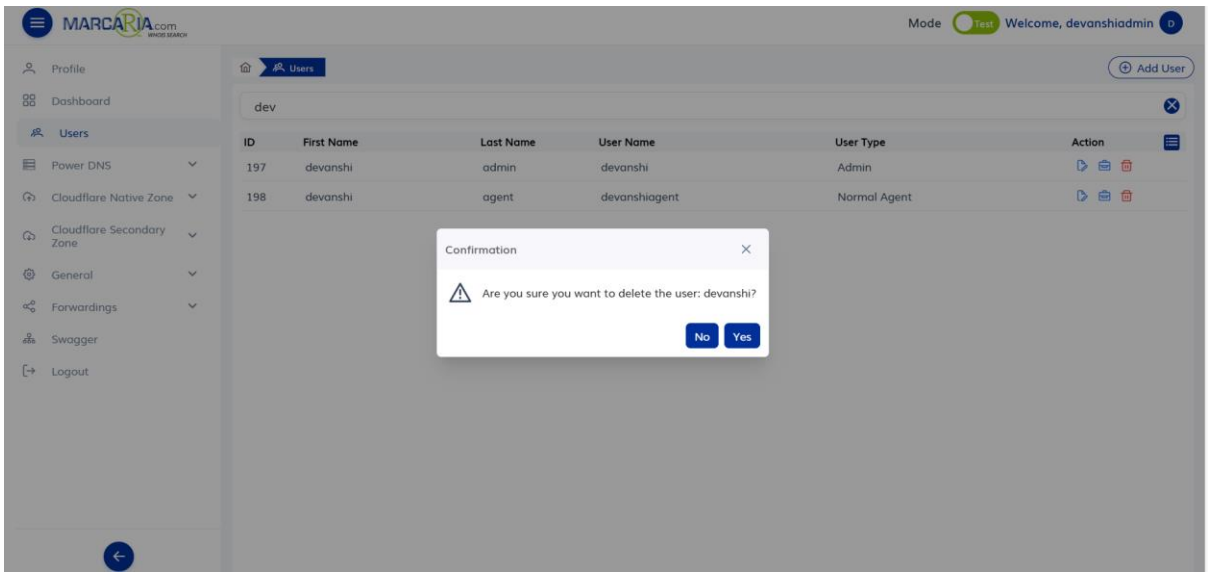
❖ Generate Token Icon:

- Click on this icon it will open the popup of Generate Token.
- The popup has information like User Name, Current Token, Generate Token button & Cancel button.
- Once a user clicks on the Generate Token button it will generate the new token and this token is used in Swagger API.
- Click on the Cancel button to close the popup.



❖ Delete Icon:

- **Delete Icon:**
  - Click on the delete icon and it will delete the user.
  - After delete action, selected user will be removed from the active user list.

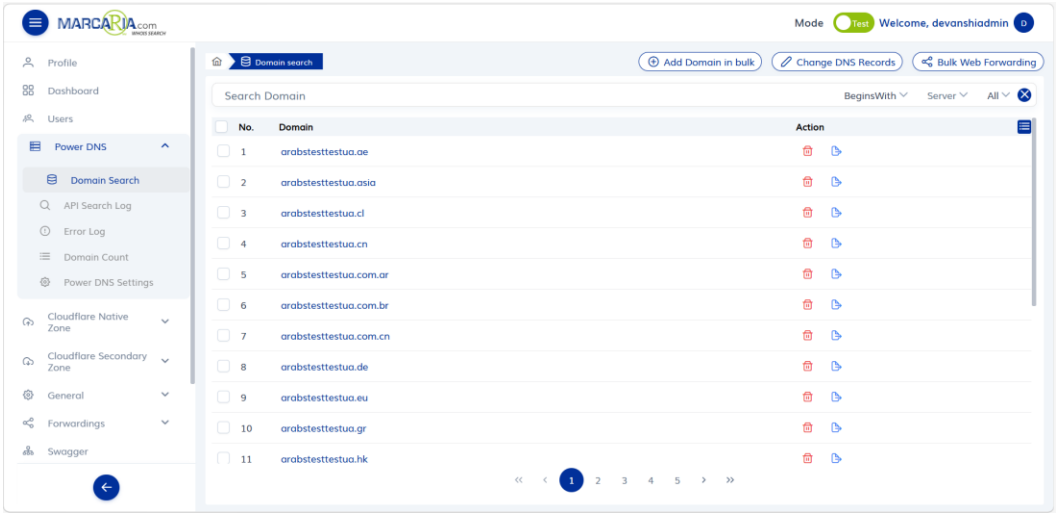


POWER DNS.

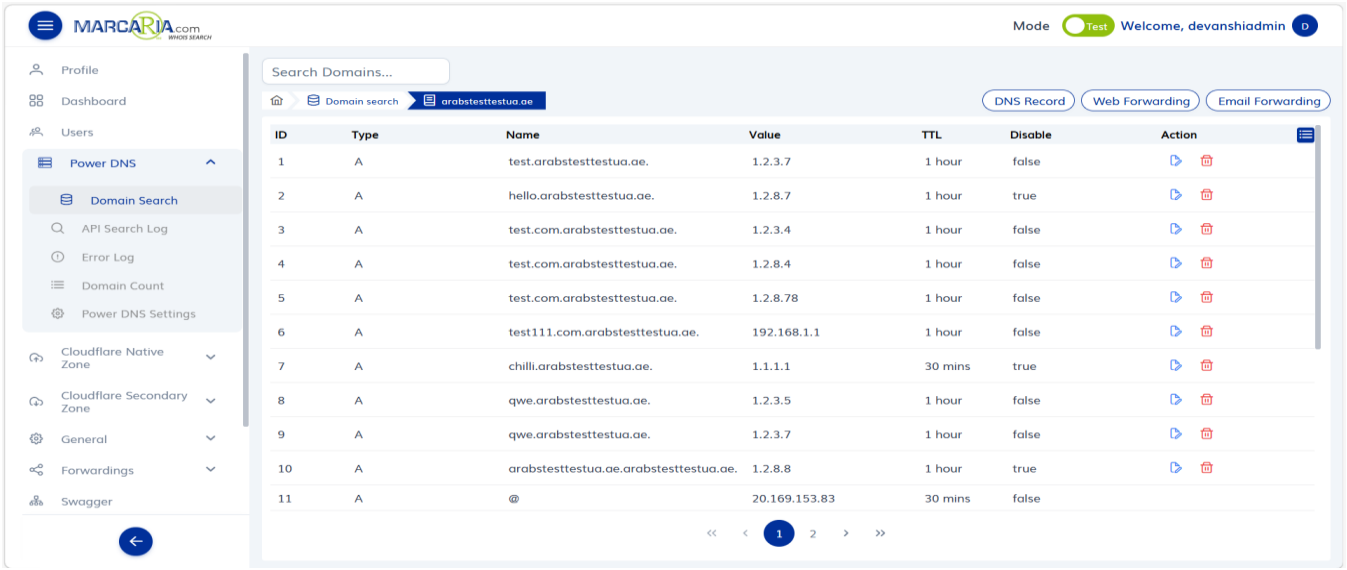
- Click on the Power DNS option from the side menu bar.
- Power DNS module is used to manage Power DNS domains, DNS records, logs, domain count and settings.
- Power DNS has 5 submodules.
    1. Domain Search.
    2. API Search Log.
    3. Error Log.
    4. Domain Count.
    5. Power DNS Settings.

Domain Search:

- Click on Domain Search option under Power DNS.
- Here, users can see the search domain search box, filter by, mode, server and domains with dropdowns, search button, add domain in bulk and change DNS records button and Bulk Web Forwarding button.
- Users can add/create domains from the Domain list menu.
- By clicking on the Domain from the side menu panel the user is able to view a list of domains which are already added.
- Users can add multiple domains using Add Domain in bulk button.
- Users can update DNS records for selected domains using the Change DNS Records button.
- Users can apply web forwarding using Bulk Web Forwarding button.

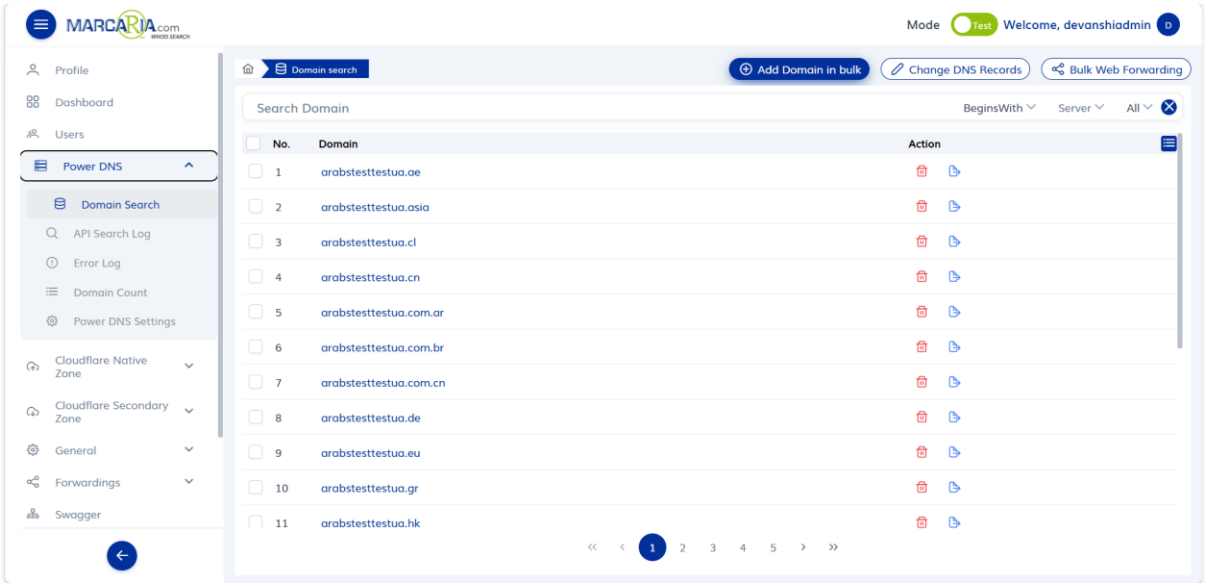


- When the user clicks on the domain name, the user can see the domain records.



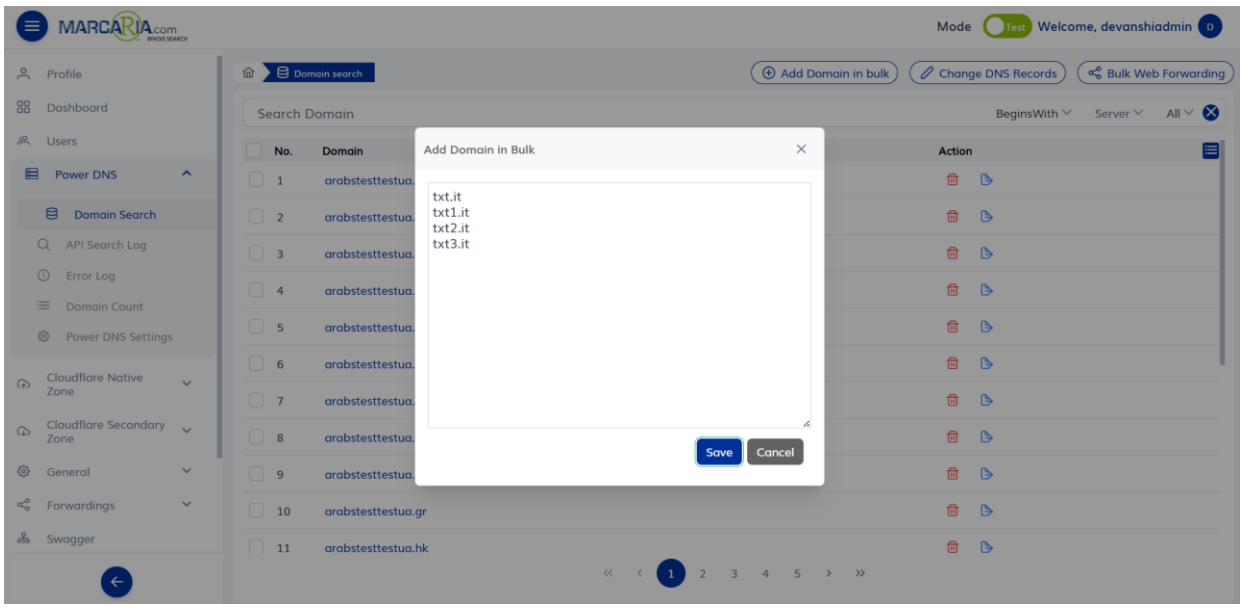
Add Domain in Bulk

- When the user clicks on the Add domain in bulk button it will open the popup with textarea ,Save and Cancel button.
- Here, users can add multiple domains in the text area separated each by entering a domain. i.e. every domain must be in a new line

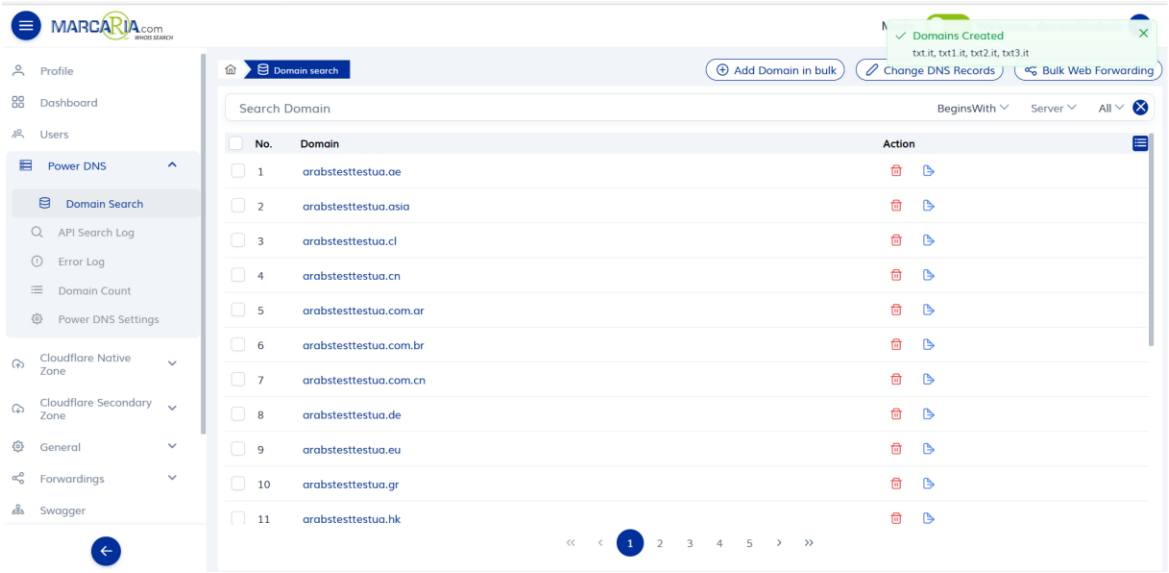


Add Domain in bulk button

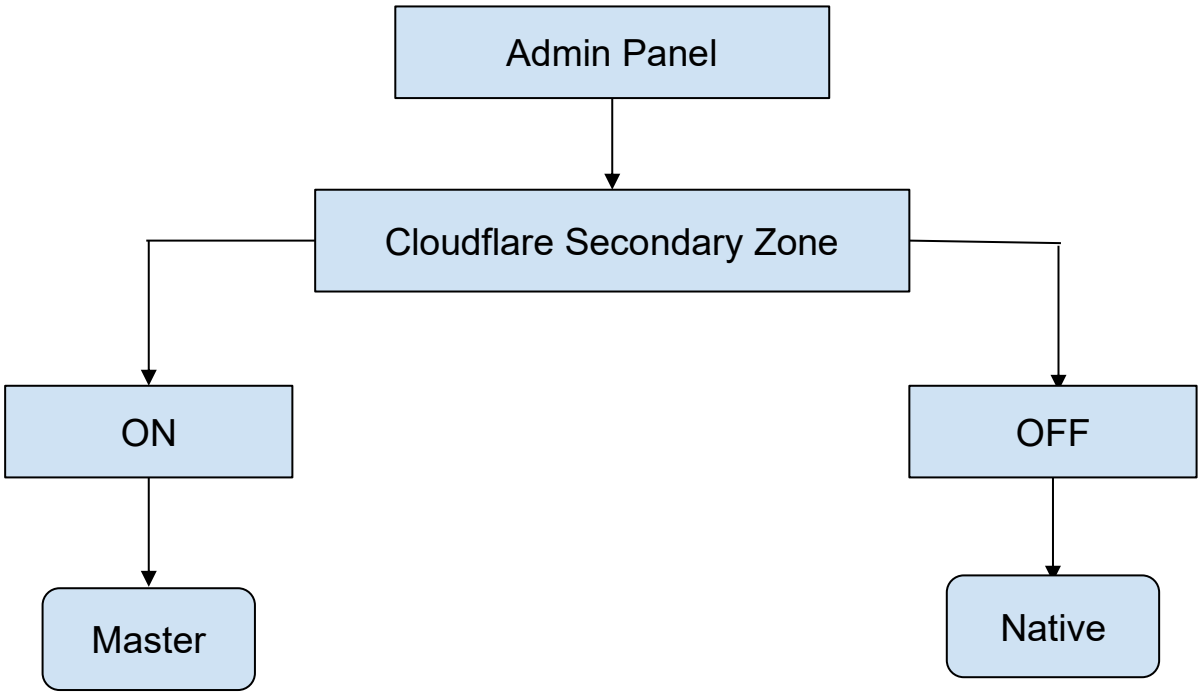
- When the user clicks on the Add domain in bulk button it will open the popup with textarea and Save, Cancel button.
- Add multiple domains in the text area separated each by entering a domain name. i.e. every domain must be in a new line.



- Click on save button it will add the domain in the list.
- Click on Cancel button it will close the popup.



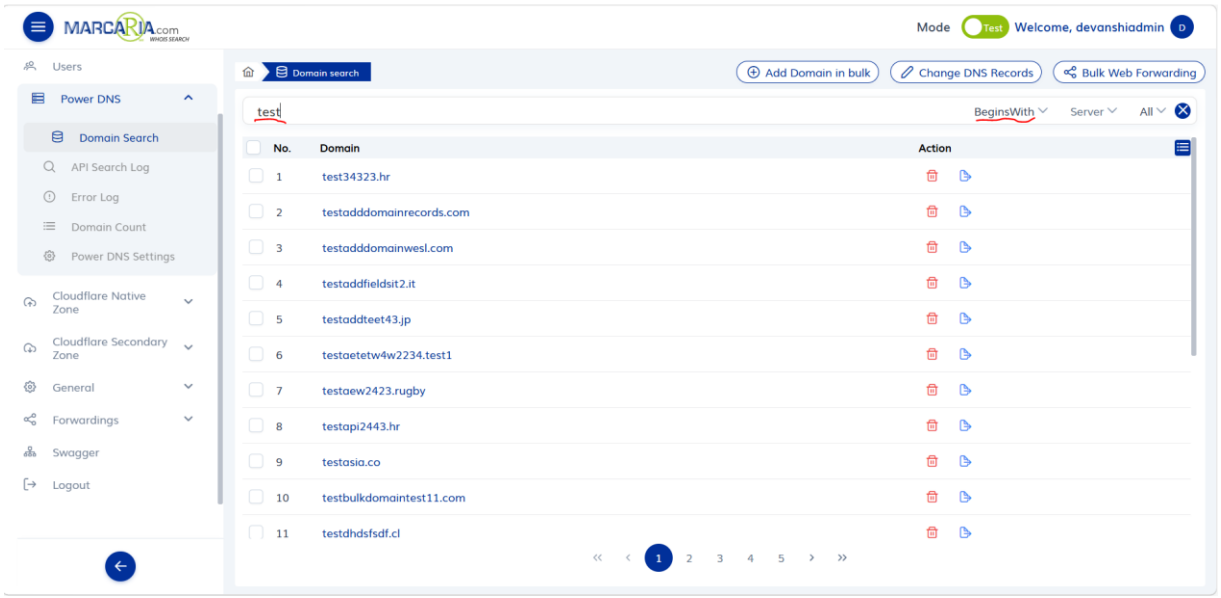
- **Scenarios When Cloudflare Secondary Zone is Enabled or Disabled.**



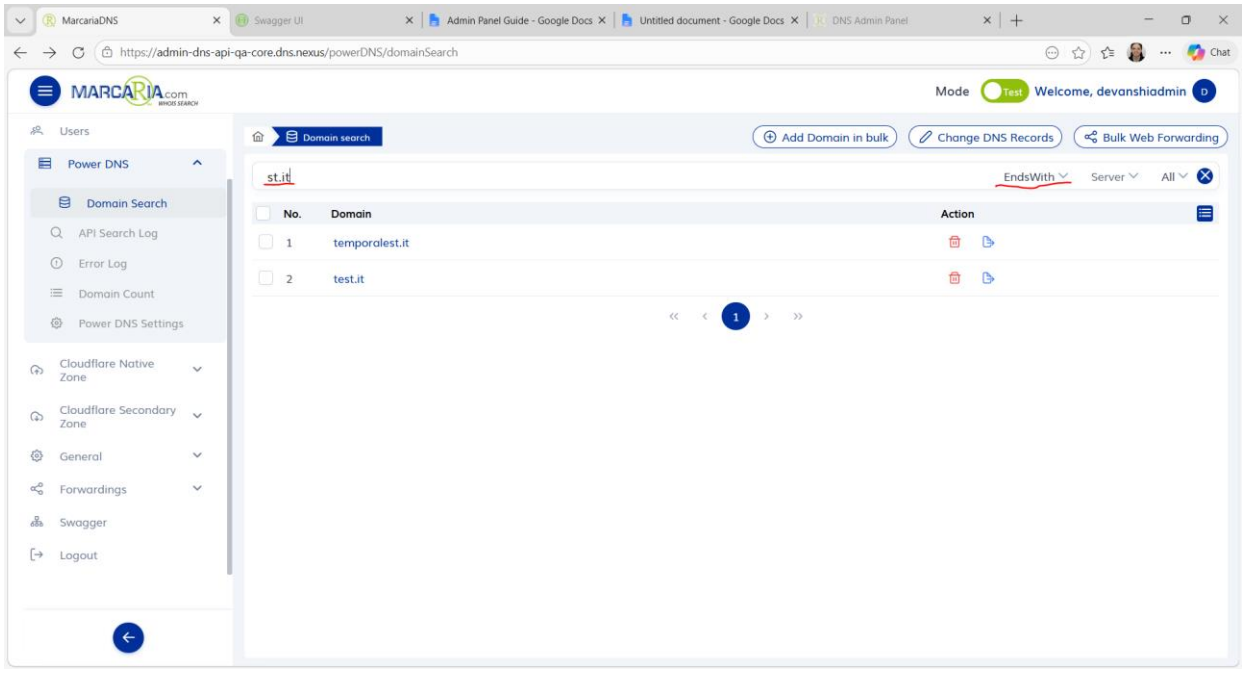
- **Cloudflare Secondary Zone Enabled**
    a. Users can view a domain that has been added with the Kind set to **"Master."**

Filter By Functionality

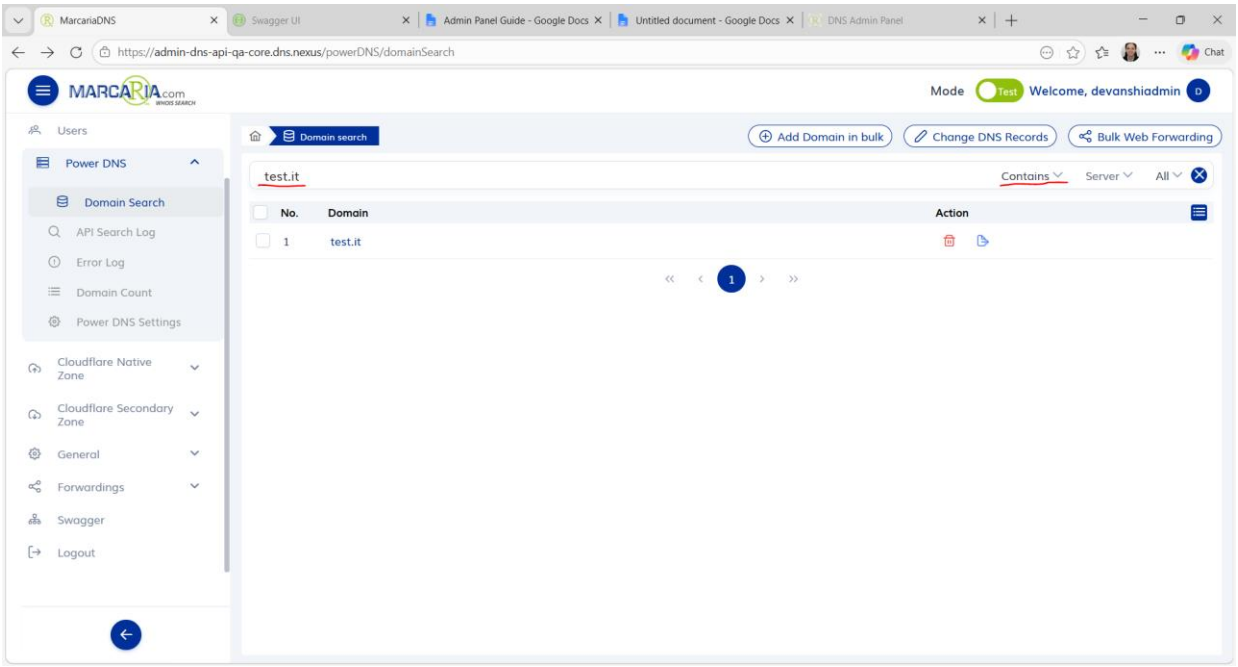
1. **Begins with :** Enter the initial letter of the domain and press enter.



2. **Ends with :** Enter the end letter of the domain and press enter.



3. **Contains :** Enter the Domain and press enter.

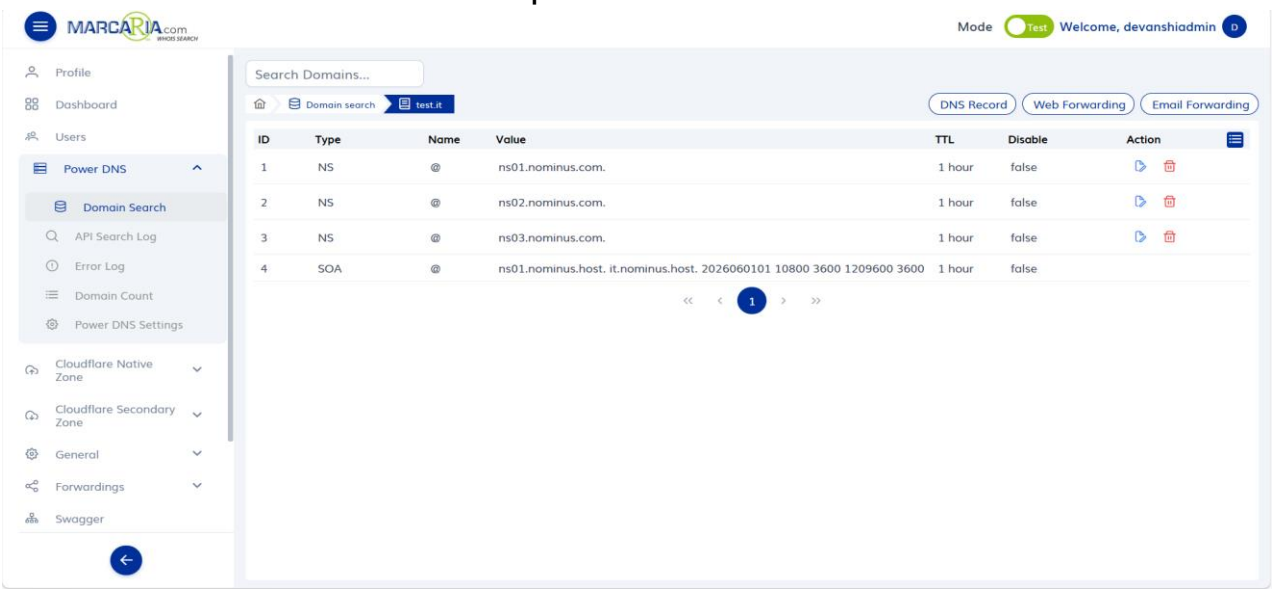


Server Dropdown Functionality

- As many IDs as you create in the forwarding server will be displayed in the server dropdown.

- Here users see Default = Yes in front of the profile which is Active. When creating a new profile at that time the newly added profile will have Default = Yes. It means the newly added profile will be considered as Active.

Click on Created Domain then open domain Record:

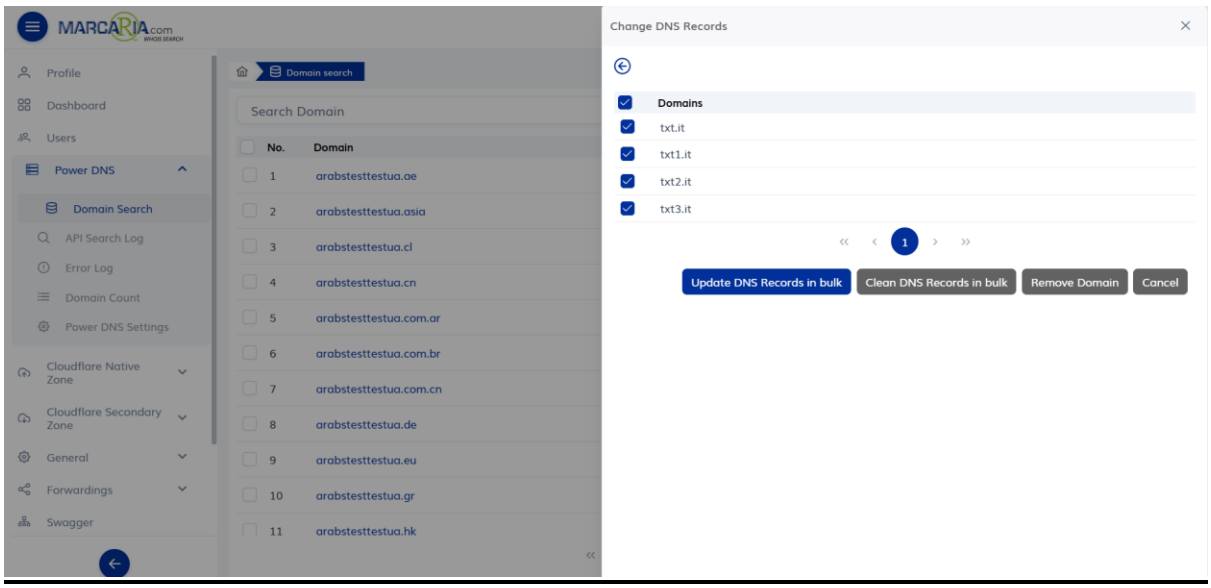


Change DNS Records Button:

- Click on the change DNS record button it will open the popup with text area and Continue & Cancel button.
- Add domain name in the text area. Users can add multiple domain names by separating it with the enter key. It means every domain name must be in the new line.
- Here, users can update records or clean dns records or remove (delete) domain.

Continue button:

- Click on the continue button it opens the popup of the Change DNS Records with following information.



- Users can see tables with the fields Available and Domains.

  - **Available field:**
    - Here in available fields if the domain is already available in the list it will have checked the checkbox.
    - Here the user is able to check/uncheck any domain as per their requirements.
    - If domain is not available in the list check box of it is in disable mode.

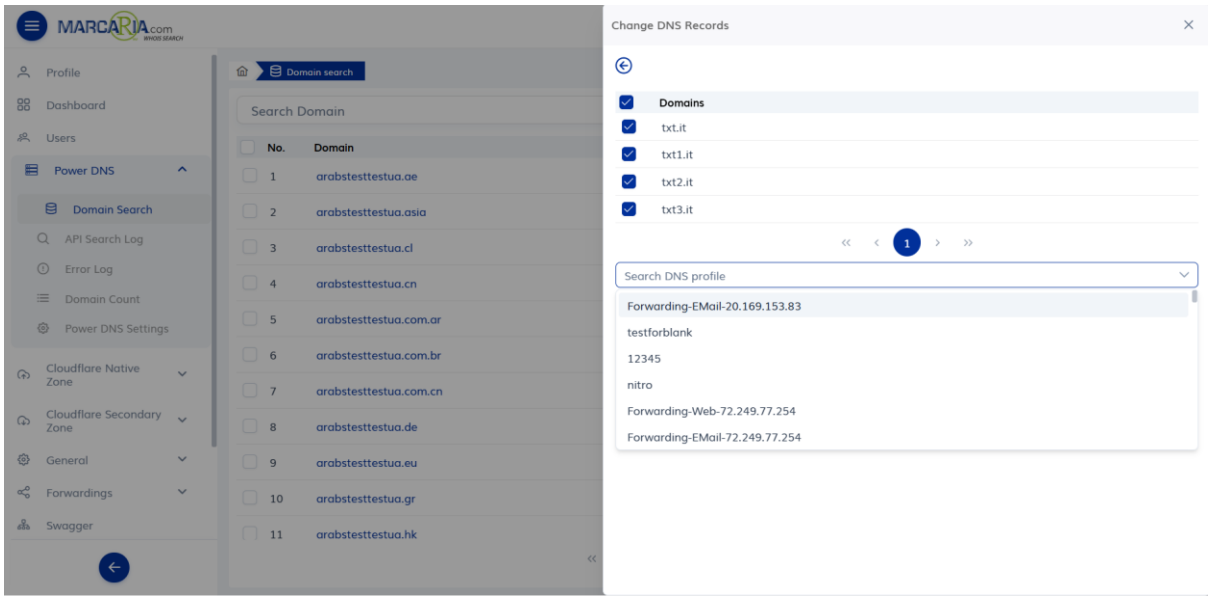
  - **Domains:**
    - It will display the name of the domains which the user entered for the change.

- **Update DNS Records in bulk button:**

  - When the user clicks on this button it will open the popup of - “ The following Domains are selected for DNS Record update ”.

- Here users can get all names of the domain which they are updated.
- Users can get one drop down for the selection of the DNS record profile that they want to update.
- In the drop down box users can view the profiles which are created under the **DNS profile and customer DNS Profile menu**.
- Click on the **Cancel button** and it will close the popup.
- Click on the Save **button and the user** is able to replace old DNS records with the selected profile records and get the notification message - “Profile Updated Successfully.

DNS Record Update : Select the DNS Record Profile and Customer Dns profile that you Want to Update to.



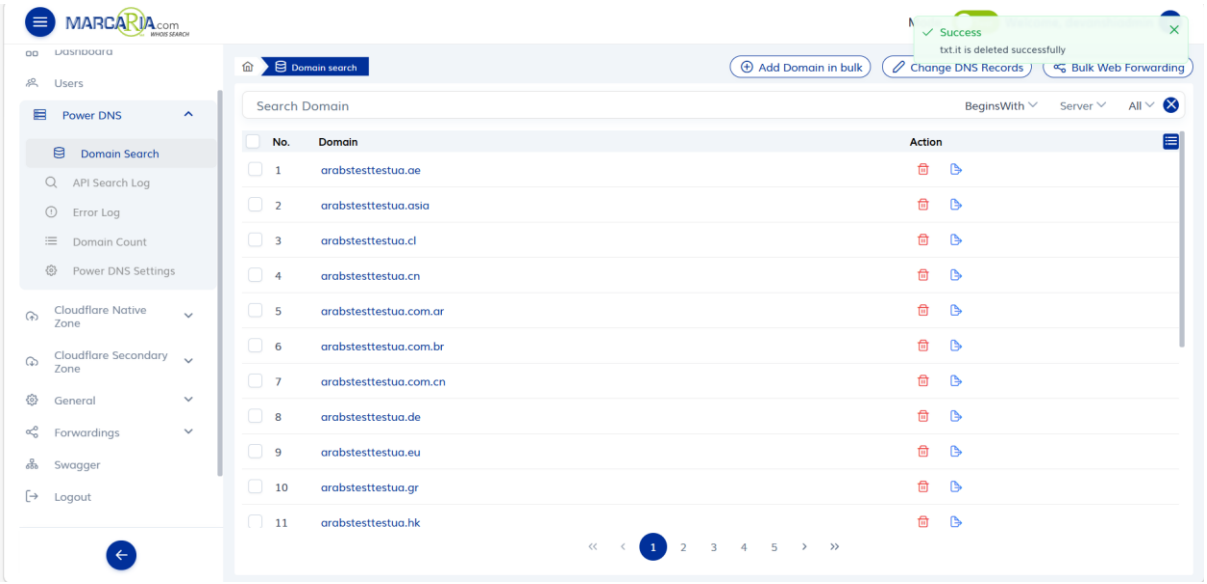
- **Clean DNS records in bulk button:**

  - Here users are able to clean bulk domains records.
  - Users can check/uncheck the domain as per the requirements which are available in the domain list.
  - By clicking on this button the user is able to clean all the records **except SOA** which are available in the selected domain.

- **Remove Domain button:**

  - Here users are able to remove multiple/bulk domains.
  - Users can check/uncheck the domain as per the requirements which are available in the domain list.

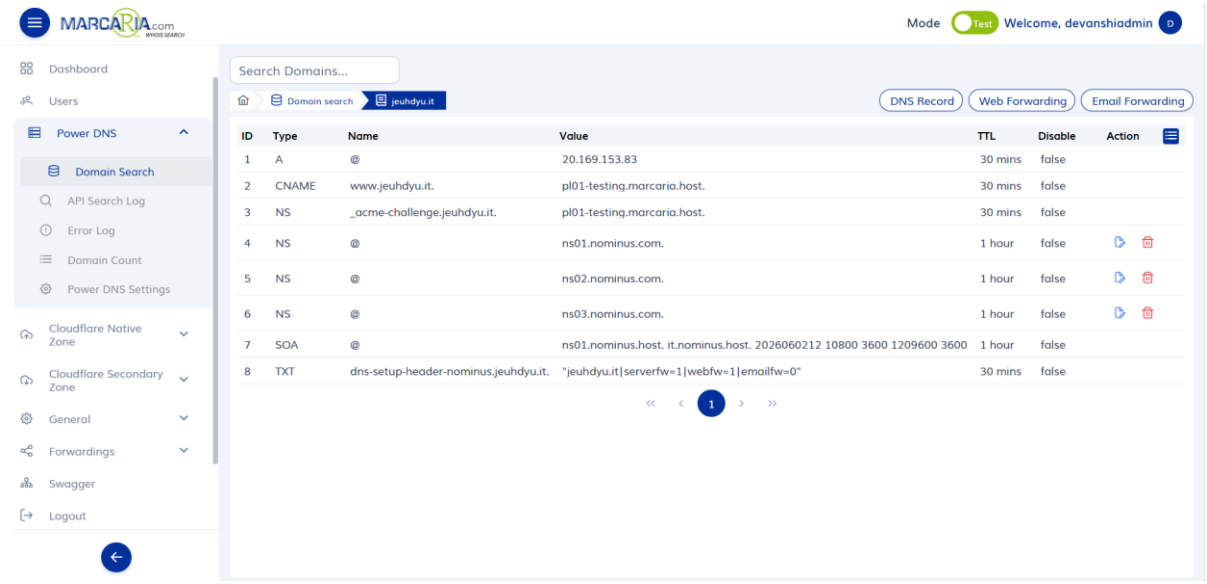
- After checking the checkbox, click on the remove button and the user can able to remove the selected domain from the domain list.



- **Cancel button:**
    - By click on the cancel button “Change DNS record popup will be close”

DNS Record List

- By click on any domain name user can able to view DNS Records.
- In this list users are able to view default records like NS records and SOA record which are automatically added at the time of the creation of the domain.
- Here users can see DNS records with fields ID, type, name, value, TTL, disable and action column.
- Users are able to see DNS records, web forwarding and email forwarding buttons.



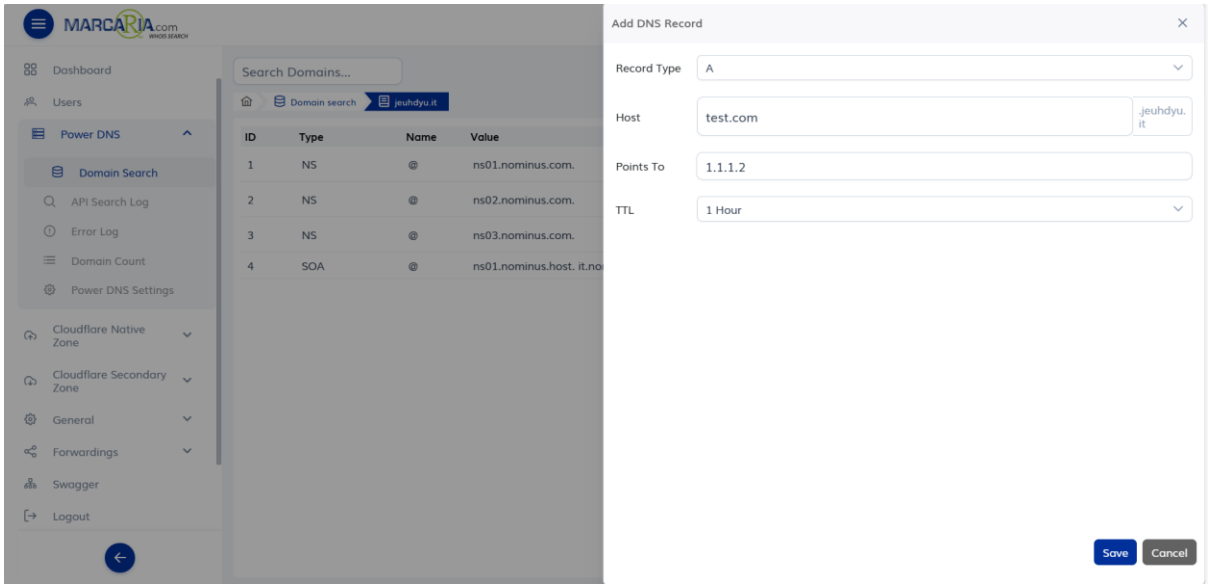
- Here users are also able to add new records by clicking on the DNS Record button.
- When the user clicks on the DNS record button it will open the section for selection of type of record from the drop down.
- Type of records are :
    1. **A** Record
    2. **AAAA** Record
    3. **MX** Record
    4. **TXT** Record
    5. **SRV** Record
    6. **CAA** Record
    7. **CNAME** Record
    8. **NS** Record

- Following are the available types.
    - **A type record :**

        - Firstly, address (A) records are the most common record type by far. In brief, A records map domain names to IPv4 addresses.

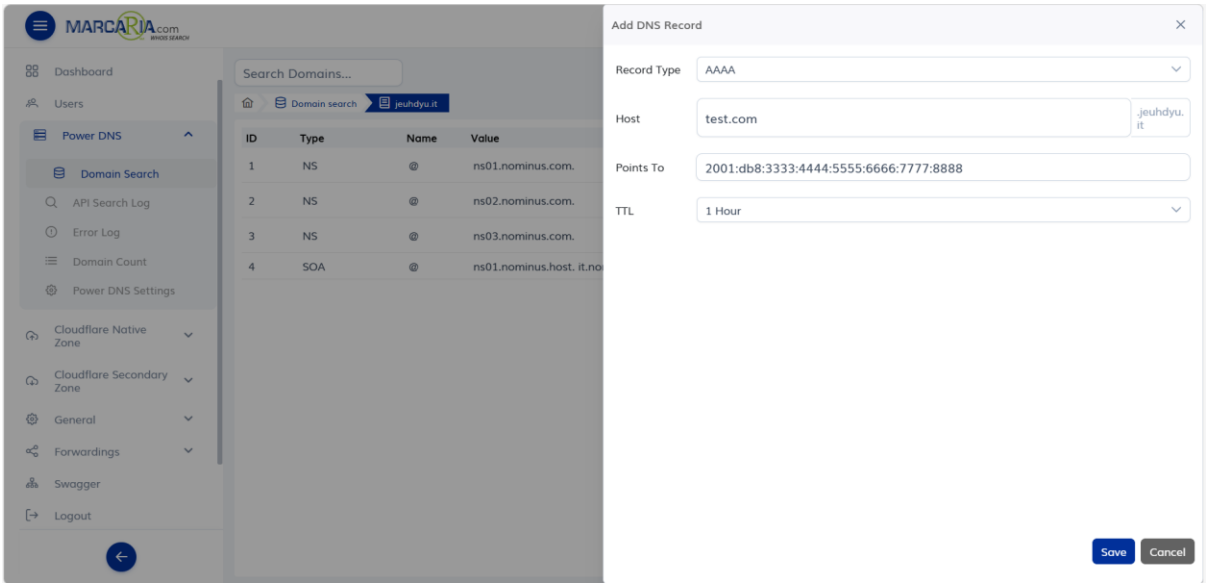
            - IPv4 notation is : **192.0.2.146**.
            - When a user selects to add A type of records it will have the following options…
            - **Host :** Provide host name.
            - **Points To :** Add IPV4 address. Here if user not enter the valid IPV4 it will generate validation message like ..”Entered value is not valid IP”
            - **TTL :** Select time duration from drop down
            - **Save button :** Click on save button it will save the record.

- **Cancel button** : Click on cancel button it will stop the process.



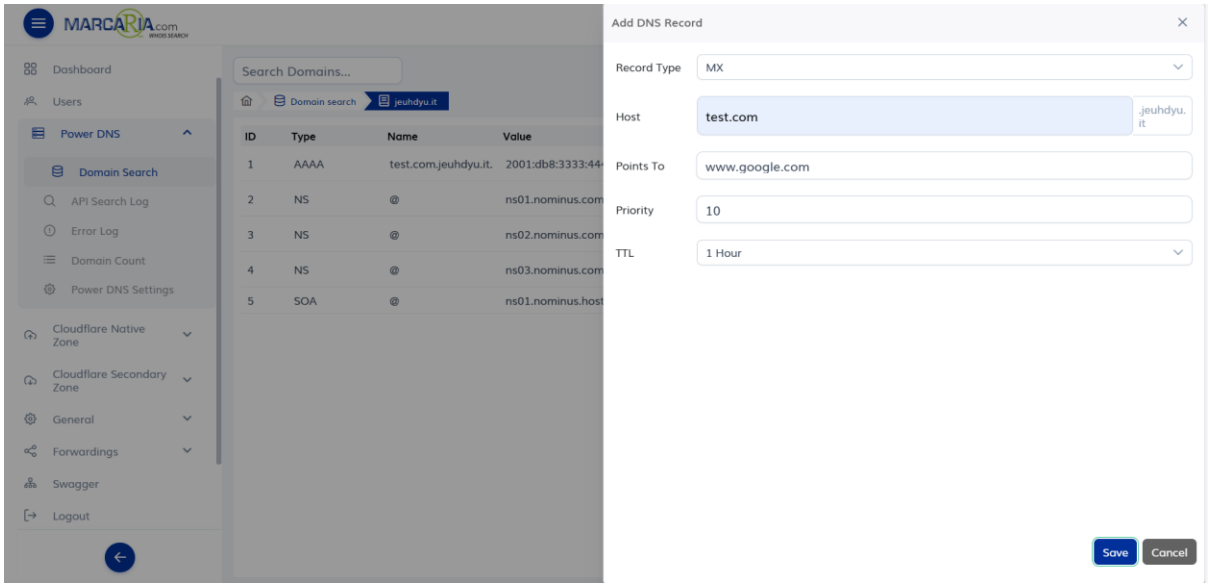
■ AAAA Type Record :

- As the internet gradually makes the transition to IPv6, there are AAAA records (spoken as “quad A”). An AAAA record is the IPv6 equivalent of an A record, responsible for mapping a domain name to an IPv6 address.
- IPv6 notation is :2001:db8:3333:4444:5555:6666:7777:8888
- When a user selects to add AAAA type of records it will have the following options…
- **Host :** Provide host name.
- **Points To :** Add IPV6 address. Here if user not enter the valid IPV6 it will generate validation message like ”Enter value in valid IPV6 format”
- **TTL :** Select time duration from drop down
- **Save button :** Click on save button it will save the record.
- **Cancel button** : Click on cancel button it will stop the process.

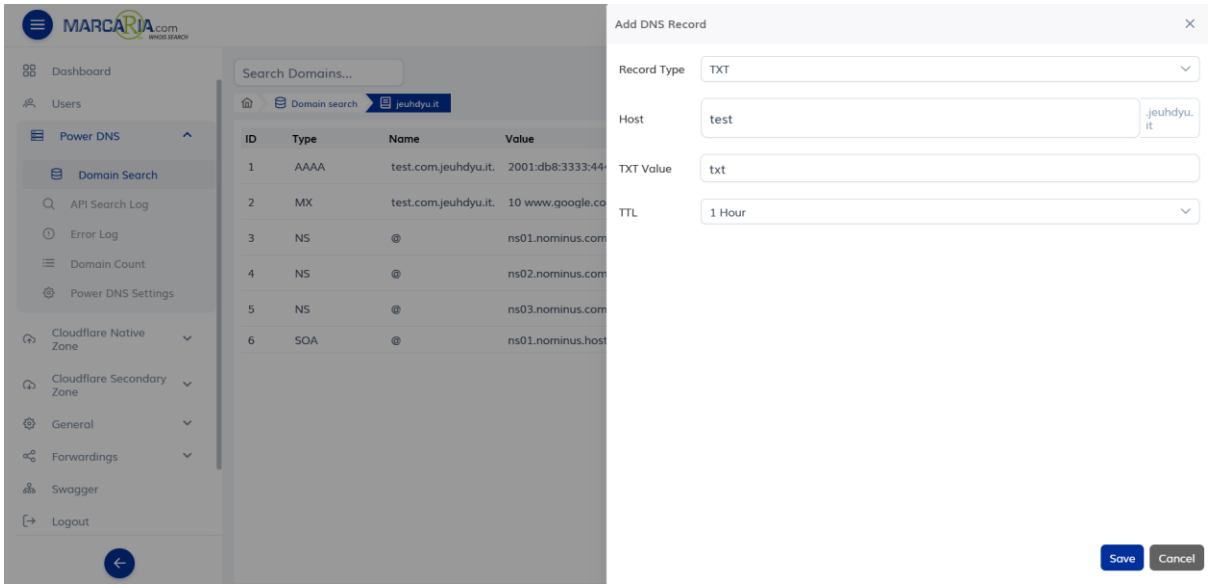


- **MX Type Record** :

  - A mail exchanger (MX) record stores the domain names of mail servers responsible for receiving emails on behalf of a domain.
  - These records can be helpful to identify authorized mail servers. An MX record is sometimes called an MX entry, particularly in configuring mail servers.
  - **Host :** Provide host name.
  - **Points To :** Value is Required.
  - **Priority :** Set the priority. The priority you set when you add an MX record set specifies the sequence that an email server receives emails. A smaller value indicates a higher priority.
  - **TTL :** Select time duration from drop down
  - **Save button :** Click on save button it will save the record.
  - **Cancel button :** Click on cancel button it will stop the process.

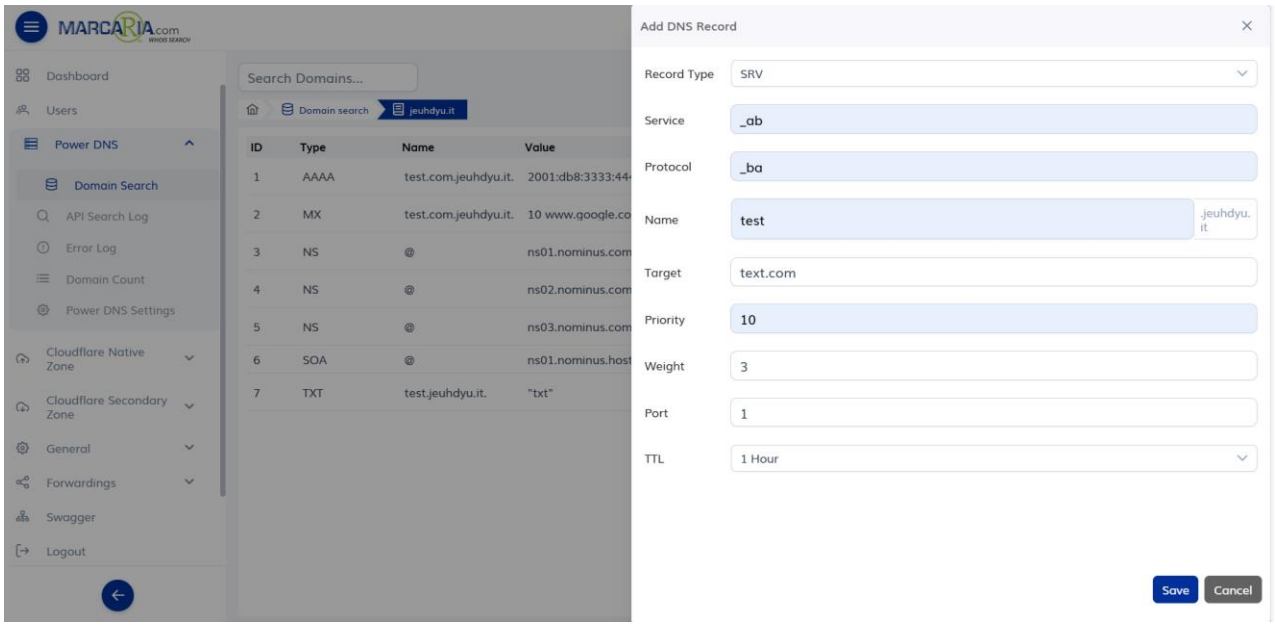


- **TXT Type Record** :
  - A text (TXT) record can store any type of descriptive information in text format. For example, network antivirus providers can use TXT records to send and receive information.
    - **Host :** Provide host name.
    - **TXT value :** Provide text value for the record.
    - **TTL :** Select time duration from drop down
    - **Save button :** Click on save button it will save the record.
    - **Cancel button :** Click on cancel button it will stop the process.



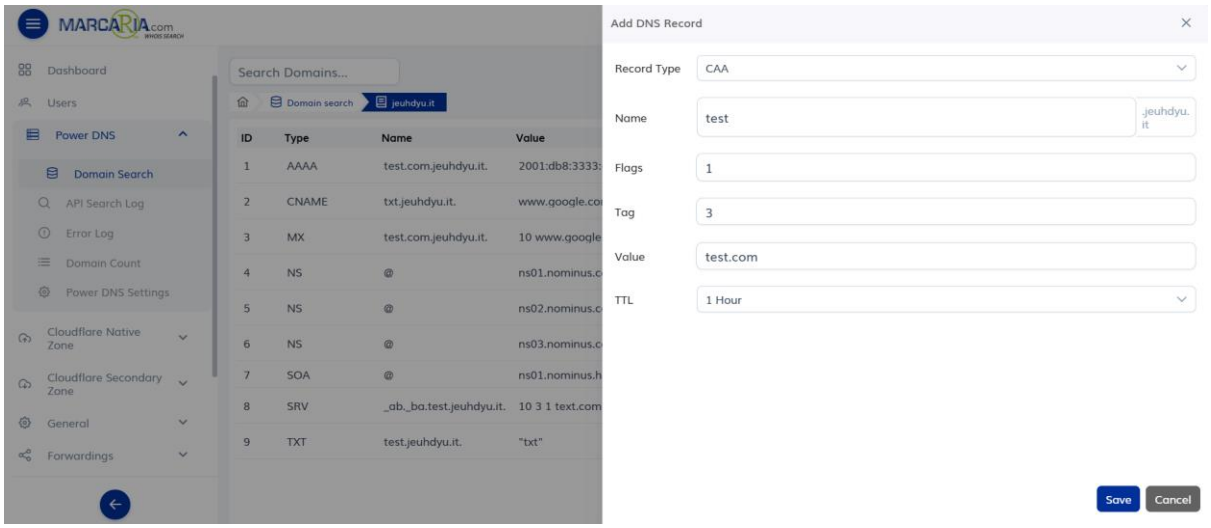
- **SRV Type Record :**

- The SRV record is a Domain Name System (DNS) resource record. It's used to identify computers hosting specific services. SRV resource records are used to locate domain controllers for Active Directory.
    - **Service :** Symbolic name for a service. Here value must be start with underscore (_)and alphanumeric
    - **Protocol :** Specifies if the service is using TCP or UDP protocols. Here value must start with underscore (_) and alphanumeric.
    - **Name :** Stores a domain name.
    - **Target :** The canonical hostname for the machine providing the service.
    - **Priority :** Determines which server is looked at first, with lower values giving higher priority.
    - **Weight :** Determines which server is looked at first if more than one has the same priority value. Higher values give more priority.
    - **Port :** The TCP or UDP port the service is running on. {Transport Layer Protocol : 1. TCP (Transmission Control Protocol) , 2. UDP(User Datagram Protocol)}
    - **TTL :** Select time duration from drop down
    - **Save button :** Click on save button it will save the record.
    - **Cancel button :** Click on cancel button it will stop the process.



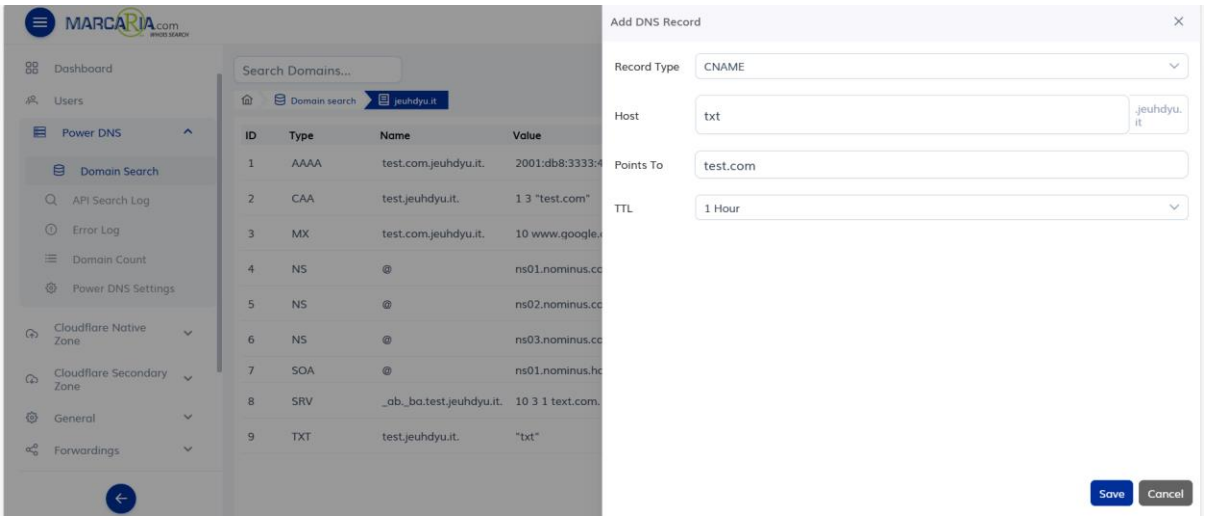
- **CAA Type Record**:

- CAA is a type of DNS record that allows site owners to specify which Certificate Authorities (CAs) are allowed to issue certificates containing their domain names.
    - **Name :** The name of the domain that is being certified.
    - **Flags :** Can be either 1 (critical) or 0 (non-critical). A critical flag means the certificate authority cannot use the CAA record if it doesn't understand the property. A non-critical flag means it can use the CAA record regardless of whether it understands the property.
    - **Tag :** The three tag options are issue (authorized to issue a single certificate), issuewild (authorized to issue a wildcard certificate), and iodef (specifies a URL for reporting policy violations).
    - **Value :** Enter value for selected tag.
    - **TTL :** Select time duration from drop down
    - **Save button :** Click on save button it will save the record.
    - **Cancel button :** Click on cancel button it will stop the process.



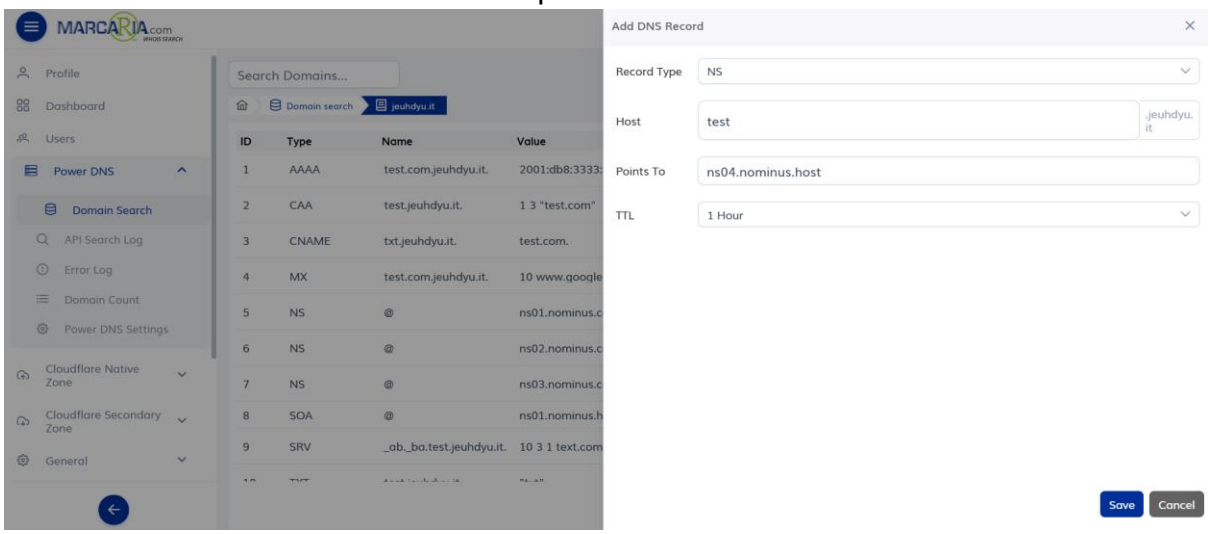
- **CNAME Type Record :**
    - Redirects a domain to a different domain. CNAME records are typically used to map a subdomain such as www or mail to the domain hosting that subdomain’s content.
        - **Host :** Provide host name.
        - **Points To :** Add target domain or host name.
        - **TTL :** Select time duration from drop down.
        - **Save button :** Click on save button it will save the record.

- **Cancel button :** Click on cancel button it will stop the process.



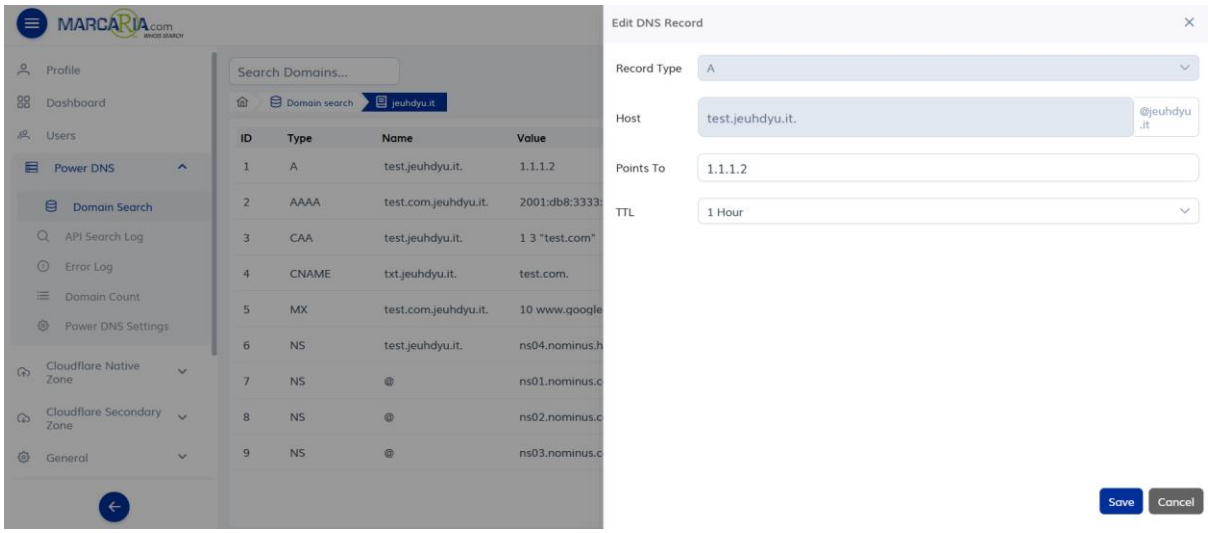
- **NS Type Record :**

- Provides a list of the authoritative name servers responsible for the domain. It's important to note that NS records point to the domain of the name server—not IP addresses.
  - **Host :** Provide host name.
  - **Points To :** Add IPV4 address.
  - **TTL :** Select time duration from drop down.
  - **Save button :** Click on save button it will save the record.
  - **Cancel button :** Click on cancel button it will stop the process.



❖ Edit Icon :

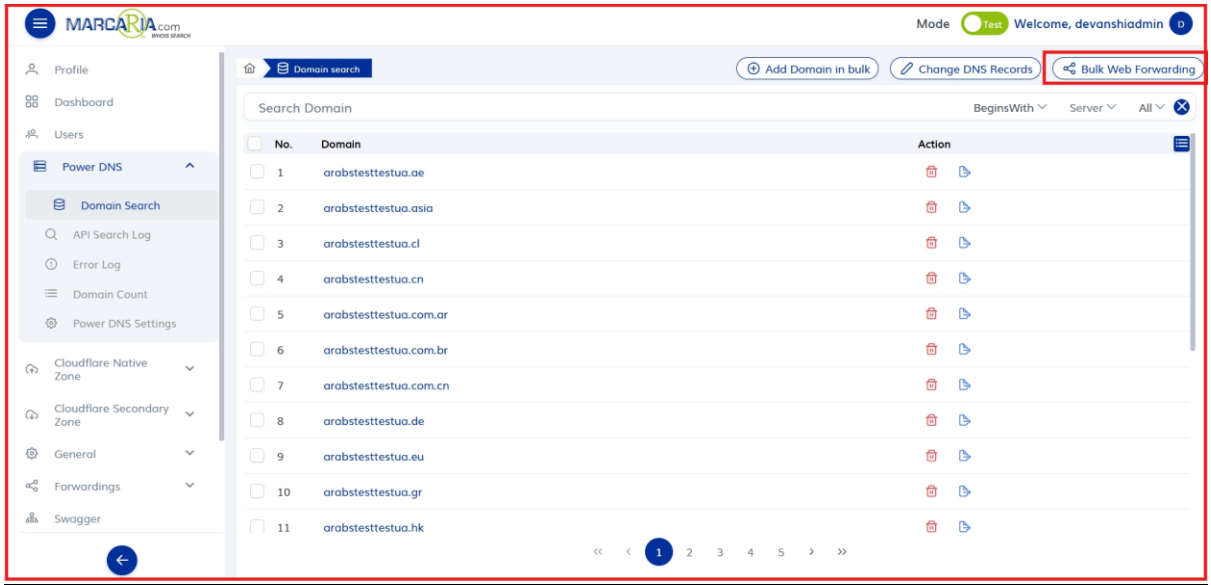
- Click on the edit icon from the Action field of the DNS records table.
- Users can get the popup with following details..
  - Fields of the record which are generated by the user.
  - Save button
  - Cancel button
- **Type** and **hostname** are disabled at the time of edit and all other fields users can modify.
- **NOTE:** For the SRV record type - Type, Service, Protocol and Name fields are disabled.



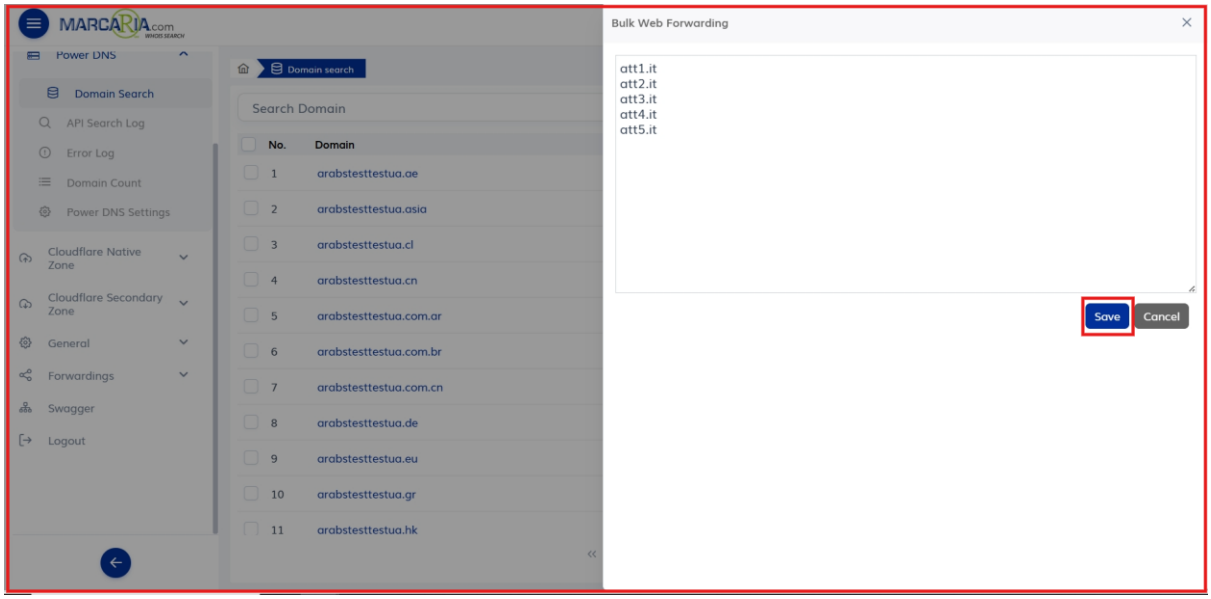
- **Save Button:** Click on the save button after doing modification it will update old data with the updated data.
- **Cancel Button:** click on the cancel button it will close the popup.
- **`NOTE :** Edit option is not available for SOA record.
- **NOTE :** Users can add @ if we want domain name as name.

Bulk Web Forwarding Button

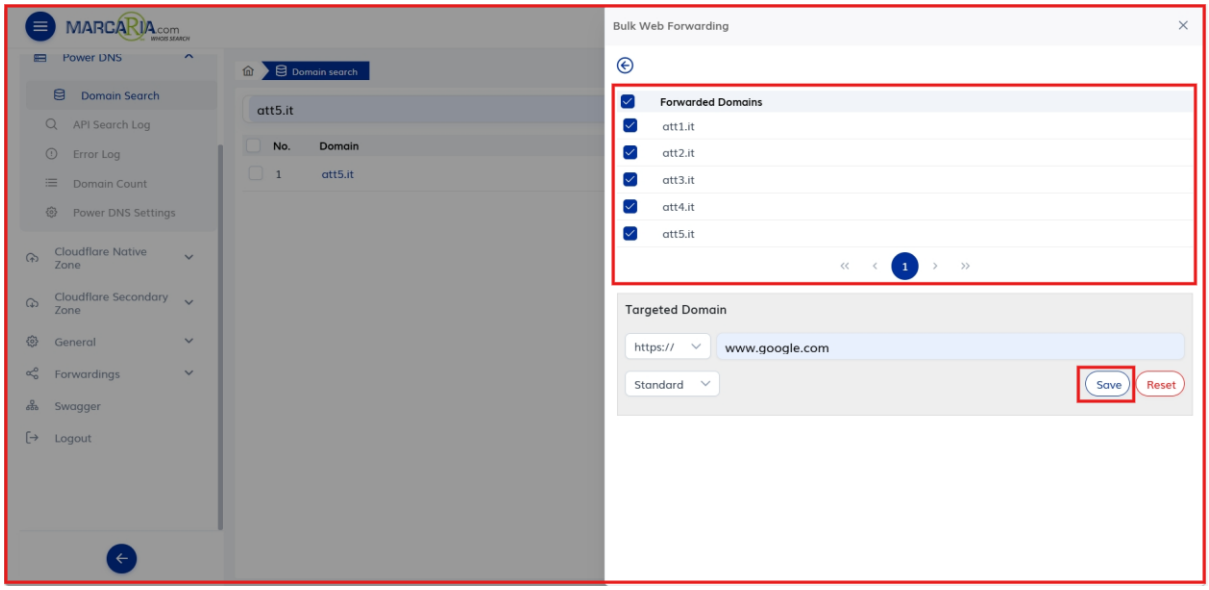
- Users can view the bulk web forwarding button domain list page.



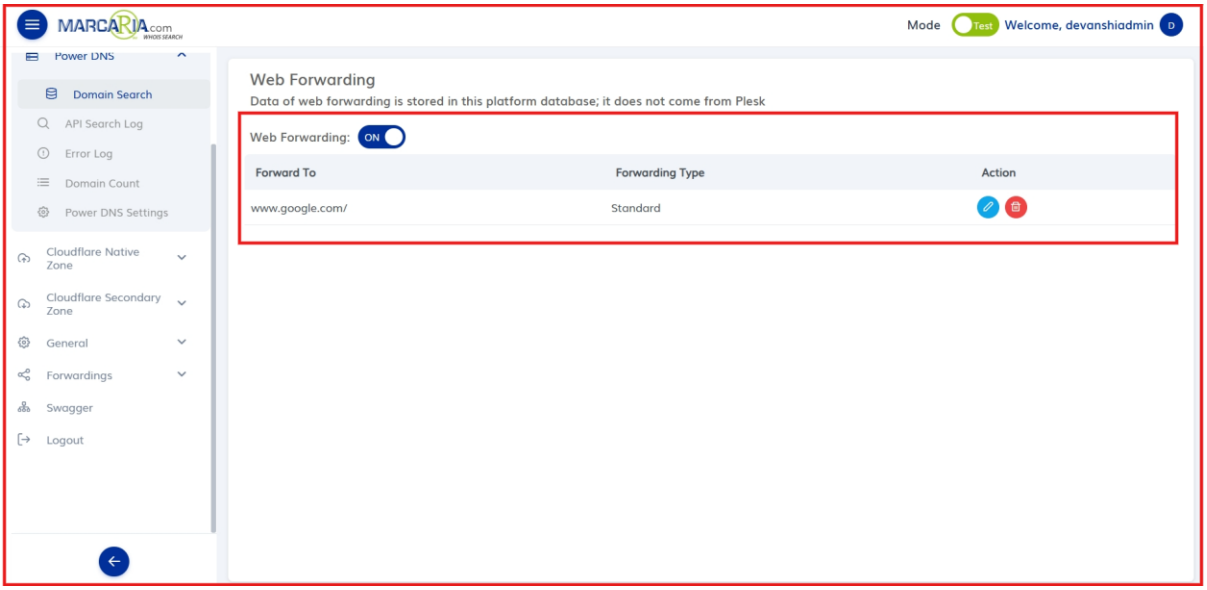
- By clicking on this button users can view the popup as below.
- Where users can enter domains to enable web forwarding in bulk and click on the continue button.



- By clicking on the Save button the user can view a popup like below, where
- users can view available domain lists and also can check / uncheck if you don’t want to enable web forwarding.

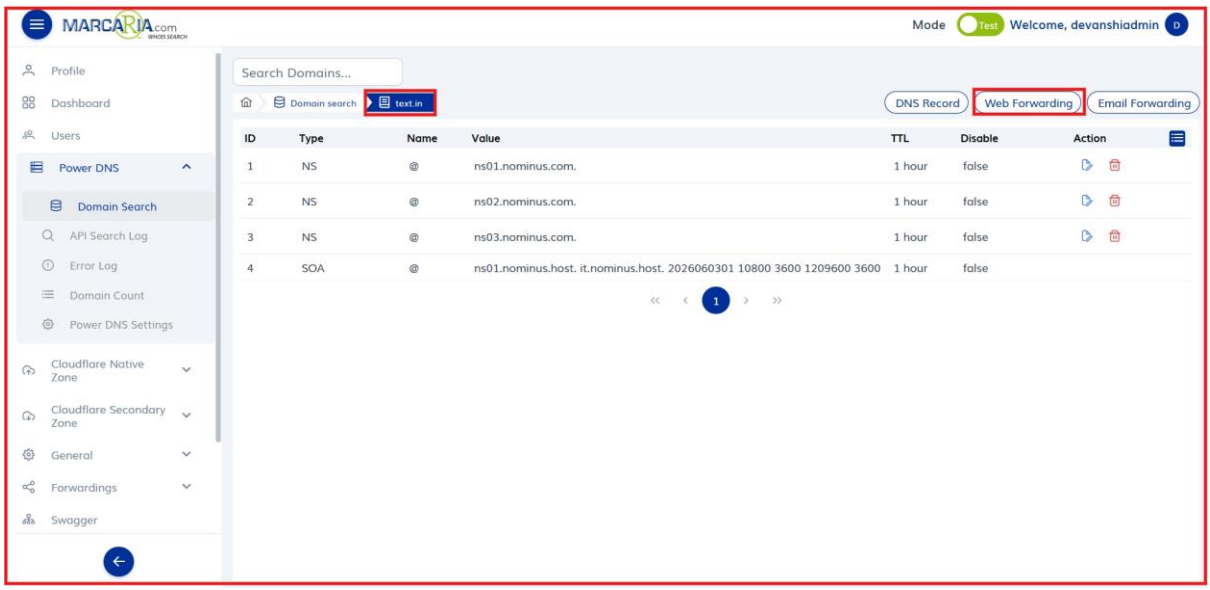


- By enabling web forwarding, users can view below messages like “Forwarding on successfully” and users also can view successfully forwarded domains.
- Users can set target domains where users have forward domain by clicking on save users can view below message like domain name and record added successfully and popup will be closed.
- Then users can view domain web forwarding enabled successfully and also can view previously set target domains.
- Firstly, need to understand this TurnOnCaddyForwarding azure app settings to determine which web forwarding service will be used.
    - **When TurnOnCaddyForwarding = True**
    - The system shall:
        - Route all web forwarding requests through the Caddy Server.
        - Use the Caddy Server APIs for both Single Domain and Bulk Web Forwarding.
        - Display the Caddy Server user interface.
    - **When TurnOnCaddyForwarding = False**
    - The system shall:
        - Continue using the existing Plesk Web Forwarding implementation.
        - Route all forwarding requests to the Plesk APIs.
        - Display the existing Plesk forwarding functionality without modification.



Web Forwarding Button

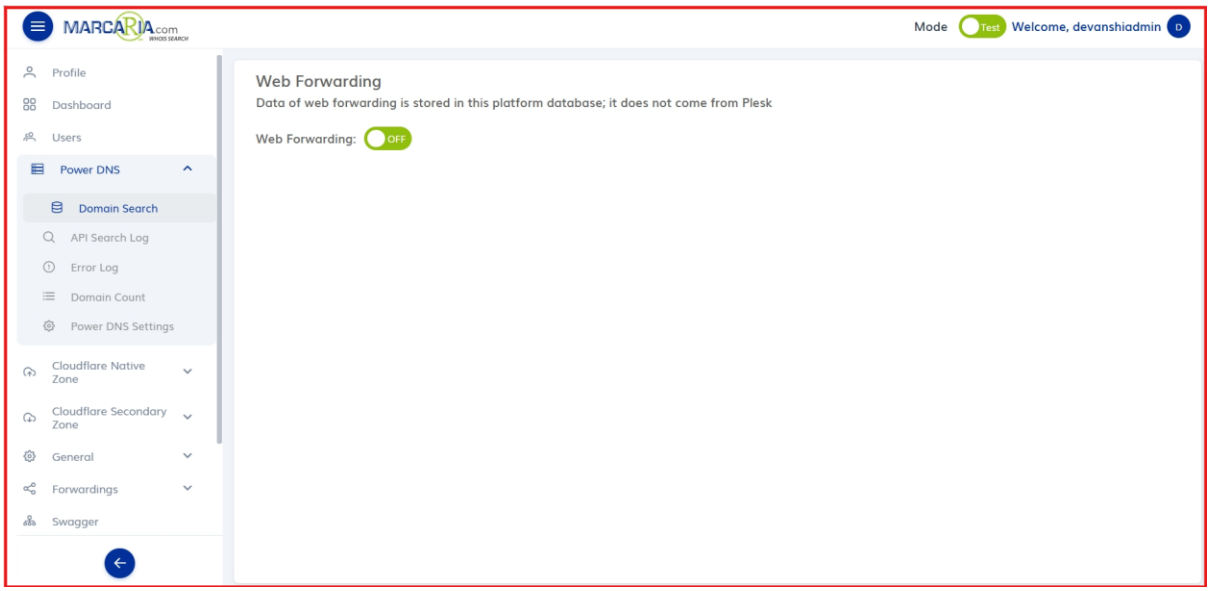
- When users click on the web forwarding button it will open the Web forwarding page for selected domain.



- TurnOnCaddyForwarding Application Setting
    - The system shall use the **TurnOnCaddyForwarding** application setting to determine which web forwarding service will be used.
- **When TurnOnCaddyForwarding = False** Here Users can enable or disable Web Forwarding using the ON/OFF switch.
    - The system shall: Continue using the existing Plesk Web Forwarding implementation.
    - Route all forwarding requests to the Plesk APIs.

- Display the existing Plesk forwarding functionality without modification.

- When TurnOnCaddyForwarding = True **[Currently this** service settings **are used in new admin panel] Here Users cannot see Web Forwarding the ON/OFF switch.**
    - The system shall:
    - Route all web forwarding requests through the Caddy Server.
    - Use the Caddy Server APIs for both Single Domain and Bulk Web Forwarding.
    - Display the Caddy Server user interface.
- When Web Forwarding is enabled, the system validates the forwarding configuration, creates the domain in Plesk if required, and adds the necessary DNS records from the configured forwarding profile.
- When Web Forwarding is disabled, the forwarding-related DNS records and configuration are removed.
- If both Web Forwarding and Email Forwarding are disabled, the forwarding configuration is removed from the forwarding server.
- The Add Web Forwarding section becomes available when Web Forwarding is enabled.
- When users add new Internationalized Domain Names (IDNs) like "superkıbrıs.com," they can choose to enable or disable web forwarding for these domains. For instance, in the example provided, the domain name "superkıbrıs.com" would be converted to "xn--superkbrs-0pbc.com" for compatibility with systems that don't support non-ASCII characters actually in logic the domain will be converted using punycode and then move forward to enable to forwarding and that will be considered for web and email forwarding both.



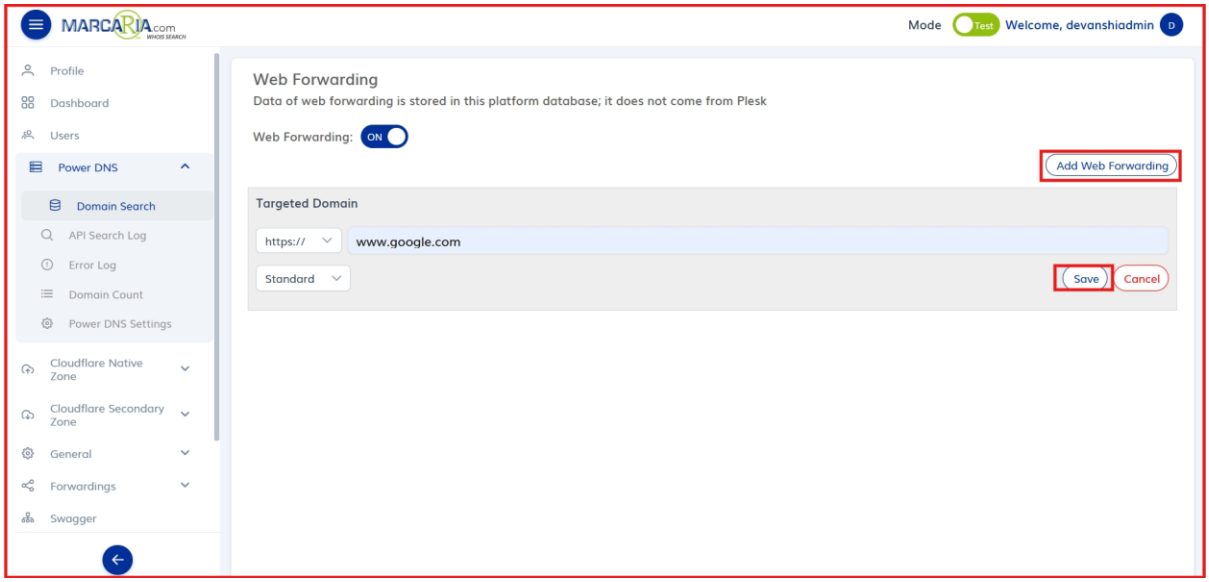
❖ **Add Web Forwarding Button:**

- Click on Add Web forwarding Button it will show the section for entering the information about Target domain and Forwarding Type.
- Added records also create on plesk as well database same for email forwarding

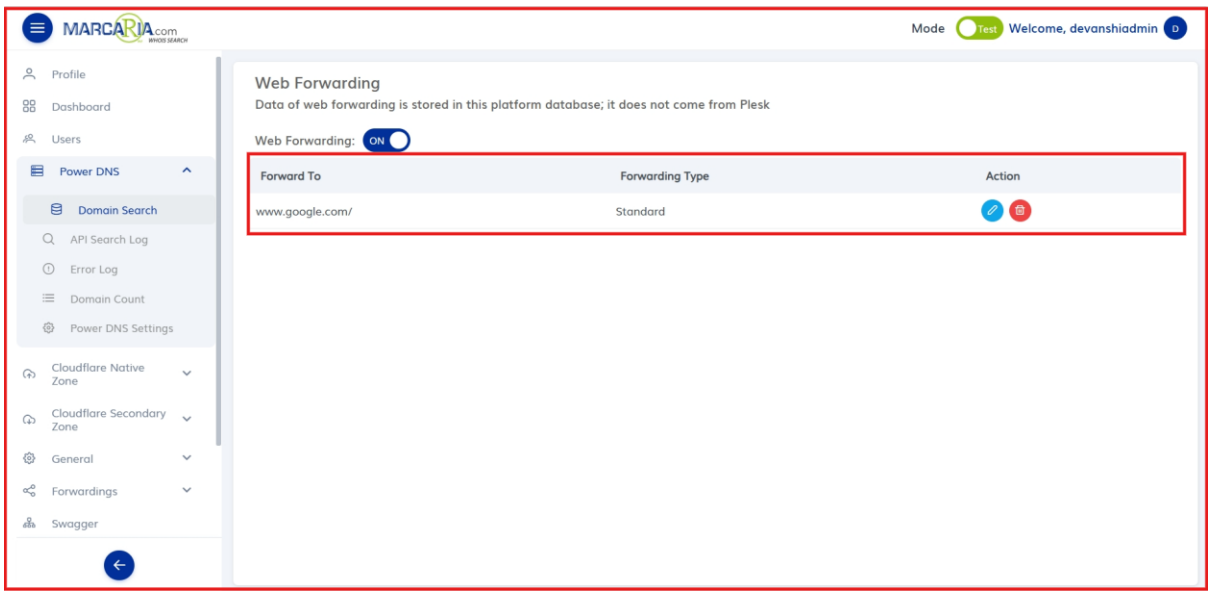
- **Target Domain:**
  - Here user can select one of the supported protocols: 1. http://
    2. https://
  - Enter the Website URL in which you want forwarding.

Forwarding Type

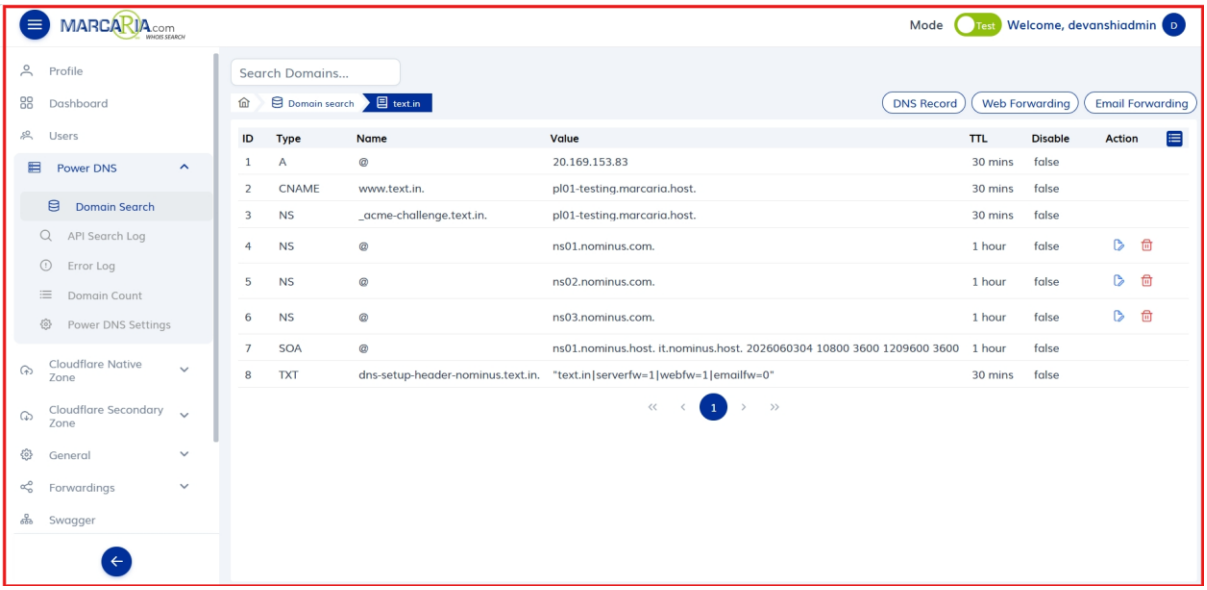
- When **TurnOnCaddyForwarding = True**, the system shall display only the following forwarding types:
  - Standard
  - Frame
  - The **None** forwarding option shall not be available.
- When **TurnOnCaddyForwarding = False**, the system shall display the following forwarding types:
  - Standard
  - Frame
  - None
- Here users get three options in **Plesk Server** such as:
- **Standard:** With the Standard forwarding, users who have been redirected to another URL can see the destination URL in the browser address bar.
- **Frame:** With the Frame forwarding, when visitors are redirected to another site, the address bar of their browsers continues to show the source URL. Thus, visitors remain unaware of the redirection.
- **None:** It means not redirecting.
- Click on save button to save the data.



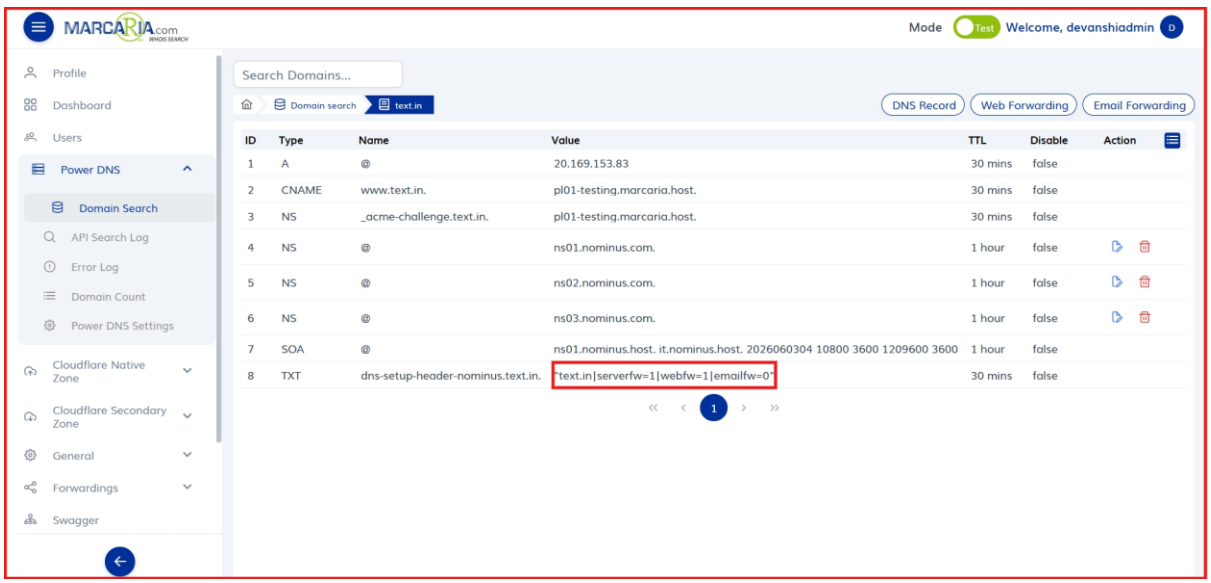
- After saving the Web Forwarding details, the record will appear in the list. Users can edit or delete the record using the available action buttons.



- After successfully Save the data Move to your domain i.e. **Domain List > Click on your domain name.**
- Here on the DNS records page, users need to check the following things.
  - If there is the same type of record already available in the list then it will make disable = true to the new record and consider the newly added record.

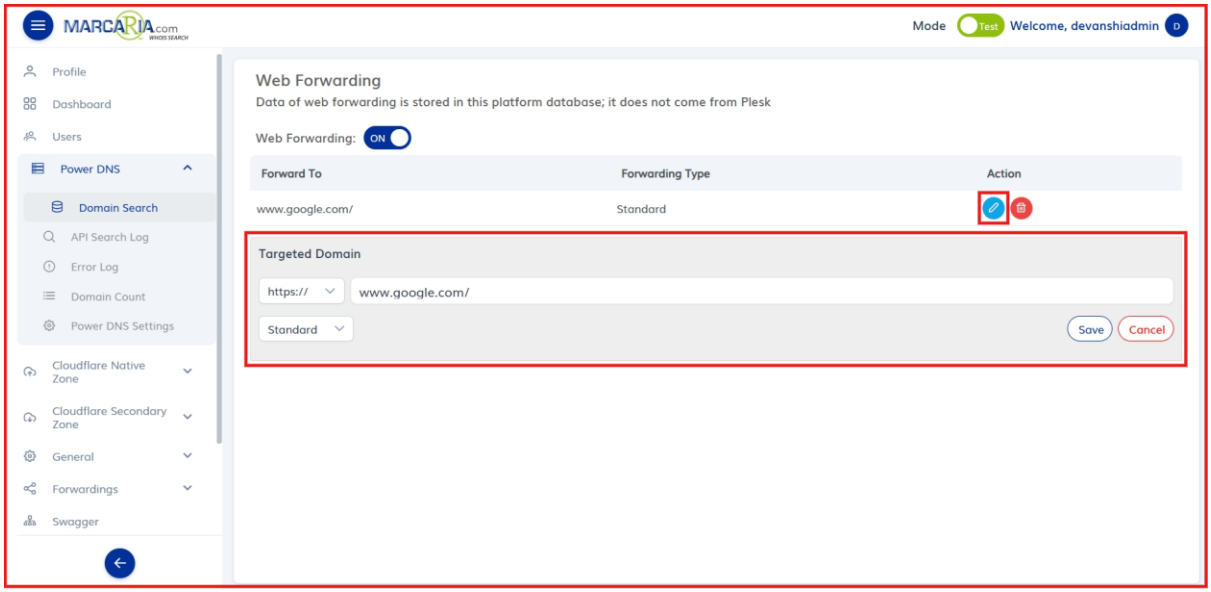


- TXT Record is by default added and it carries information like ..
  - ServerFW=’1’ means “1” is Server ID
  - WebFW=’1’ means “1” for Web forwarding ON and “0” for OFF
  - EmailFW=’0’ means email forwarding is OFF and “1” for ON.



❖ Edit Web forwarding Icon:

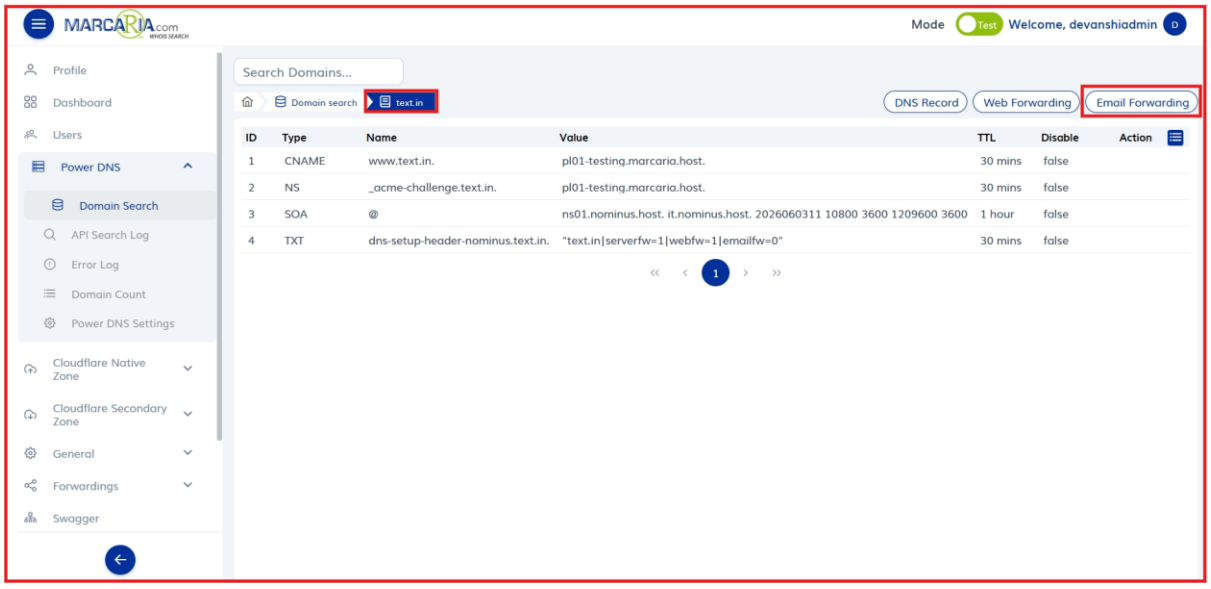
- Once domain forwarding is created it will create a table with fields like Forward To, Forwarding Type, Action - Edit Icon.
- **Forward To:** It will display the name of the domain on which you want to redirect.
- **Forwarding Type:** It will display the selected type for the domain.
- **Edit Icon:** Click on the Edit icon it will open the section which allows us to edit the data.



- **Save Button:** Click on the Save button it will save the modification.
- **Delete Button:** Click on the Delete button it will delete the record.
- **Cancel Button:** It will Cancel the process.

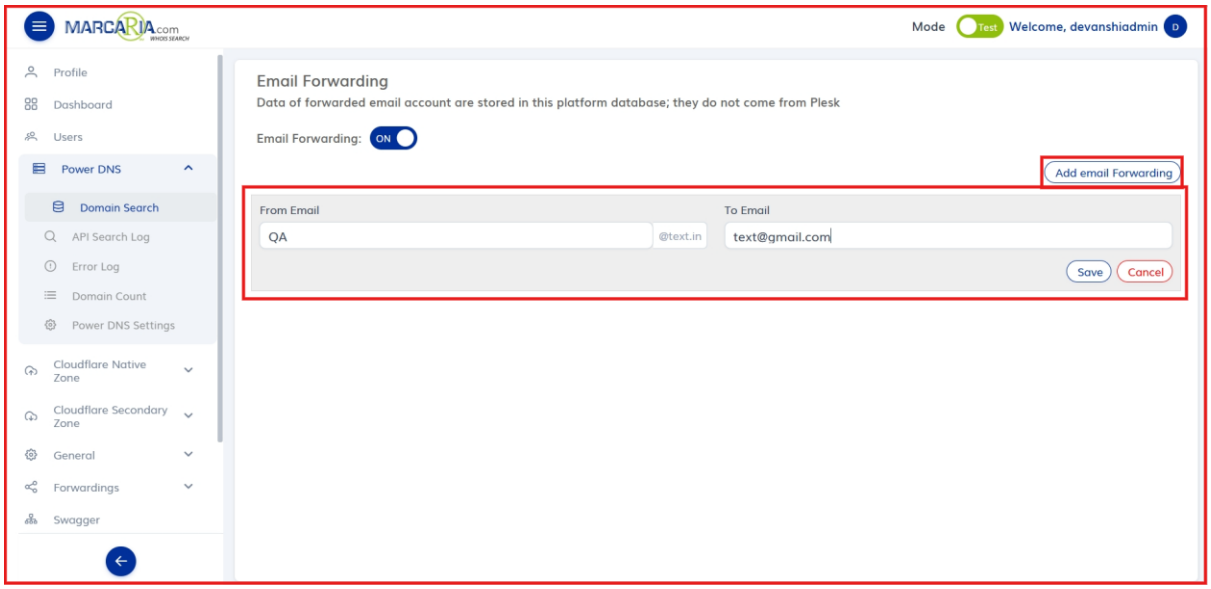
❖ Email Forwarding Button:

- Click on the Email Forwarding button to open the Email Forwarding page for the selected domain.

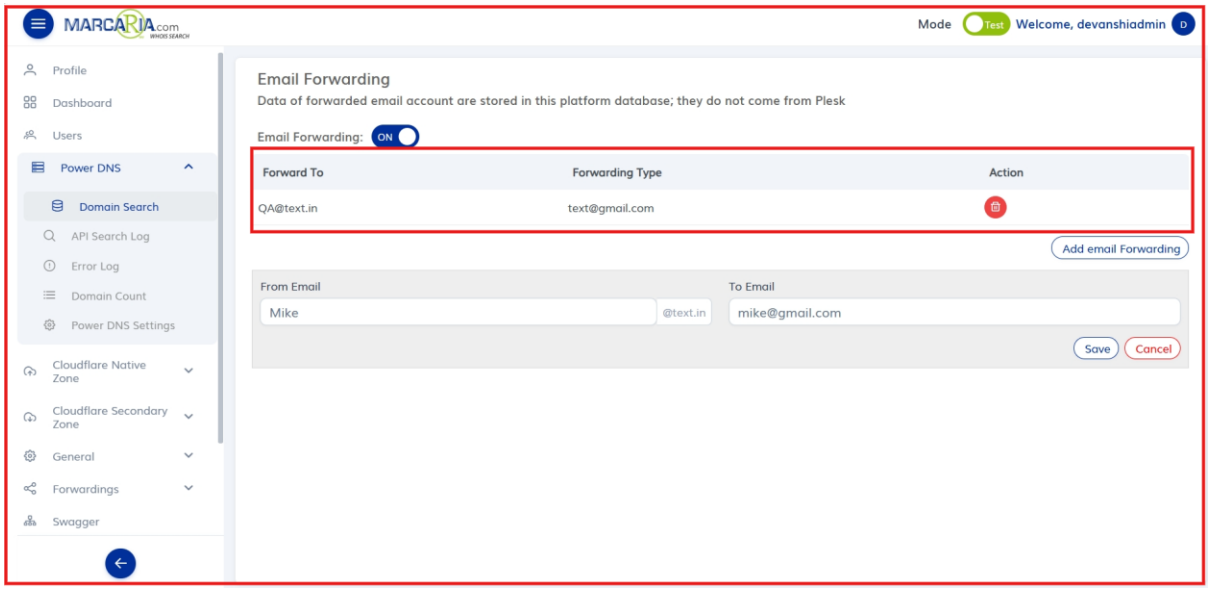


- Here Users can enable or disable Email Forwarding using the ON/OFF switch.
- Here users can ON the switch to enable email Forwarding.
- And users can OFF the switch to disable the email Forwarding.

- By doing ON email forwarding it will open the section for the entering details of From Email and To Email.
- Here Users can add multiple records up to the configured maximum limit.
- **From Email :** Here users need to add a source email address from which you want to forward the emails.
- **To Email :** Here users need to add a destination email address in which they want the email address.



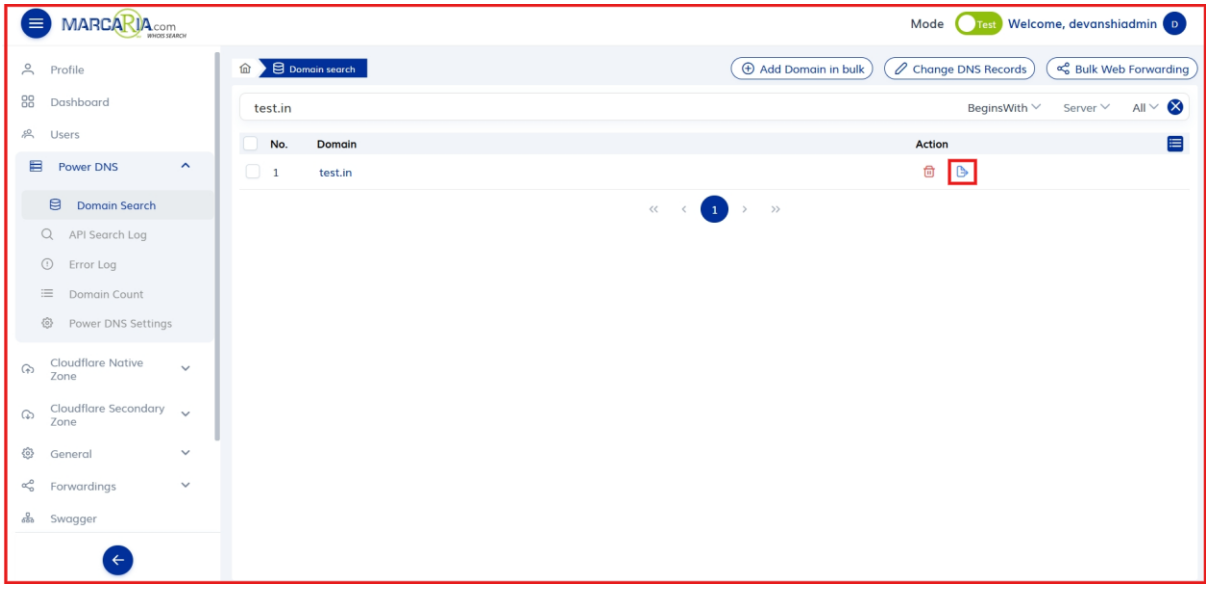
- **Save:** Click on the Save button record is added and display it on the screen.
- **Cancel:** Click on the Cancel button, the process will stop.



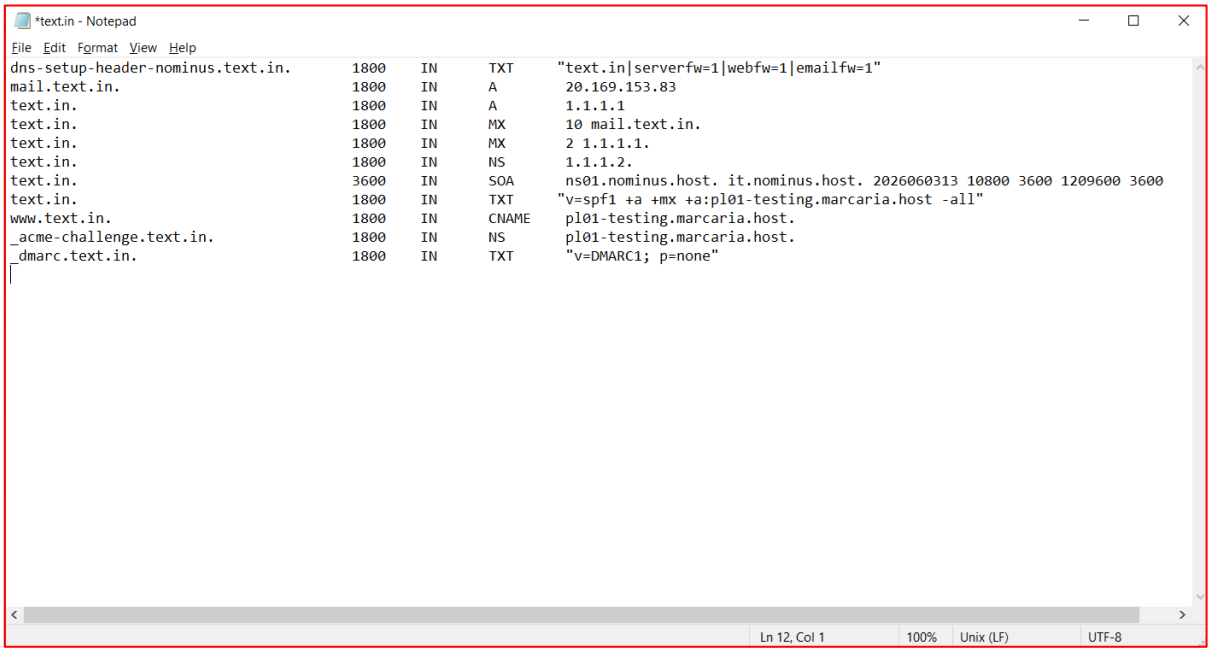
- **Delete Icon:** Users are able to view the delete icon at the table of the list of email addresses. Click on it and the user will be able to delete the record.

❖ Export Zone Button:

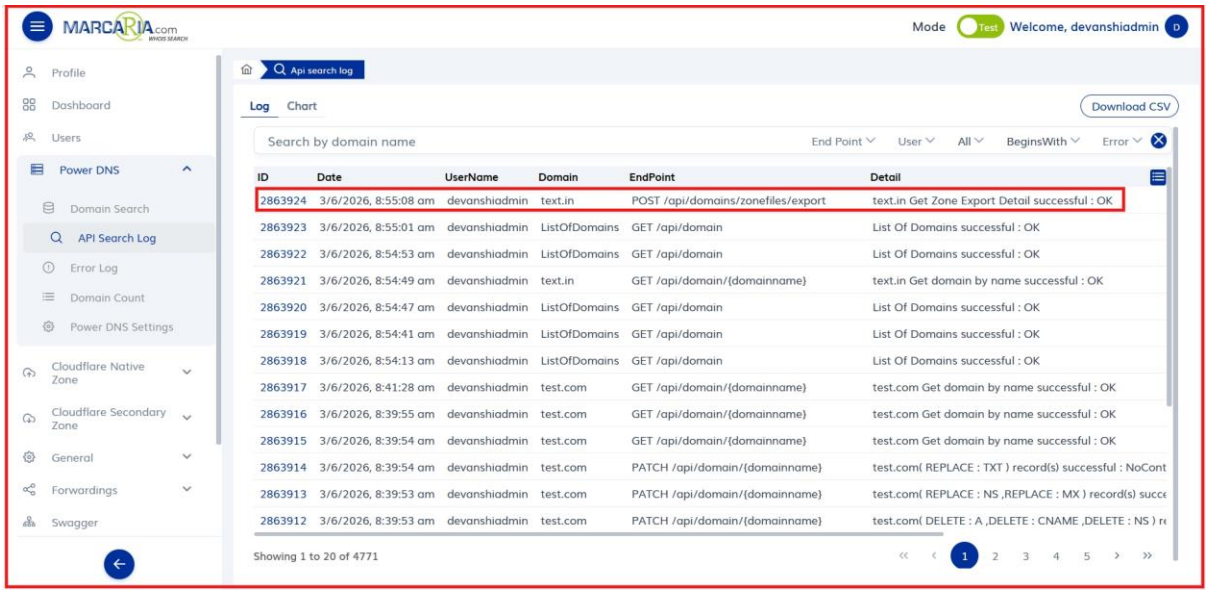
- Users can view the Export Zone icon in the Action column of the Domain Search page.



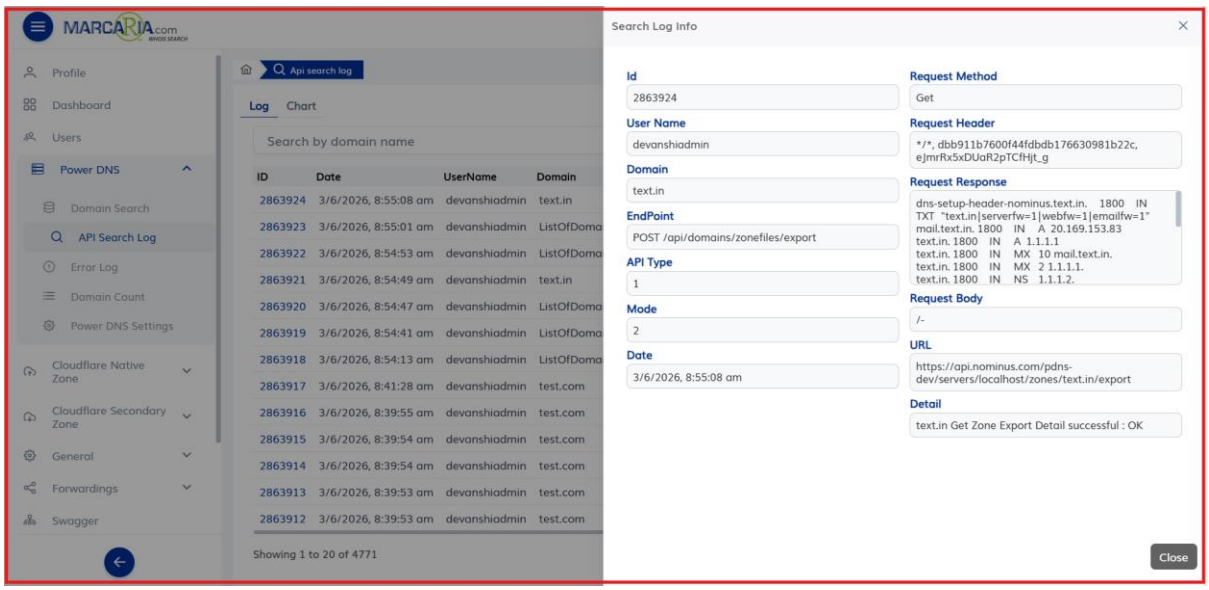
- Click on the **Export Zone** icon to download the DNS zone file for the selected domain.
- The downloaded file contains the DNS records of the selected domain.
- The exported file name will be the same as the domain name.



- Users can view log in API search log



- Users can view API search log info by clicking on log ID.

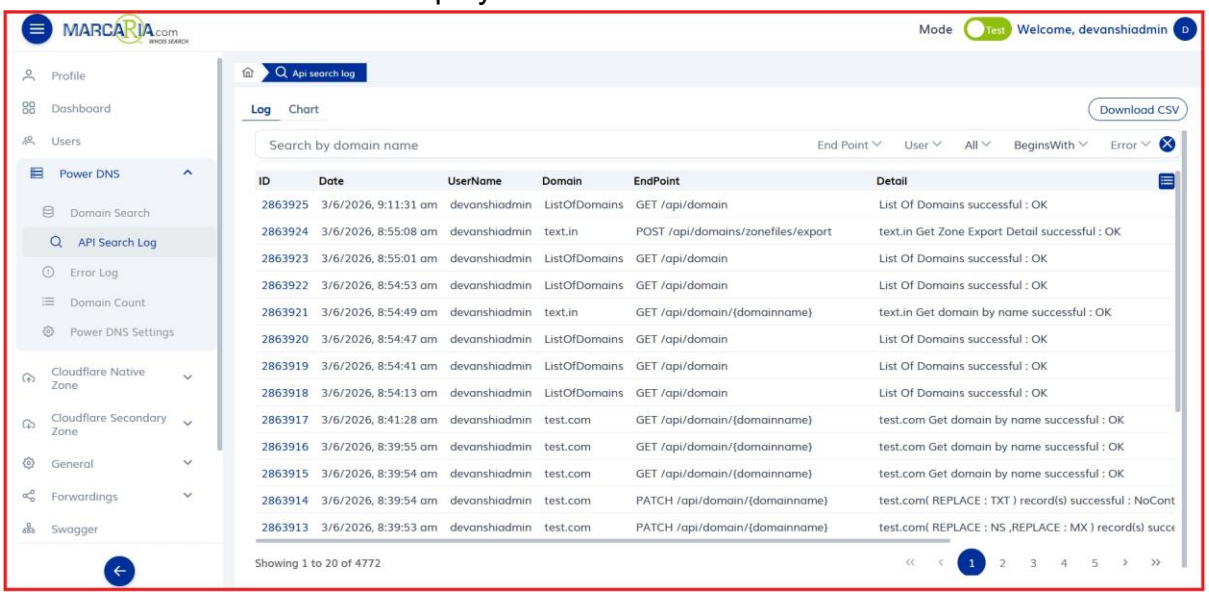


API Search Log:

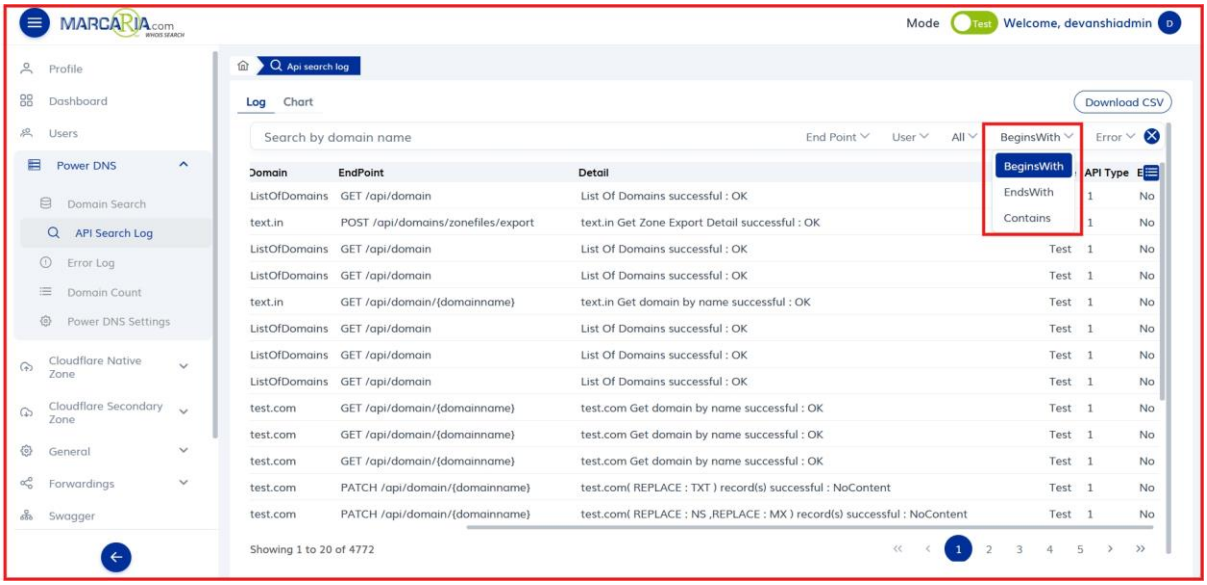
a. Here users are able to view API log details.
b. Users are able to download the CSV file of the API log.

c. Users are able to see different filter options such as…
   i. **User:** Can filter according to the Users.
   ii. **Mode:** Can filter according to the Mode.
   iii. **API Type:** Can filter according to the API Type.
   iv. **Error:** Can filter according to the Error Type.
   v. **Domain:** Search by entering the domain name.
   vi. **FilterBy:** Can filter according to the beginning with, end with and contains.
   vii. **Endpoint:** Can filter according to the Endpoint.

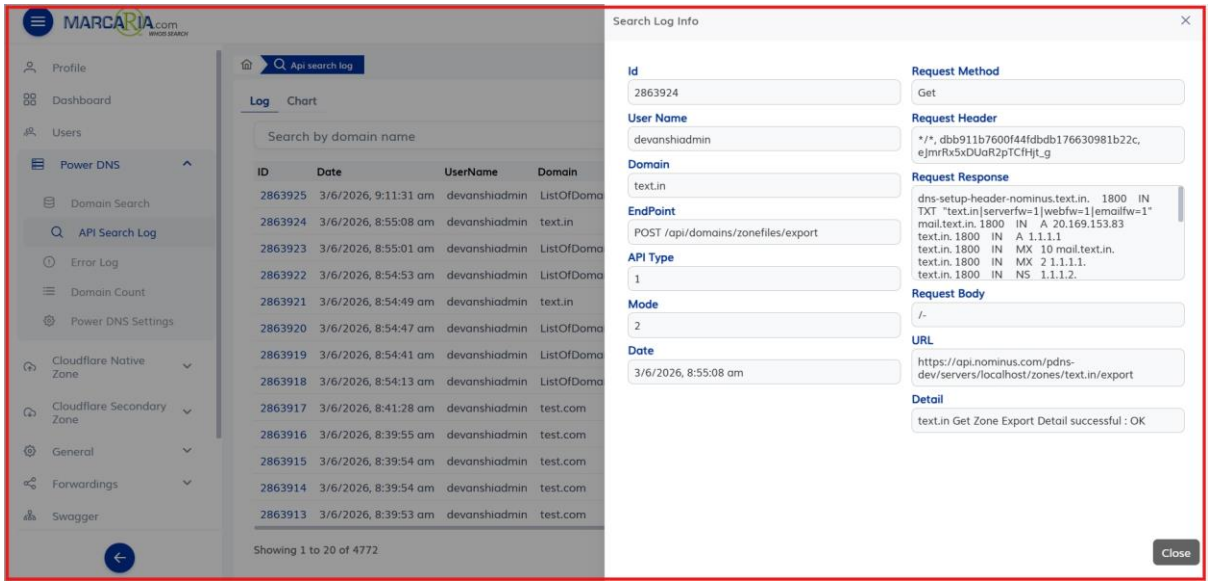
d. Users are able to view information like….
   i. **ID:** Unique identification number.
   ii. **Date:** Timestamp of the error log.
   iii. **User Name:** Display user name. (Login with that name - admin)
   iv. **Domain:** Display domain name
   v. **End Point:** Display API endpoint.
   vi. **Detail:** Provides information.
   vii. **Mode:** Display information of mode.
   viii. **APIType:** Display type of API.
   ix. **Error:** Display status if error occurs then Yes else No.



   x. **FilterBy :** Display Dropdown with three option : Beginswith, Endswith, Contains **(New Implementation)**



- Click on the unique identification number (**ID**) and the user will get a popup with the information about the Search Log.
- All search log information will be displayed as disabled mode.
- In search Log Info popup users will get the following information.
    - **Id:** Unique identification number.
    - **User Name:** Information about User Name.
    - **Domain:** Information about domain.
    - **End Point:** Information about End Point.
    - **URL:** Information about URL.
    - **Detail:** Provides Information.
    - **Request Methods:** Information about method name.
    - **Request Header:** Information about requested header.
    - **Requested Response:** Information about requested response.
    - **Request Body:** Information about request body.
    - **API Type:** Information about API type.
    - **Mode:** Information about Mode.
    - **Date:** Information about time stamp.

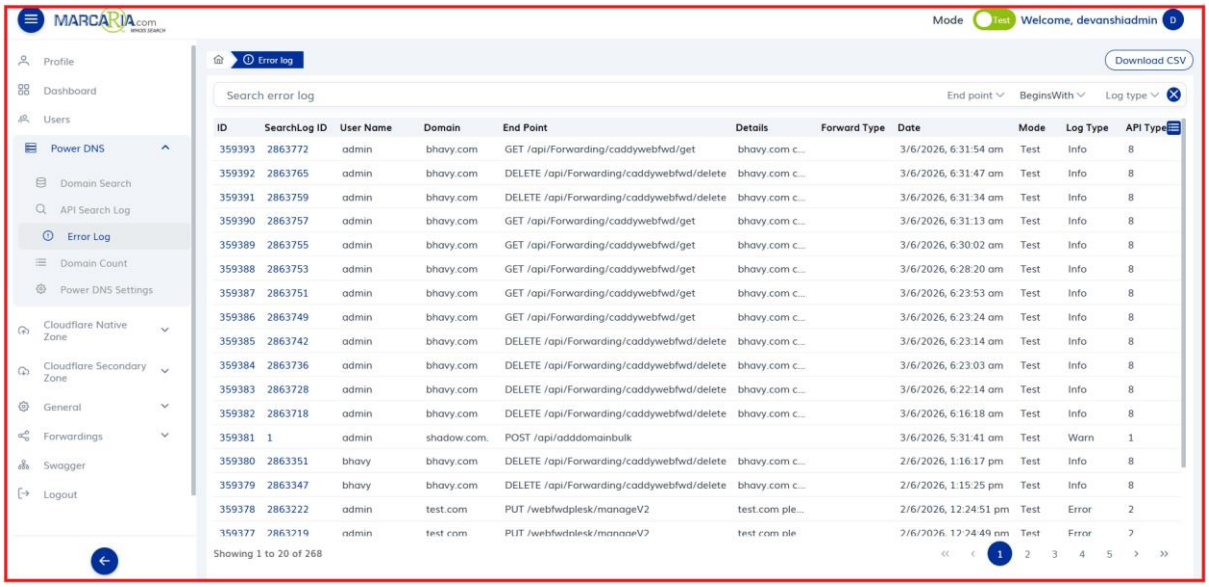


Error Log:

- Here users are able to view the error log if any errors occurred.
- Users can have the option to apply filters based on..
  - **Search By Mode:** It have three option Live, Test, All
  - **Search Error Log:** Enter text to search error logs.
  - **End Point:** Select the endpoint from the dropdown list.
  - **FilterBy:** Display Dropdown with three options : Beginswith, Endswith, Contains.
  - **Log Type:** Filter records based on log type (**Info, Warn, Error**).

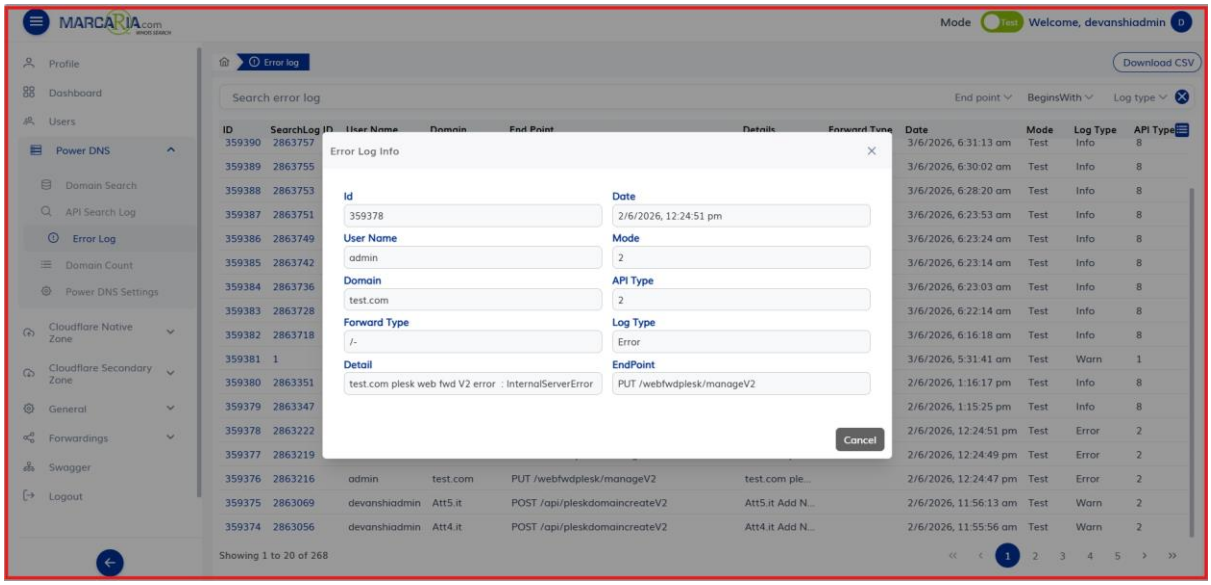
- Users are able to view the following information..
  - **ID:** Unique Identification number.
  - **SearchLog ID:** It is display the unique ID of the Search Log i.e. API Search Log.
  - **User Name:** Displays the username.
  - **Domain:** Name of the domain in which error occurred.
  - **EndPoint:** Name of the End Point.
  - **Detail:** It will display the errors in details
  - **Forward Type:** Display the forward type.
  - **Date:** Timestamp is displayed. (Display Date & Time)
  - **Mode:** Display information about Mode. (Live :1 , Test : 2)
  - **Log Type:** Displays the log category (Info, Warn, Error).
  - **API Type:** Display information about API type.

- Here users are able to download CSV files of the error log.

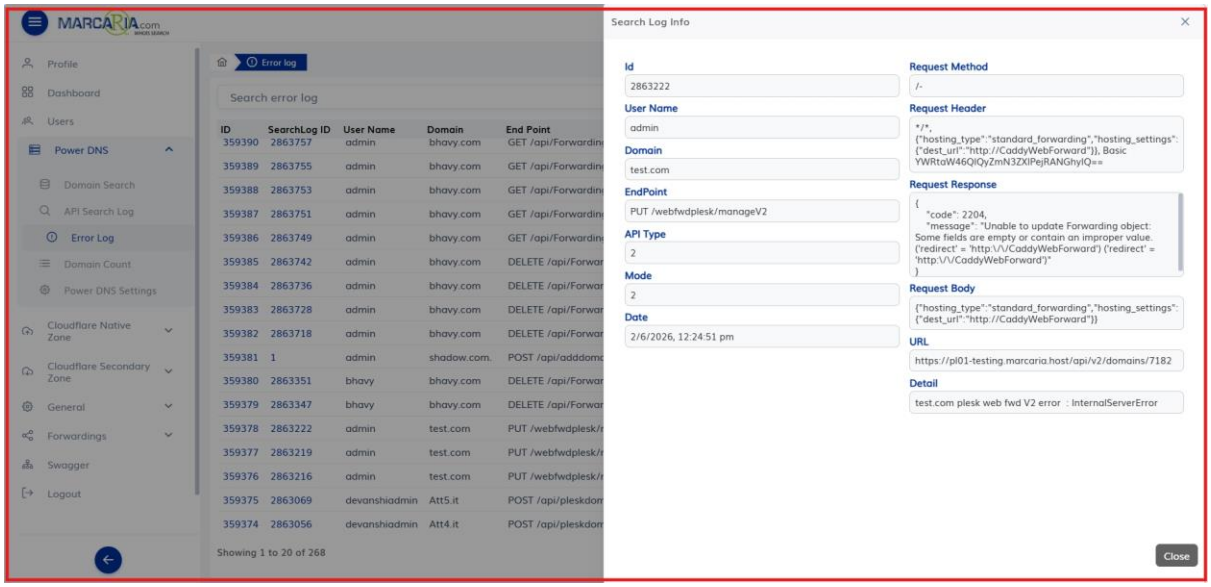


- Click on the ID user will be able to view the Error Log Info popup with following details.
  - **ID:** Unique identification number.
  - **Domain:** Display domain name.
  - **User Name:** Information about the user name.
  - **End Point:** Display End Point name.
  - **Message:** It will print the exception message which has occurred.
  - **Detail:** Display details about error.
  - **Forward Type:** Display the forward type.
  - **Date:** Display timestamp.
  - **Mode:** Display mode information.
  - **Log Type:** Displays the log category.
  - **API Type:** Details of API type.

- All error log information will be displayed as disabled mode.



- Click on the **Search Log ID** user will be able to view Search Log Info popup with following information…
    - **Id:** Unique identification number.
    - **User Name:** Information about User Name.
    - **Domain:** Information about domain.
    - **End Point:** Information about End Point.
    - **URL:** Information about URL.
    - **Detail:** Provides Information.
    - **Request Methods:** Information about method name.
    - **Request Header:** Information about requested header.
    - **Requested Response:** Information about requested response.
    - **Request Body:** Information about request body.
    - **API Type:** Information about API type.
    - **Mode:** Information about Mode.
    - **Date:** Information about time stamp.
- All search log information will be displayed as disabled mode.



Note Error Log :

- **Error Log :**

1. **Message :** NULL

- If there is an error in the coding, the message will be displayed in the Message Column. (**Message :** Decoded String is not a valid)
- Id will not be displayed in Search Log Id. (Blank Display)

2. **Detail :**

- If there is any error in the Api then the message will be displayed in the Detail column. (**Message :** System.ArgumentException)
- The ID will be displayed in the search log ID.

4. API Search Chart:

- It will display the total of API searches in the chart form.
- It will display date on X axis and count on Y axis
- Here users also can see the data according to the user by selecting the user available from the drop down.



- When the user hovers over the graph, the month name and the monthly API search count will be displayed.



Domain Count:

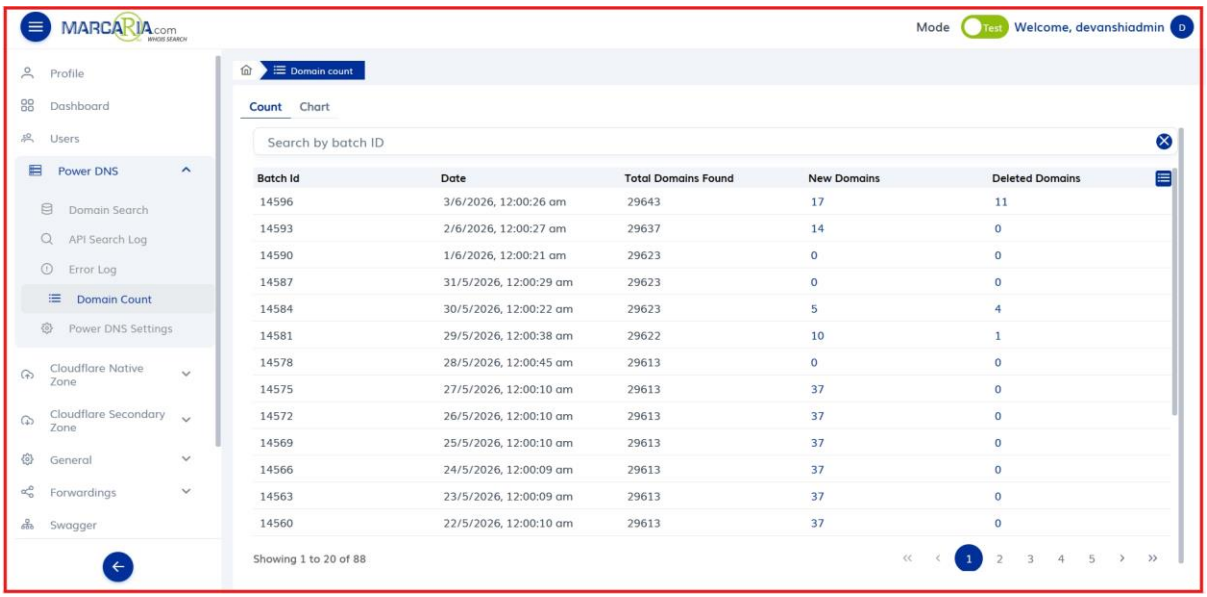
a. Here users can check the log of the created domain according to the date.
b. The Latest created domain will be displayed **Tomorrow**. (Domain Count)
c. Users are able to find the table with the following fields..
    i. **Batch ID:** It is a unique identification number.
    ii. **Date:** The timestamp of the domain creation.
    iii. **Total Domain Found :**It will provide information about the total domain creation till date.

iv. **New Domain:** It will provide information about the newly created domain of the specific date.

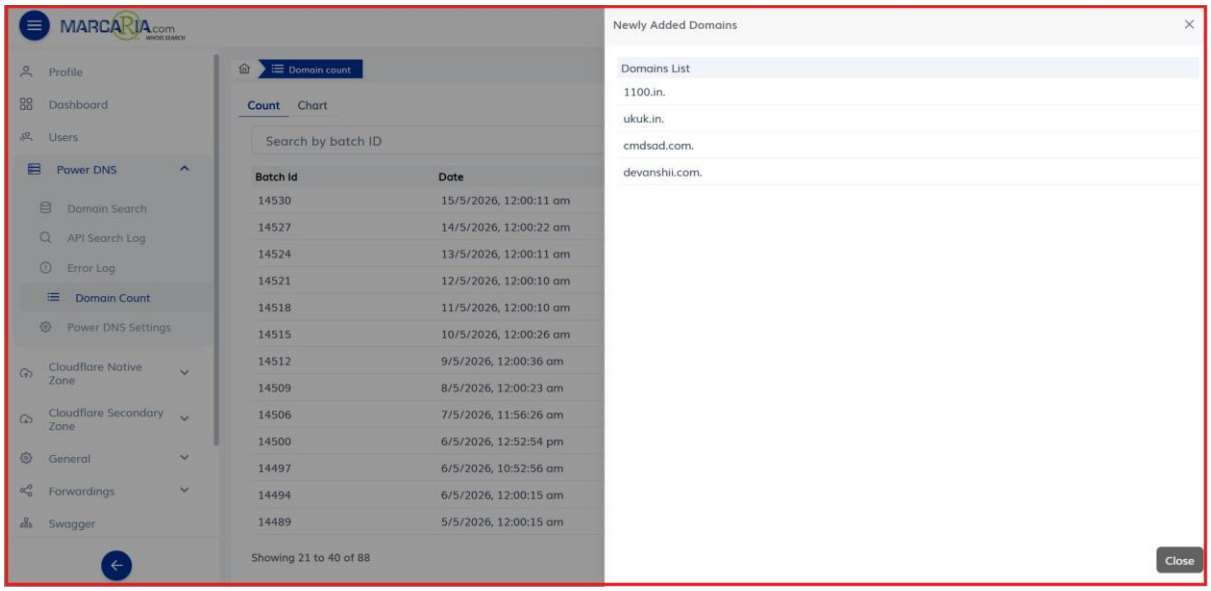
v. **Deleted Domain:** It will provide information about the deleted domain for the specific date.

d. Using a search box, users are able to search the records.

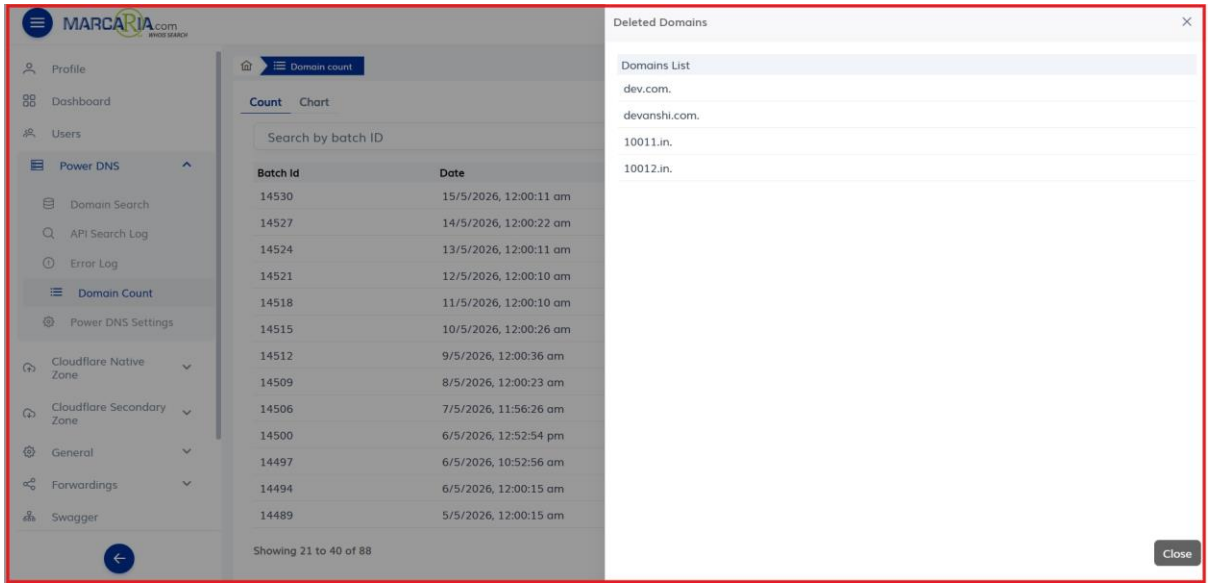
e. Using sorting functionality users are able to sort the data.



f. When the user clicks on the **new domain count**, the list of newly created domains will be shown

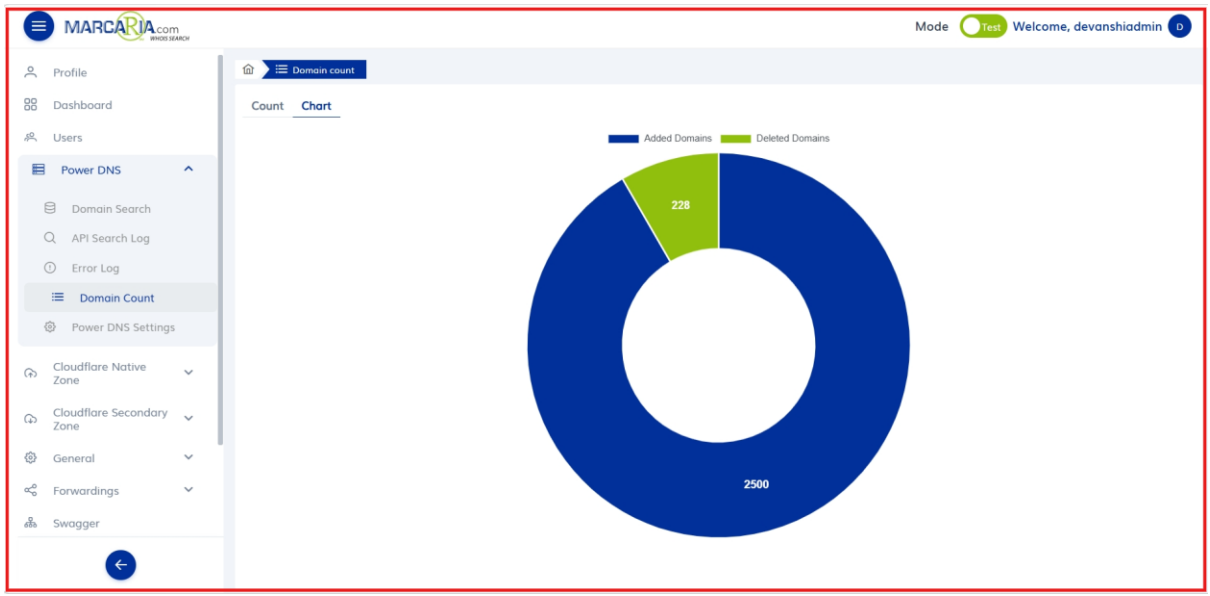


g. When the user clicks on the **deleted domain count**, the list of deleted domains will be shown.



Domain Count Chart:

- Here user can view the data in the chart form.
- Here users can see the overall domain count in a donut chart format.
- Blue represents **Added Domains (Total Domains)** and green represents **Deleted Domains**.
- The chart displays only total summary data instead of date-wise data.
- When the user hovers over the blue segment, the total added domain count will appear.
- When the user hovers over the green segment, the deleted domain count will appear.

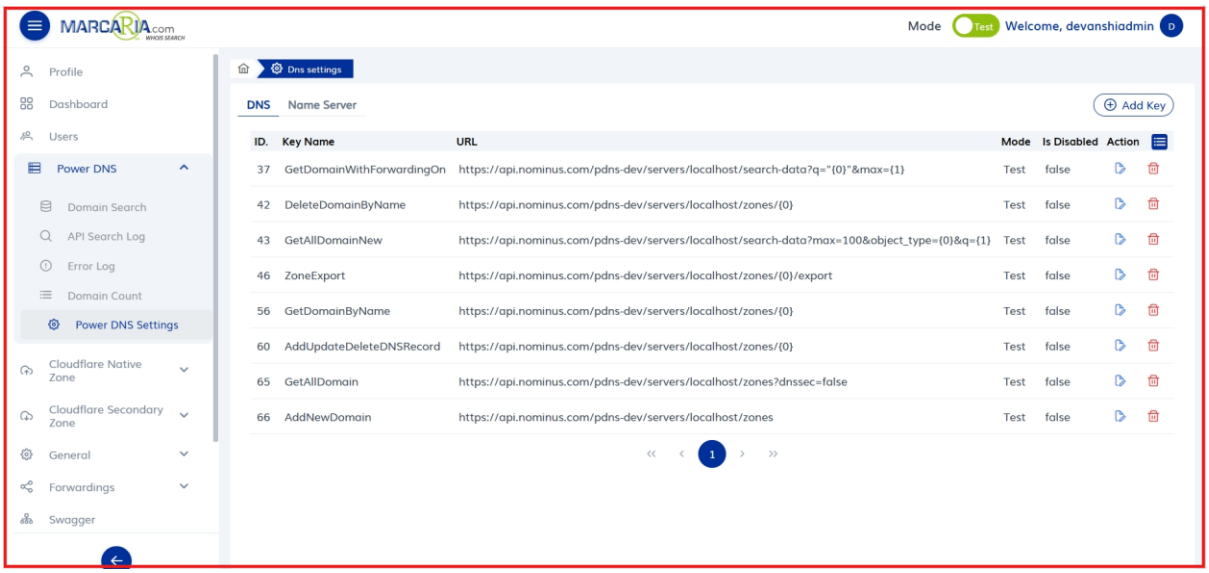


Power DNS Settings:

- Power DNS details have two modules.
    1. DNS.
    2. Name Server.

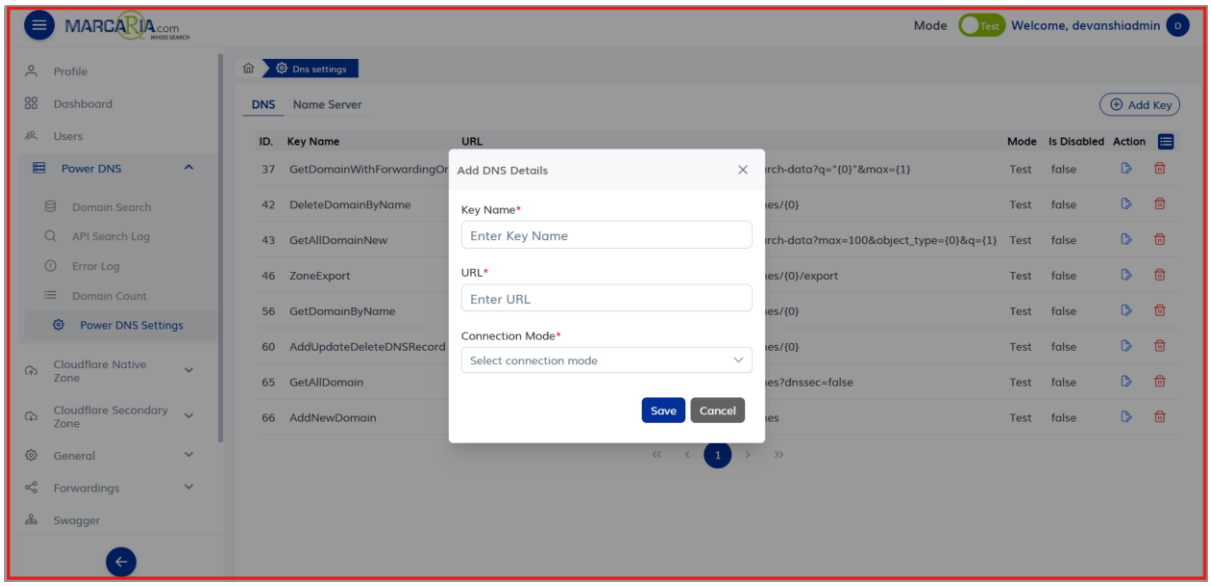
1) DNS:

- Users can map the base (PowerDNS) API with the key name and this key name is used in coding to identify the Base (PowerDNS) API.
- Here users are able to view DNS settings records in a grid format along with pagination support.
- Here users can add power DNS details using the add button.
- Here users are able to view details like.
    - **ID:** Unique Identification number.
    - **Key Name:** It is the name which is used for creating reference of the API in the coding part.
    - **URL:** Display the URL which is assigned to the key. I.e. when a key is used in code it will indirectly call the URL.
    - **Mode:** Display the Mode information.
    - **Is Disabled:** Here if the user does not check the checkbox its value is N otherwise its value is Y. Here if users don't want to use the API longer at that time they can do Is Disabled checkbox checked instead of hard delete.
    - **Action:** User is able to Edit the information or delete it.



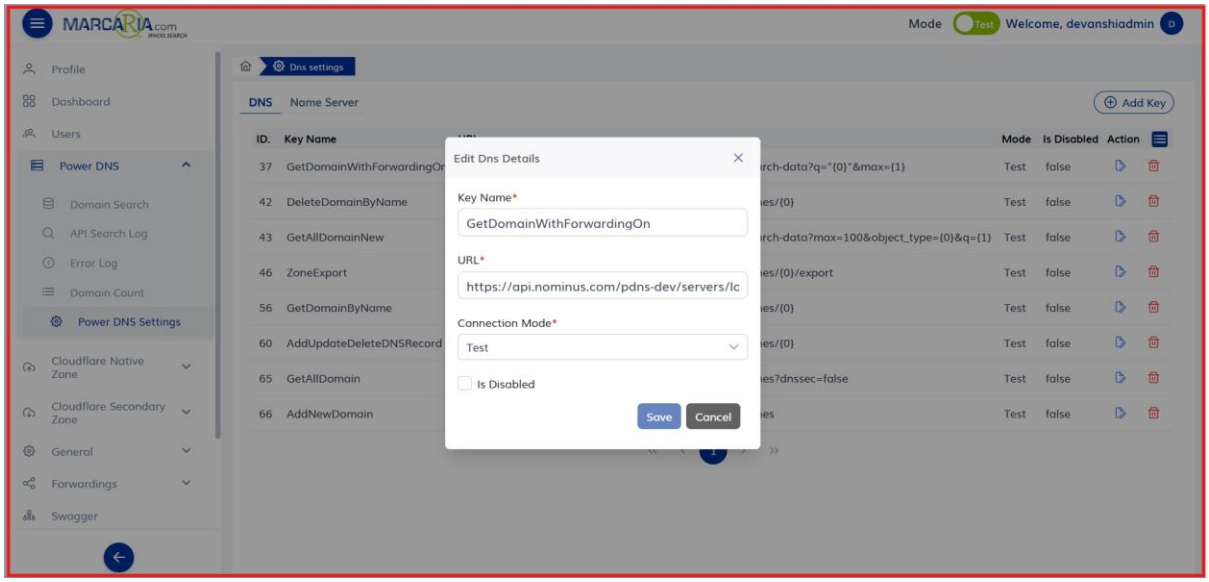
❖ Add Button:

- Once the user clicks on the Add Key button it will show Key Name, URL and Mode fields.
- **Submit Button:** Click on Submit button it will add DNS detail
- If a record with the same Key Name and Mode already exists, the system will display the message "Entry already exists for the same Key and Mode."
- **Cancel Button:** Click on Cancel button pop will closed



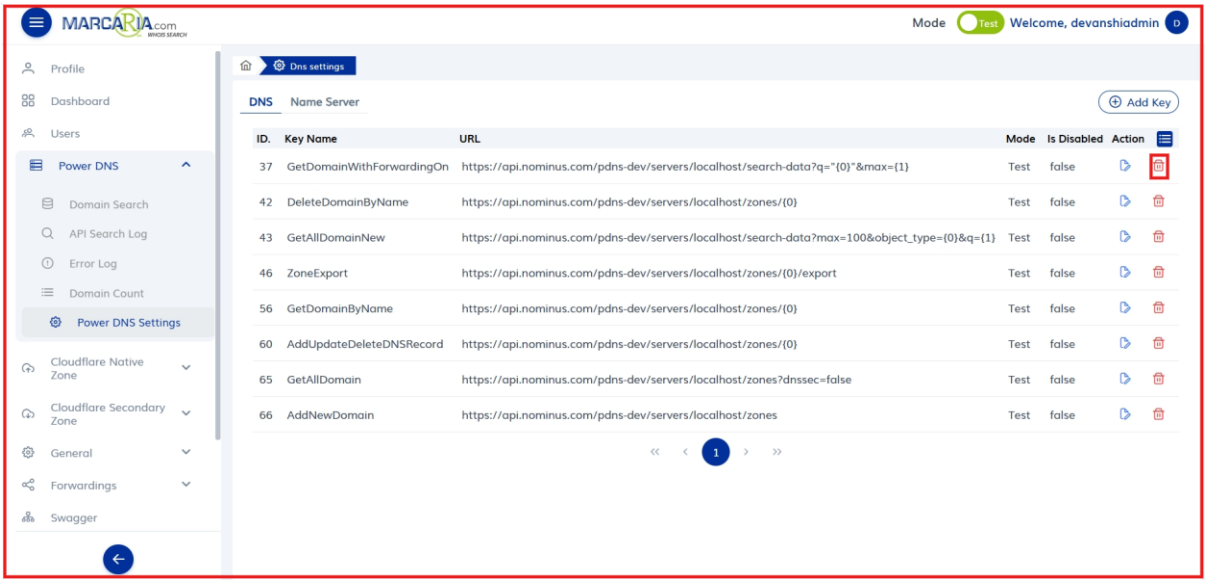
❖ Edit Button:

- Click on the Edit icon it will open a popup containing Key Name, URL, Connection Mode and Is Disabled fields which allows users to edit the DNS setting details.
- Users can modify the Key Name, URL, Connection Mode and Is Disabled status and save the changes.



❖ Delete Button:

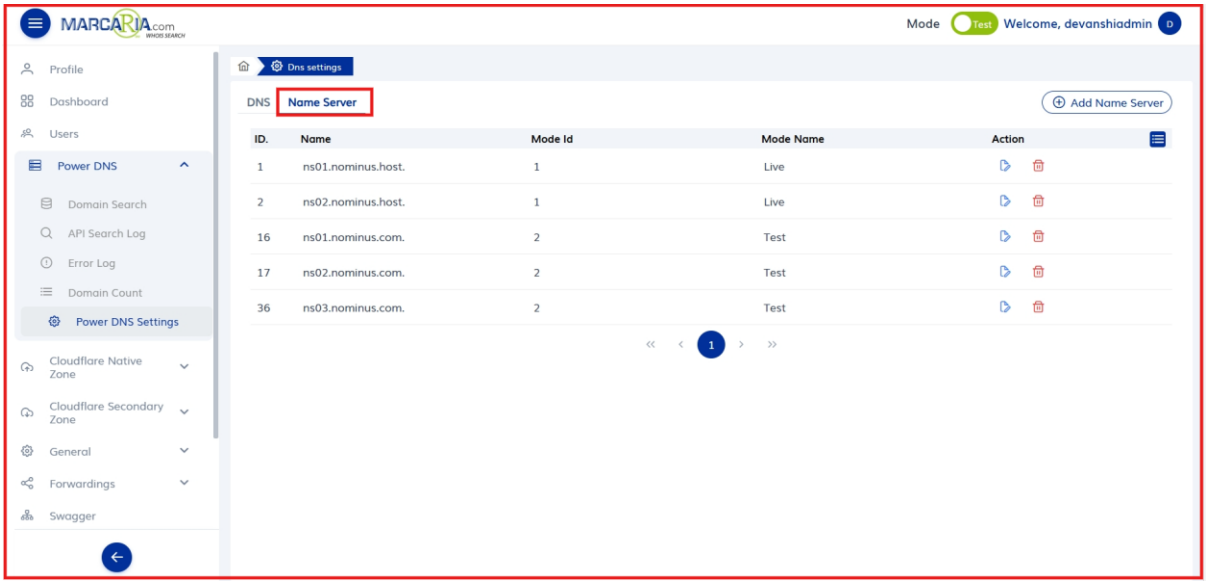
- Click on the Delete Button user is able to delete the Power DNS Detail and get the message like - **“Power DNS Record Successfully Deleted”.**



2) Name Server:

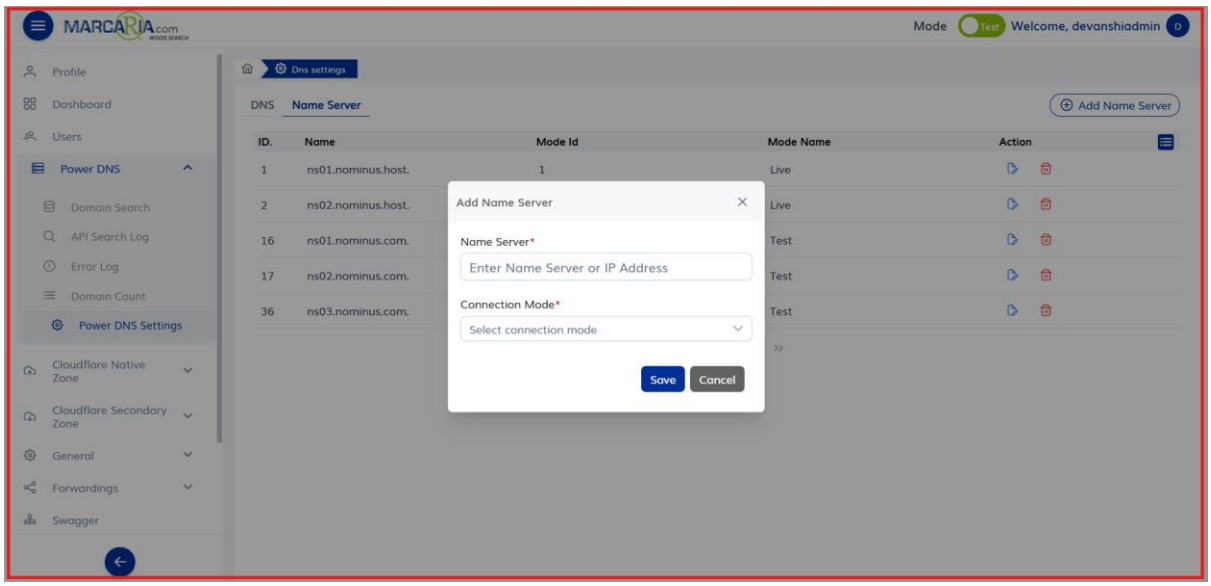
- When a user creates a new domain at that time the records which are added by default will be managed from the PowerDNS settings -> Name Server (Tab).
- Here, users are able to view information like..
  - **ID:** Unique IDentification number.

- **Name Server:** Information about Name Server.
- **Mode ID:** Information about mode id.
- **Mode Name:** Information about mode name.
- **Action field:** Provide Edit and Delete option.



❖ Add Name Server Icon:

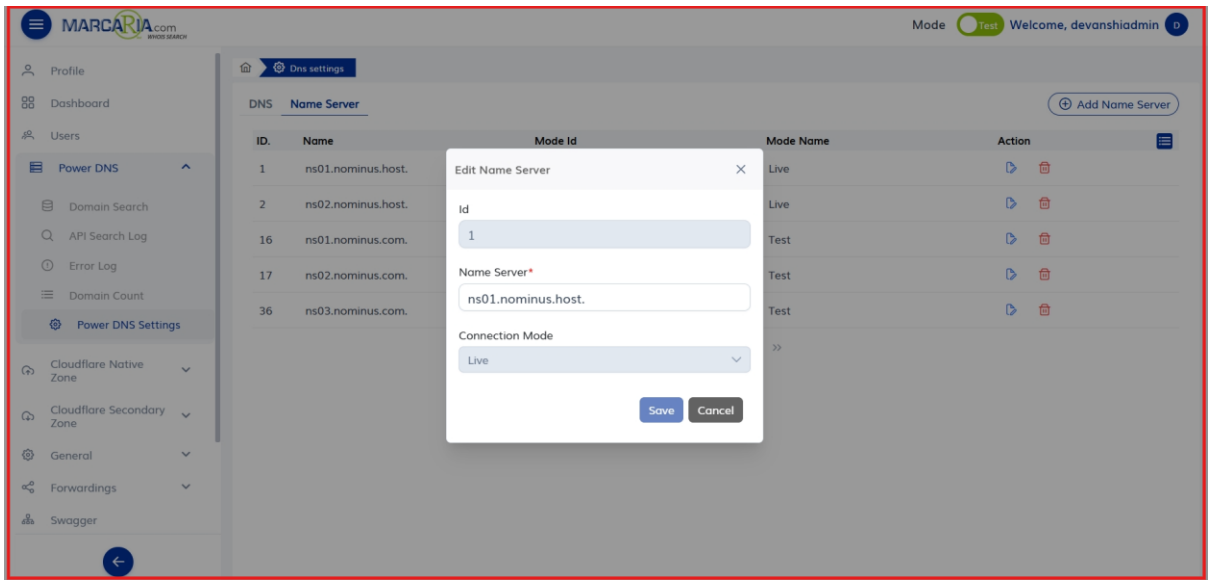
- Click on the Add Name Server icon it will open the popup and the user can see the name server text area and connection mode dropdown for the adding data.



- **Save Button:** Click on save button it will add DNS server
- **Cancel Button:** Click on Cancel button pop will closed

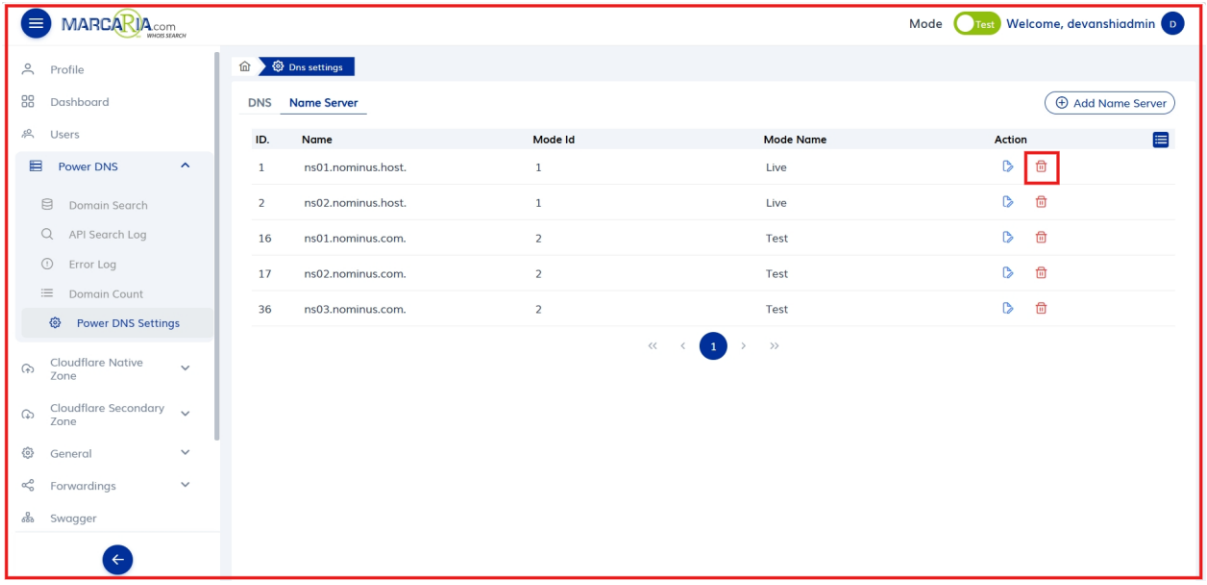
❖ Edit Button:

- Click on the Edit icon popup will open with ID, Name Server and Connection Mode fields.
- Here users are able to edit the Name Server and Connection Mode information.



❖ Delete Button:

- Click on the Delete Button user is able to delete the name server and get the message like - **“Name Server Successfully Deleted”.**

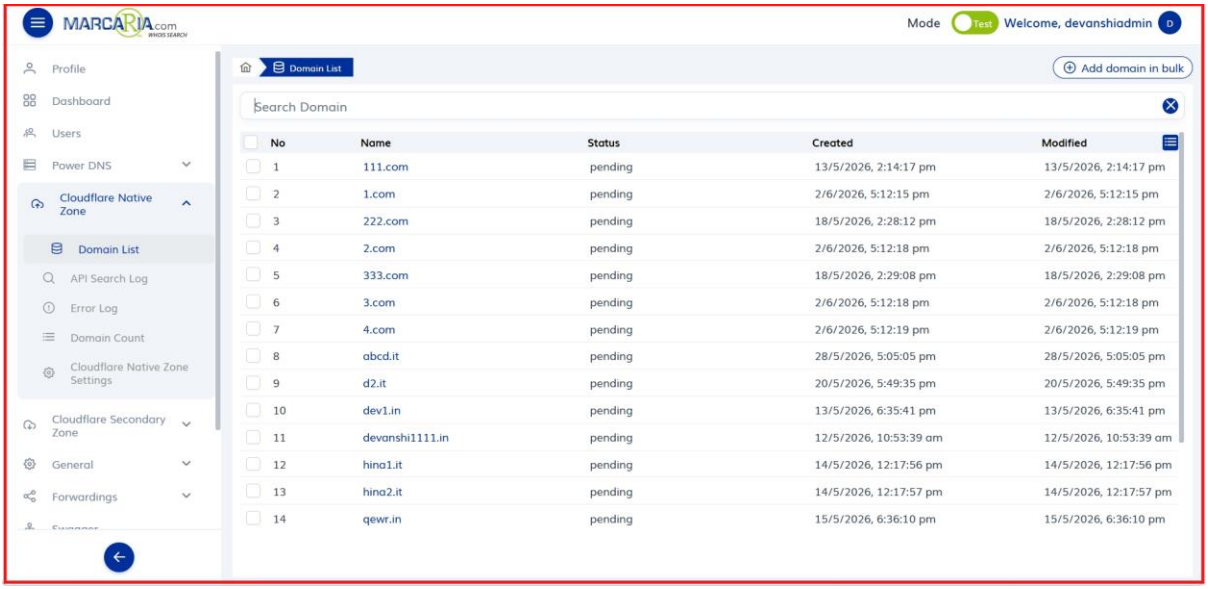


CLOUDFLARE NATIVE ZONE

- Cloudflare Native Zone has 5 submodules.
    1. Domain Search.
    2. API Search Log.
    3. Error Log.
    4. Domain Count.
    5. Power DNS Settings.

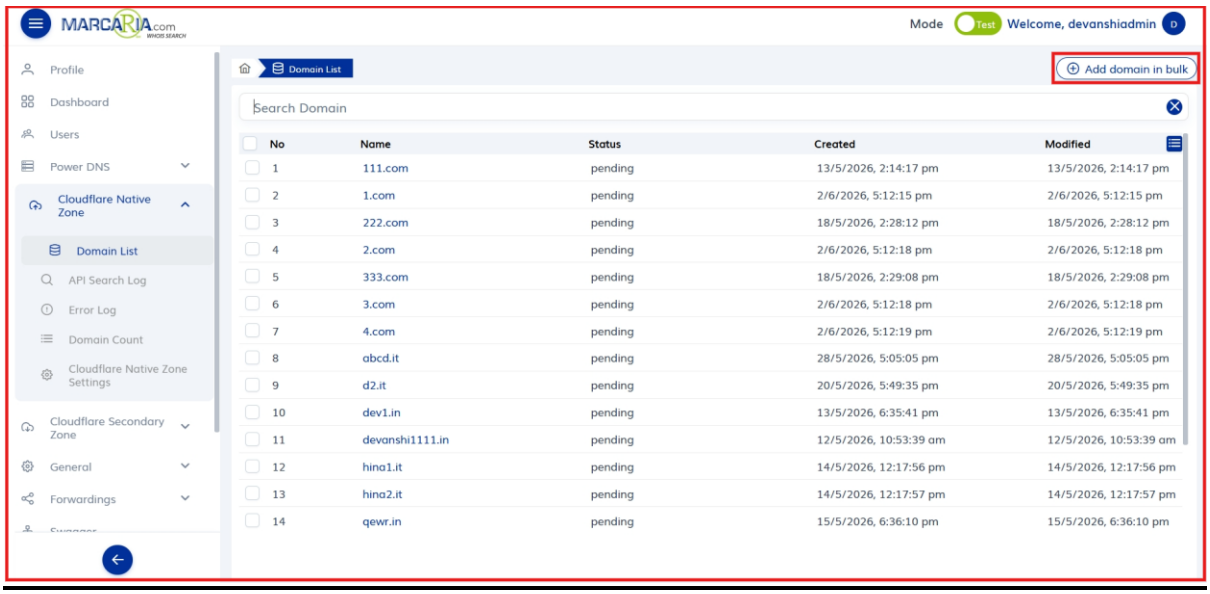
Domain List :

- Here, users can see the domain list, Add Domain in Bulk button and Search Domain search box.
- Here, users are able to see No, checkbox functionality, domain name, status, created date and modified date.
- Users can add/create domains from the Domain List menu.
- By clicking on the Domain from the side menu panel the user is able to view a list of domains which are already added.
- Users can search domains using the Search Domain textbox.
- Users can select multiple domains using the checkbox and perform bulk operations.



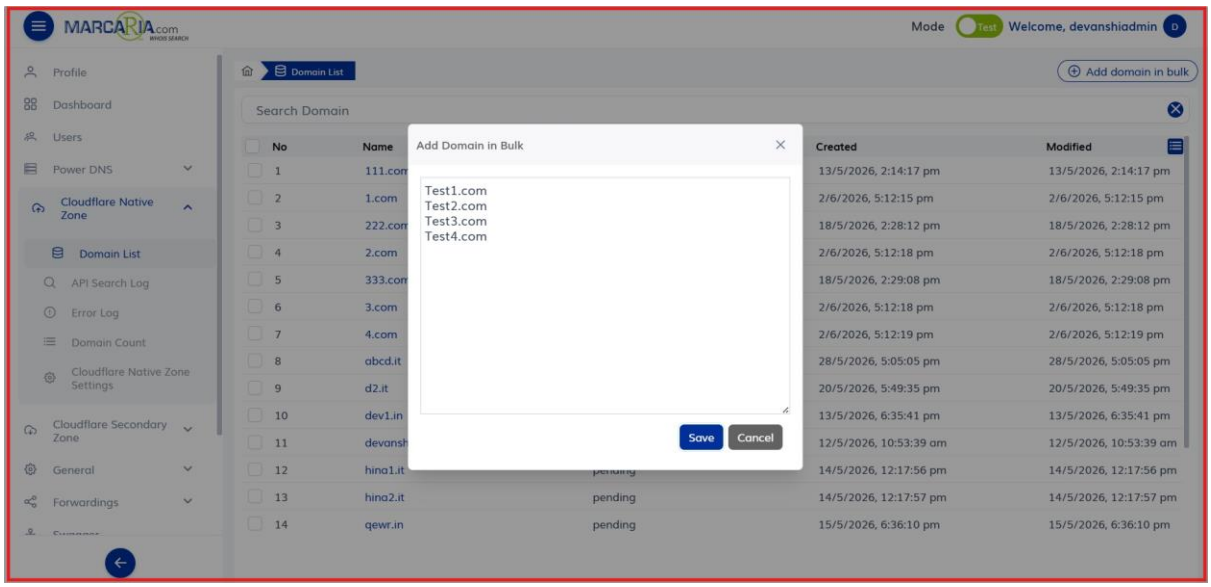
❖ Add Domain in bulk button:

- When the user clicks on the Add domain in bulk button it will open the popup with text area and Save, Cancel button.



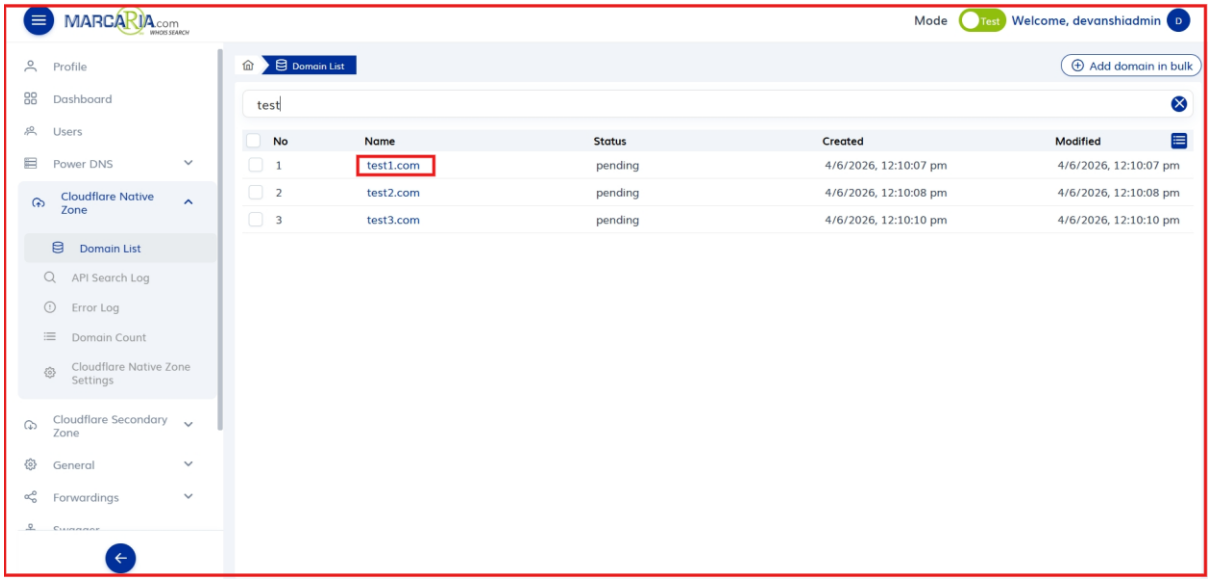
- Add multiple domains in the text area separated each by entering a domain name. i.e. every domain must be in a new line.
- Multiple domains can be added at the same time by entering one domain per line.

NOTE: Entered domain name must be unique. If a user entered name which is already available in the list it will give validation of the conflict domain name.

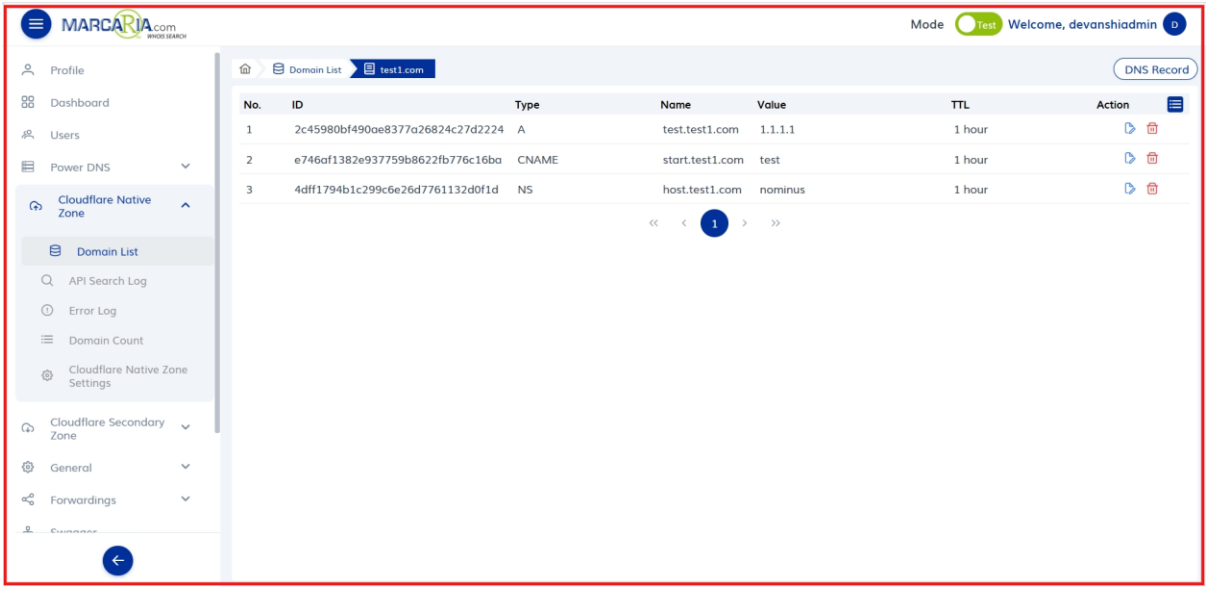


❖ DNS Record List :

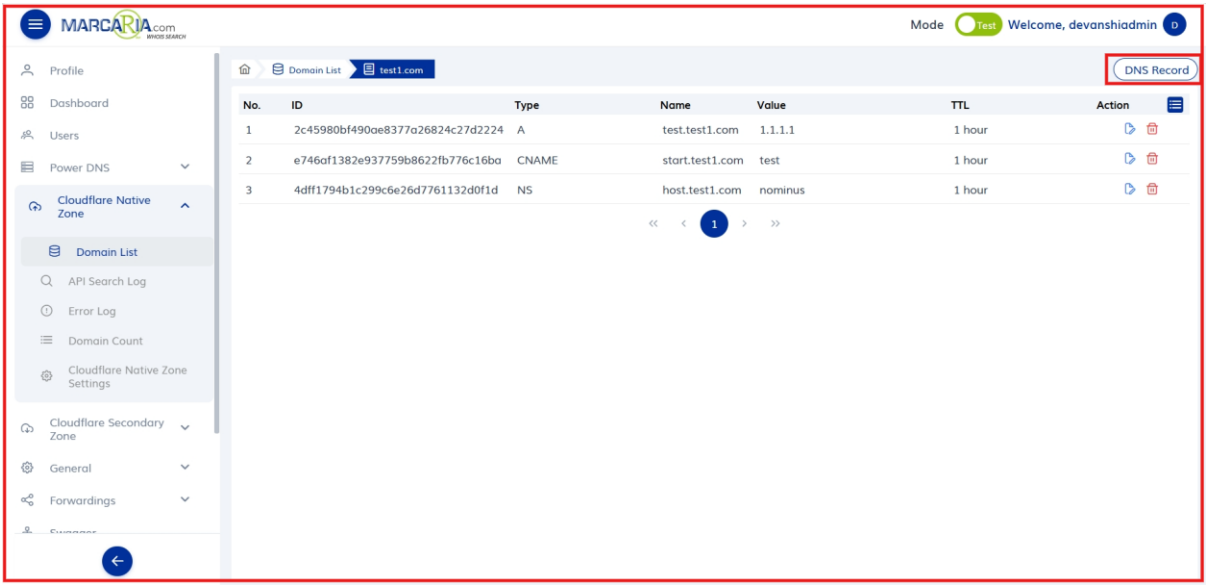
- By clicking on a domain name, users are able to view DNS records associated with the selected domain.



- Here users can see ID, Type, Name, Value, TTL and Action column with Edit and Delete functionality.



- Here users are also able to add new records by clicking on the DNS Record button.
- When the user clicks on the DNS Record button it will open the section for selection of record type from the dropdown.



- Type of records are :

1. **A** Record
2. **AAAA** Record
3. **MX** Record
4. **TXT** Record
5. **SRV** Record
6. **CAA** Record
7. **CNAME** Record

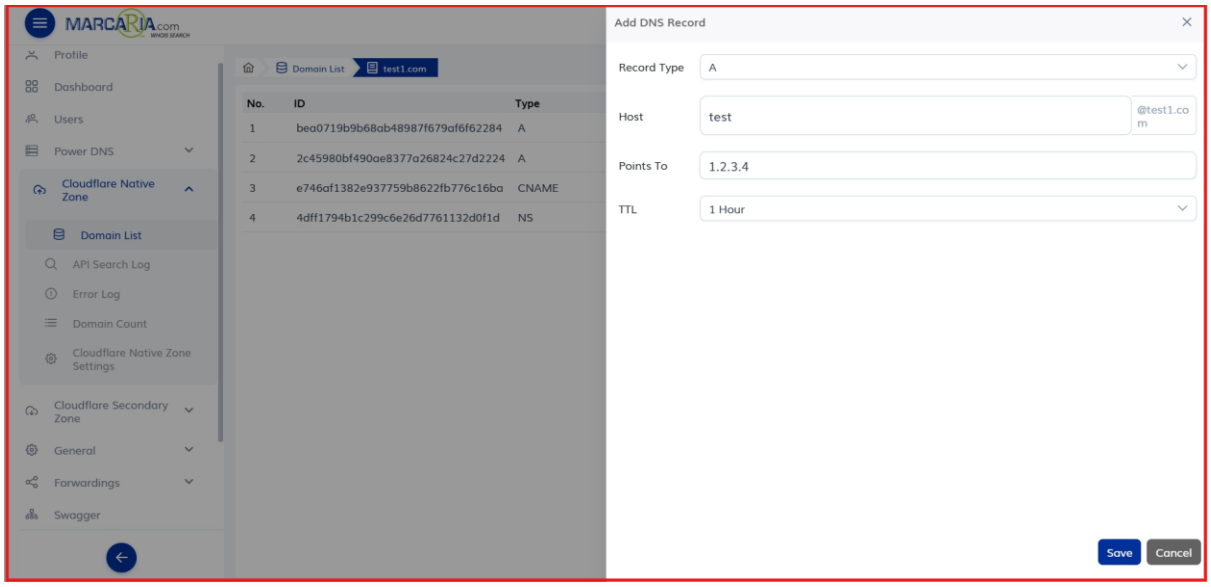
8. **NS** Record.

- Following are the available types.

- **A type record :**

  - Firstly, address (A) records are the most common record type by far. In brief, A records map domain names to IPv4 addresses.

    - IPv4 notation is : **192.0.2.146**.
    - When a user selects to add A type of records it will have the following options…
    - **Host :** Provide host name.
    - **Points To :** Add IPV4 address. Here if user not enter the valid IPV4 it will generate validation message like ..”Entered value is not valid IP”
    - **TTL :** Select time duration from drop down
    - **Save button :** Click on save button it will save the record.
    - **Cancel button :** Click on cancel button it will stop the process.

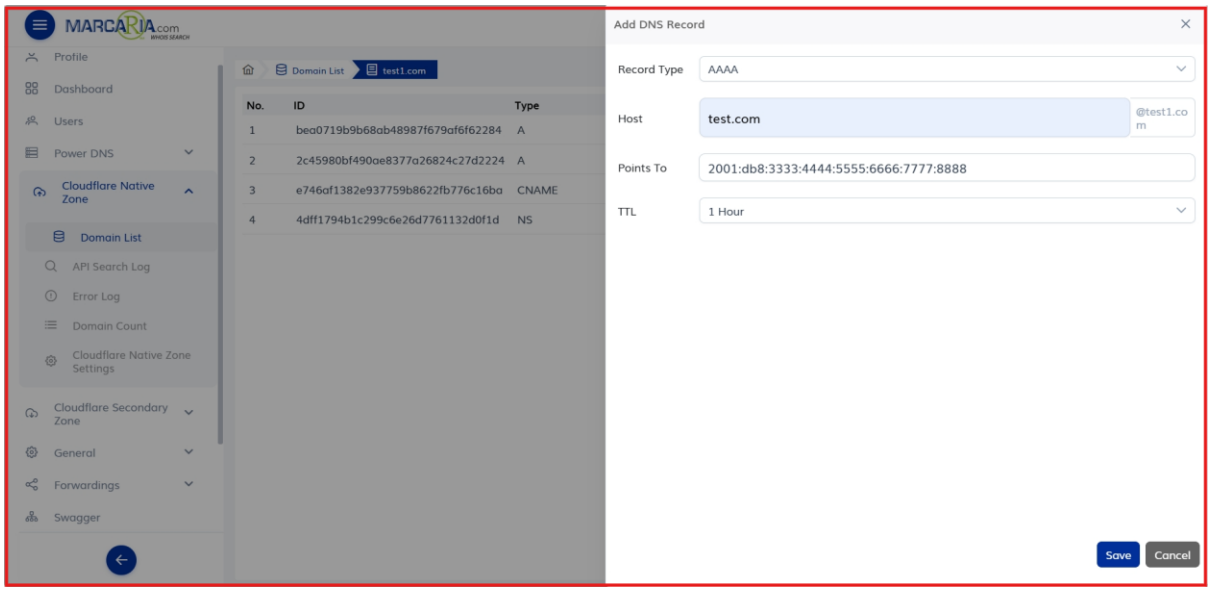


- **AAAA Type Record :**

  - As the internet gradually makes the transition to IPv6, there are AAAA records (spoken as “quad A”). An AAAA record

is the IPv6 equivalent of an A record, responsible for mapping a domain name to an IPv6 address.

- IPv6 notation is :2001:db8:3333:4444:5555:6666:7777:8888
- When a user selects to add AAAA type of records it will have the following options…
    - **Host :** Provide host name.
    - **Points To :** Add IPV6 address. Here if user not enter the valid IPV6 it will generate validation message like ”Enter value in valid IPV6 format”
    - **TTL :** Select time duration from drop down
    - **Save button :** Click on save button it will save the record.
    - **Cancel button :** Click on cancel button it will stop the process.

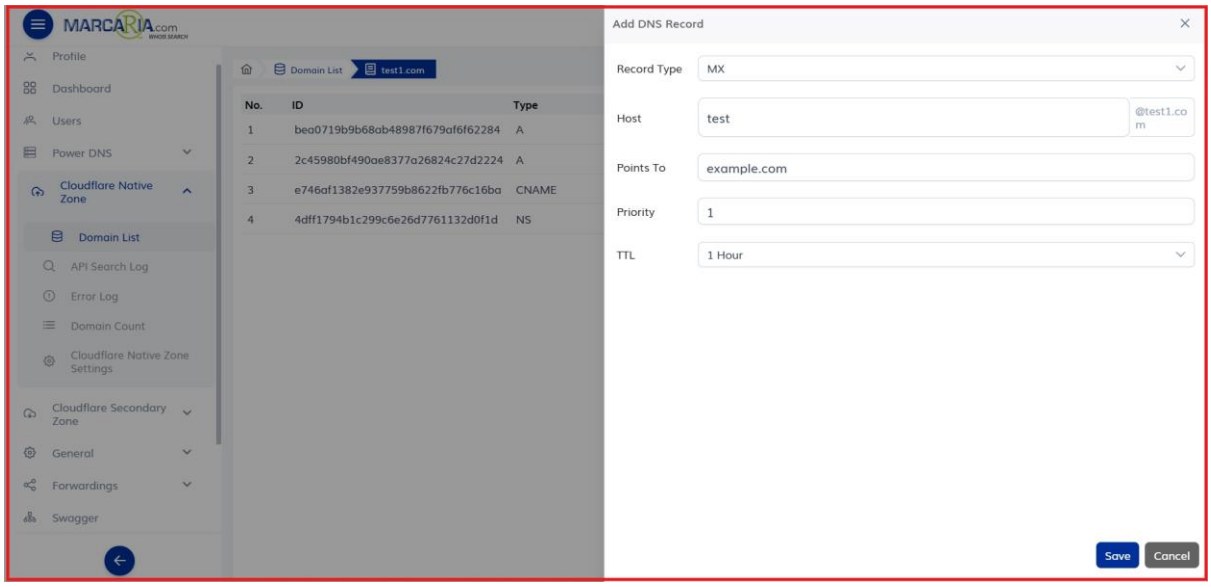


- **MX Type Record** :

- A mail exchanger (MX) record stores the domain names of mail servers responsible for receiving emails on behalf of a domain.
- These records can be helpful to identify authorized mail servers. An MX record is sometimes called an MX entry, particularly in configuring mail servers.
    - **Host :** Provide host name.
    - **Points To :** Value is Required.
    - **Priority :** Set the priority. The priority you set when you add an MX record set specifies the sequence

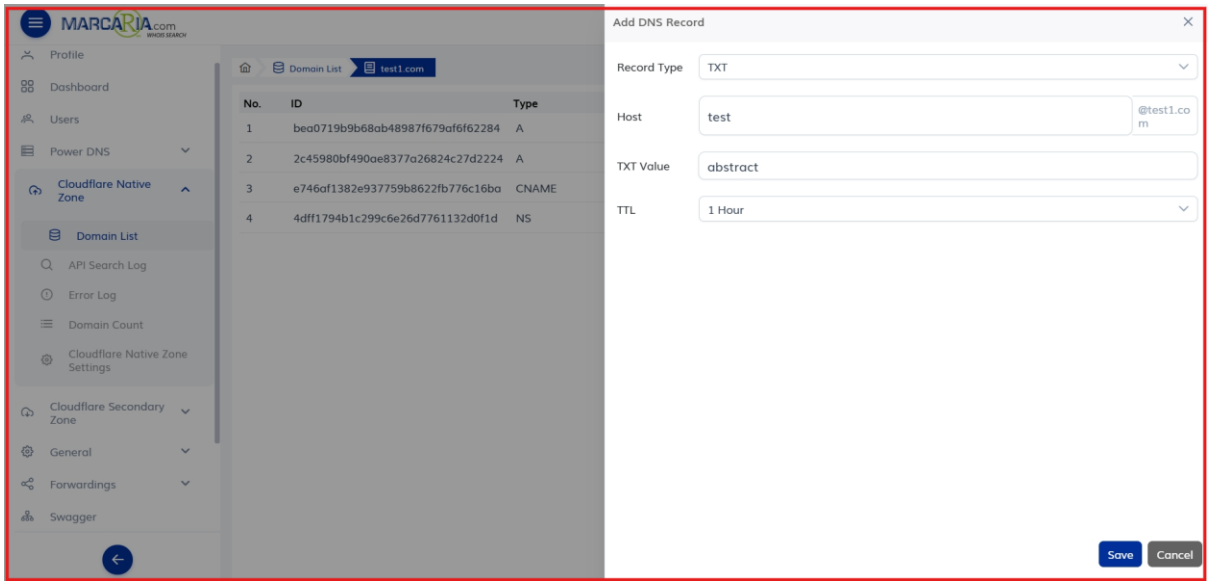
that an email server receives emails. A smaller value indicates a higher priority.

- **TTL :** Select time duration from drop down
- **Save button :** Click on save button it will save the record.
- **Cancel button :** Click on cancel button it will stop the process.



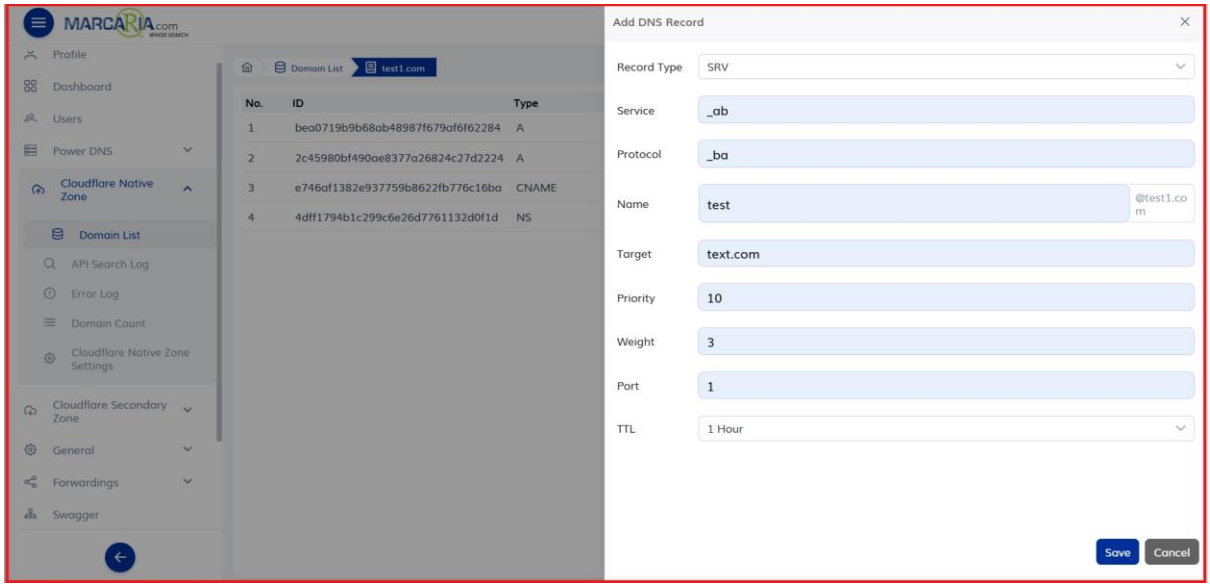
- **TXT Type Record** :

  - A text (TXT) record can store any type of descriptive information in text format. For example, network antivirus providers can use TXT records to send and receive information.
    - **Host :** Provide host name.
    - **TXT value :** Provide text value for the record.
    - **TTL :** Select time duration from drop down
    - **Save button :** Click on save button it will save the record.
    - **Cancel button :** Click on cancel button it will stop the process.



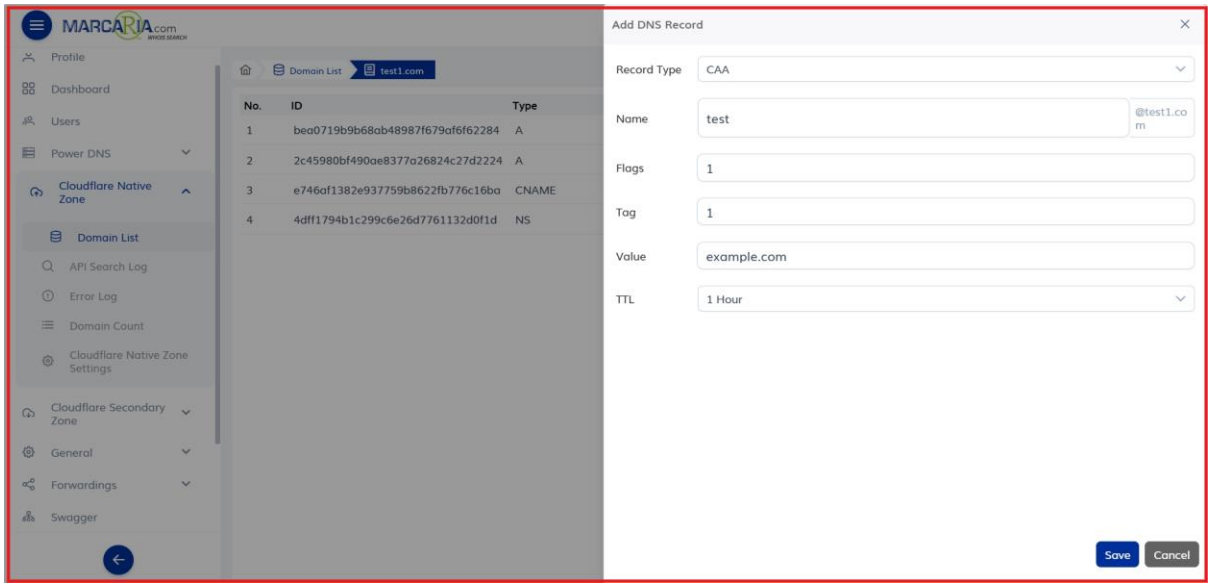
- **SRV Type Record :**

- The SRV record is a Domain Name System (DNS) resource record. It's used to identify computers hosting specific services. SRV resource records are used to locate domain controllers for Active Directory.
    - **Service :** Symbolic name for a service. Here value must be start with underscore (_)and alphanumeric
    - **Protocol :** Specifies if the service is using TCP or UDP protocols. Here value must start with underscore (_) and alphanumeric.
    - **Name :** Stores a domain name.
    - **Target :** The canonical hostname for the machine providing the service.
    - **Priority :** Determines which server is looked at first, with lower values giving higher priority.
    - **Weight :** Determines which server is looked at first if more than one has the same priority value. Higher values give more priority.
    - **Port :** The TCP or UDP port the service is running on. {Transport Layer Protocol : 1. TCP (Transmission Control Protocol) , 2. UDP(User Datagram Protocol)}
    - **TTL :** Select time duration from drop down
    - **Save button :** Click on save button it will save the record.
    - **Cancel button :** Click on cancel button it will stop the process.



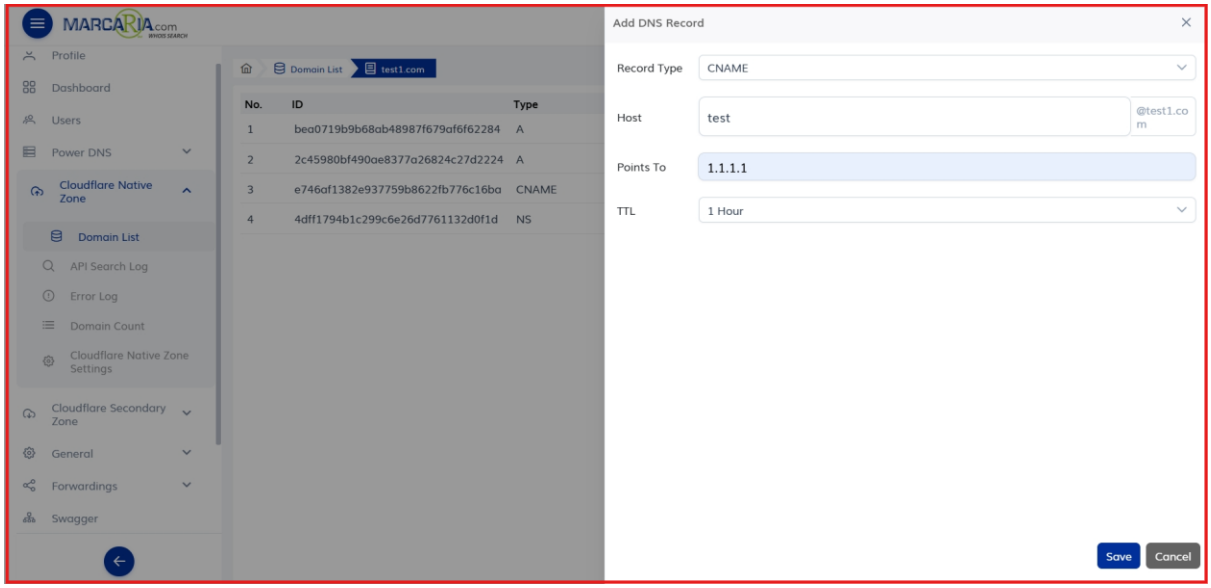
- **CAA Type Record**:

  - CAA is a type of DNS record that allows site owners to specify which Certificate Authorities (CAs) are allowed to issue certificates containing their domain names.
    - **Name :** The name of the domain that is being certified.
    - **Flags :** Can be either 1 (critical) or 0 (non-critical). A critical flag means the certificate authority cannot use the CAA record if it doesn't understand the property. A non-critical flag means it can use the CAA record regardless of whether it understands the property.
    - **Tag :** The three tag options are issue (authorized to issue a single certificate), issuewild (authorized to issue a wildcard certificate), and iodef (specifies a URL for reporting policy violations).
    - **Value :** Enter value for selected tag.
    - **TTL :** Select time duration from drop down
    - **Save button :** Click on save button it will save the record.
    - **Cancel button :** Click on cancel button it will stop the process.



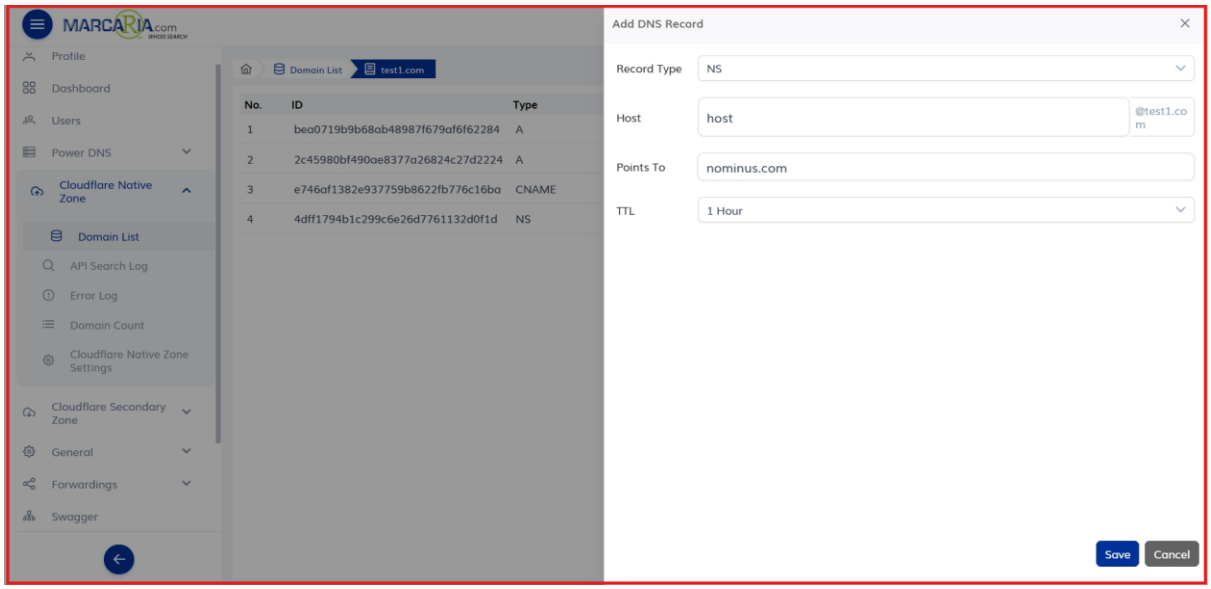
- **CNAME Type Record :**

  - Redirects a domain to a different domain. CNAME records are typically used to map a subdomain such as www or mail to the domain hosting that subdomain's content.
    - **Host :** Provide host name.
    - **Points To :** Add target domain or host name.
    - **TTL :** Select time duration from drop down.
    - **Save button :** Click on save button it will save the record.
    - **Cancel button :** Click on cancel button it will stop the process.



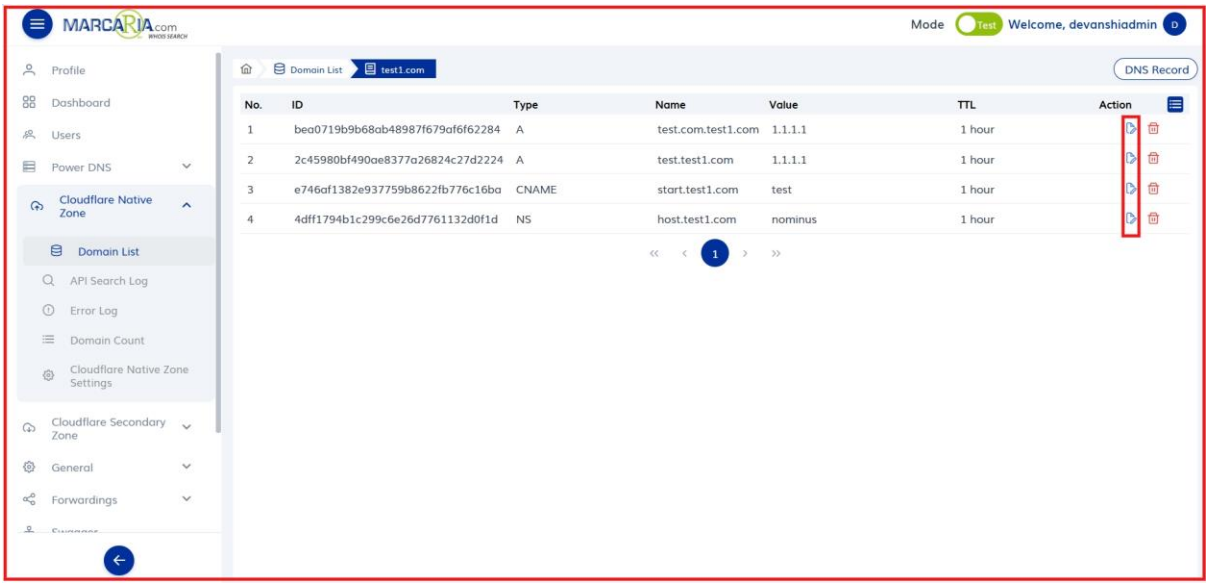
■ **NS Type Record :**

- Provides a list of the authoritative name servers responsible for the domain. It's important to note that NS records point to the domain of the name server—not IP addresses.
  - **Host :** Provide host name.
  - **Points To :** Add IPV4 address.
  - **TTL :** Select time duration from drop down.
  - **Save button :** Click on save button it will save the record.
  - **Cancel button :** Click on cancel button it will stop the process.



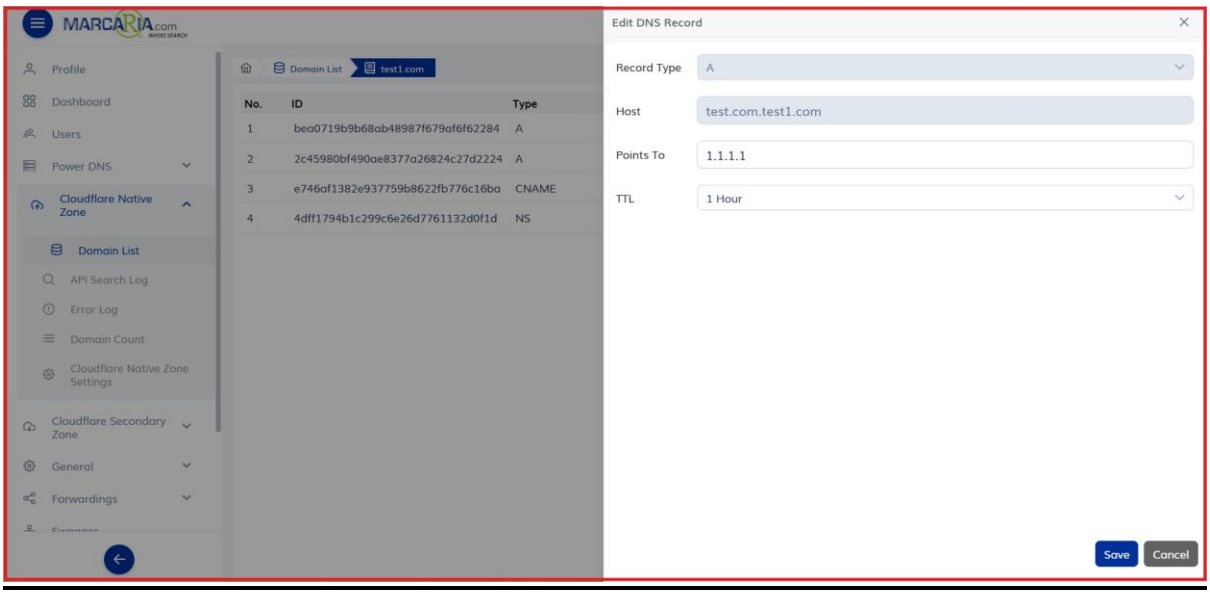
❖ Edit Icon :

- Click on the edit icon from the Action field of the DNS records table.



- Users can get the popup with following details..
  - Fields of the record which are generated by the user.
  - Save button
  - Cancel button
- **Type** and **hostname** are disabled at the time of edit and all other fields users can modify.

NOTE: For the SRV record type - Type, Service, Protocol and Name fields are disabled.

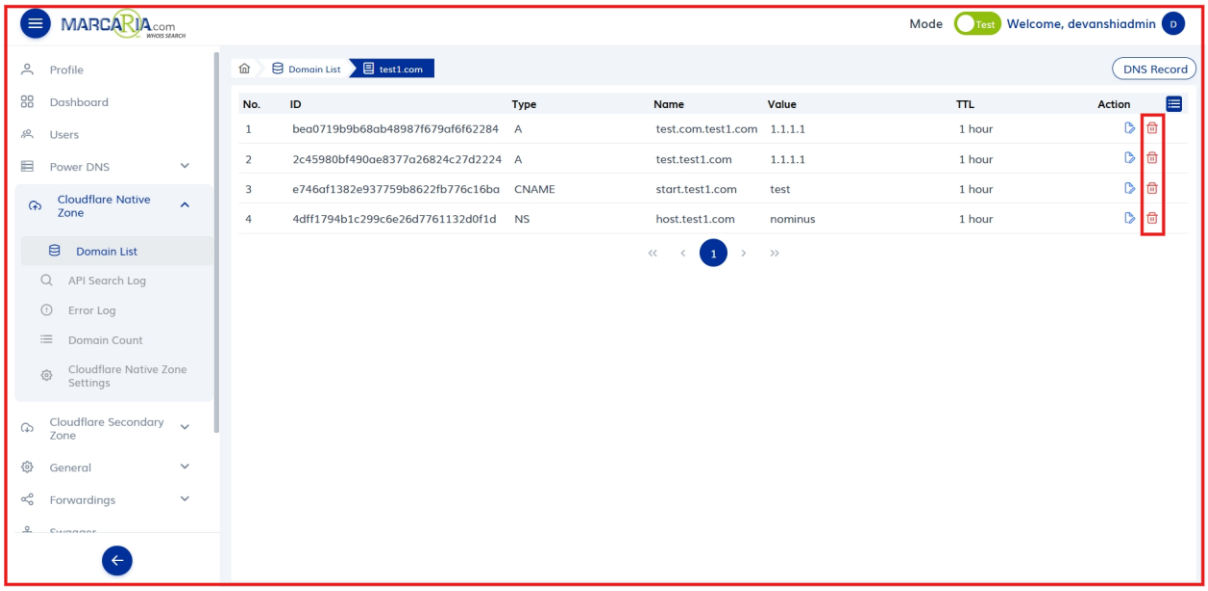


- **Save Button:** Click on the save button after doing modification it will update old data with the updated data.
- **Cancel Button:** click on the cancel button it will close the popup.

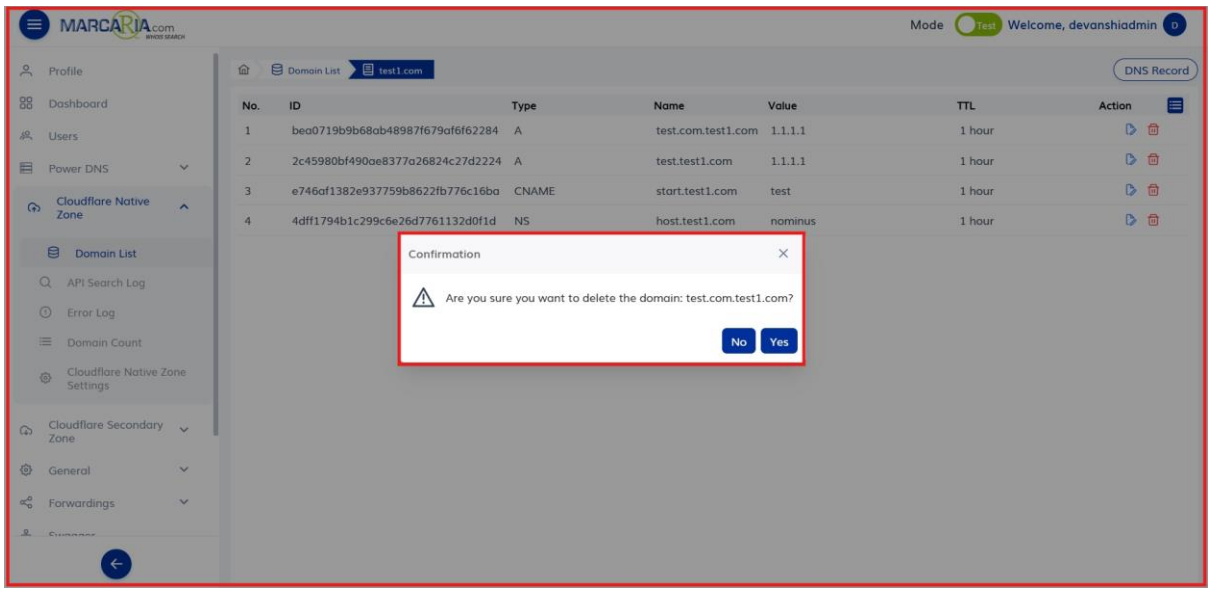
NOTE : Edit option is not available for SOA record.
NOTE : Users can add @ if we want domain name as name.

❖ Delete button :

- Click on the Delete icon from the Action field.

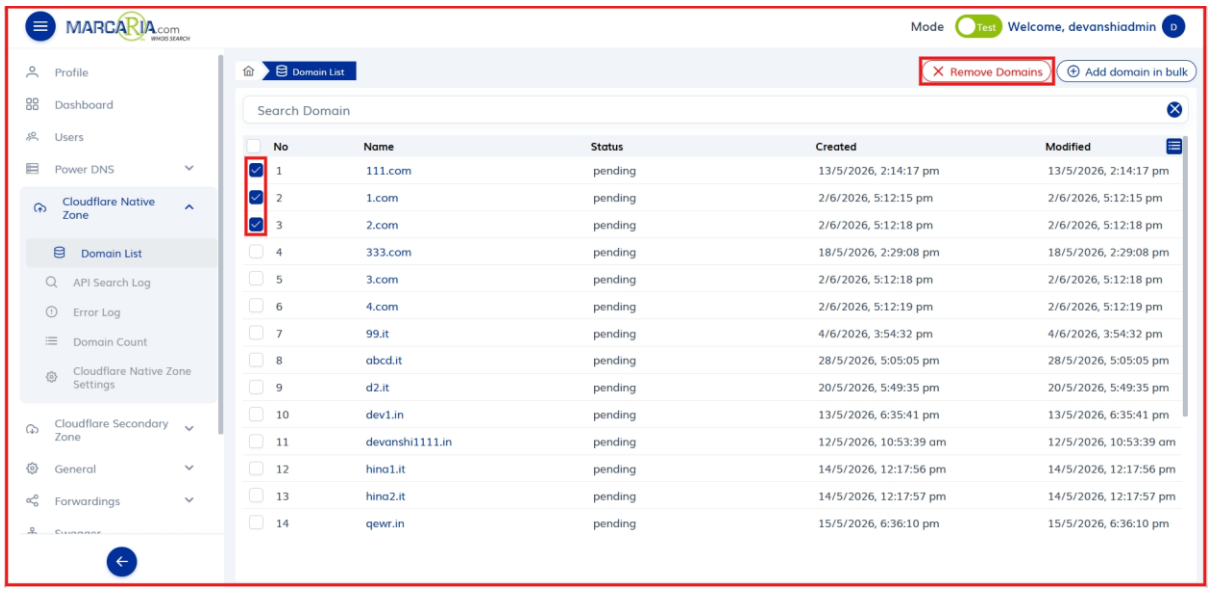


- A confirmation popup will be displayed asking the user to confirm the deletion of the selected domain.
- Click on the Yes button to delete the selected domain or click No to cancel the operation.



❖ **Remove Domain button:**

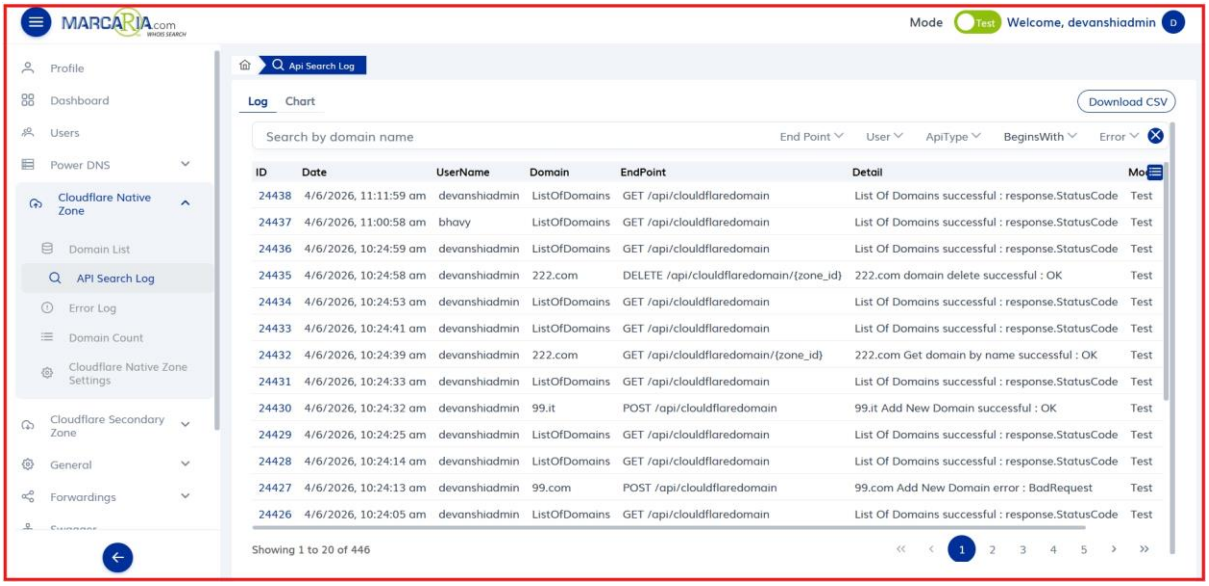
- Here users are able to remove multiple/bulk domains.
- Users can check/uncheck the domain as per the requirements which are available in the domain list.
- After checking the checkbox, click on the remove button and the users are able to remove the selected domain from the domain list.



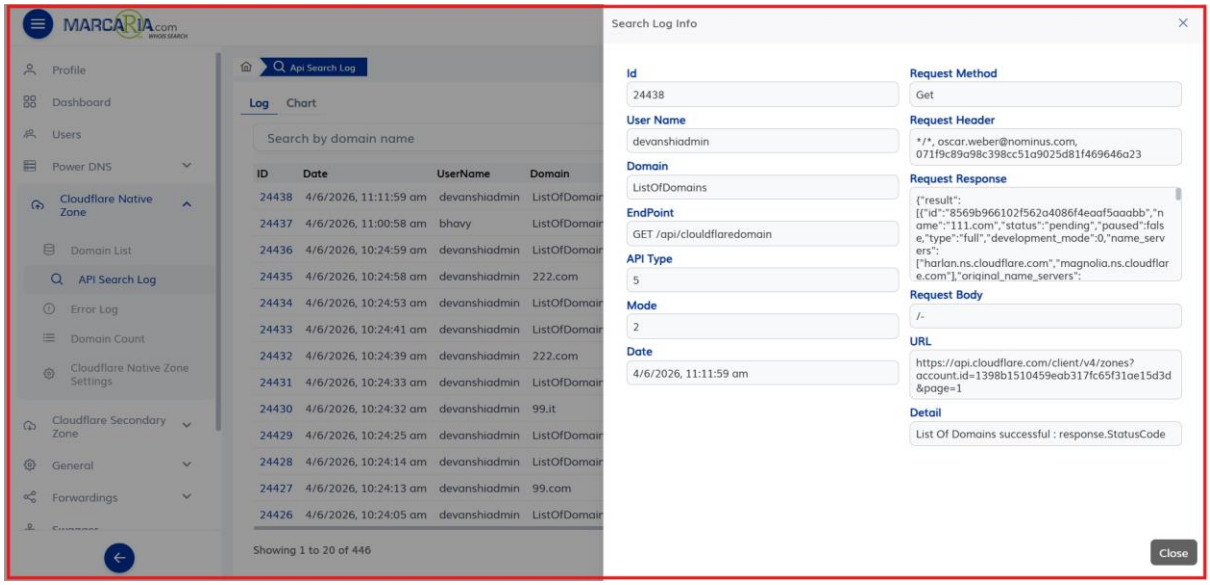
API Search Log:

a. Here users are able to view API log details.
b. Users are able to download the CSV file of the API log.
c. Users are able to see different filter options such as…
   i. **User:** Can filter according to the Users.
   ii. **Mode:** Can filter according to the Mode.
   iii. **API Type Dropdown:** Can filter according to the API Type.
   iv. **Error Dropdown:** Can filter according to the Error Type.
   v. **Domain:** Search by entering the domain name.
   vi. **Endpoint Dropdown:** Can filter according to the Endpoint.
   vii. **FilterBy:** Can filter according to the beginning with, end with and contains.

d. Users are able to view information like….
    i. **ID:** Unique identification number.
    ii. **Date:** Timestamp of the API log.
    iii. **User Name:** Display user name. (Login with that name - admin)
    iv. **Domain:** Display domain name
    v. **End Point:** Display API endpoint.
    vi. **Detail:** Provides information.
    vii. **Mode:** Display information of mode.
    viii. **APIType:** Display type of API.
    ix. **Error:** Display status if error occurs then Yes else No.



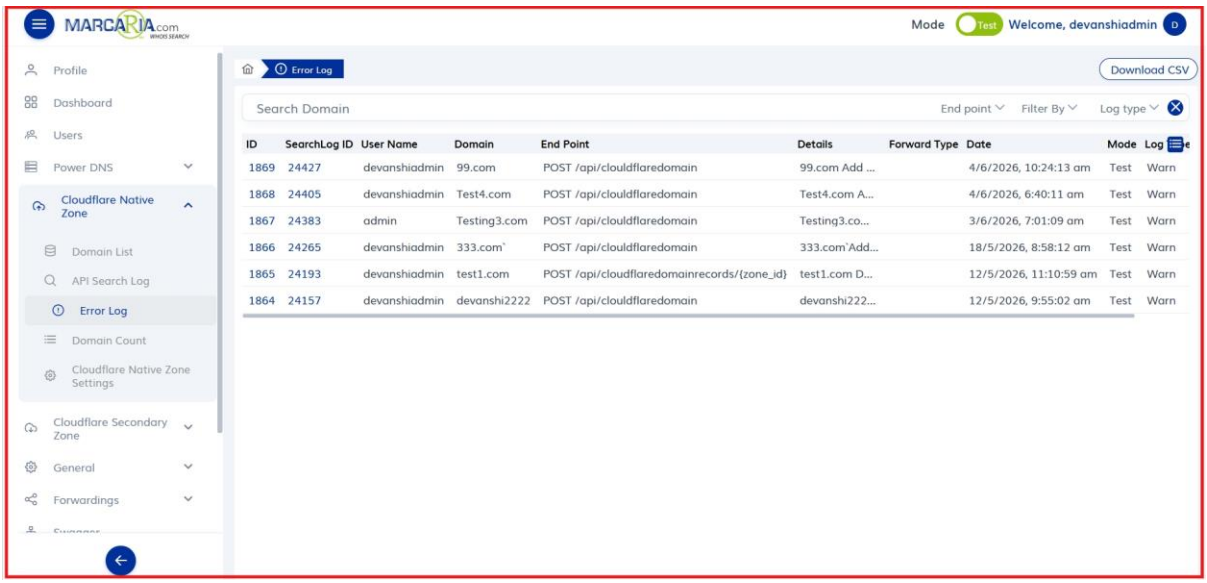
e. Once clicked on the unique identification number (**ID**) and the user will get a popup with the information about the Search Log.
f. All search log information will be displayed as disabled mode.
g. In search Log Info popup users will get the following information.
    i. **Id:** Unique identification number.
    ii. **User Name:** Information about User Name.
    iii. **Domain:** Information about domain.
    iv. **End Point:** Information about End Point.
    v. **URL:** Information about URL.
    vi. **Detail:** Provides Information.
    vii. **Request Methods:** Information about method name.
    viii. **Request Header:** Information about requested header.
    ix. **Requested Response:** Information about requested response.
    x. **Request Body:** Information about request body.
    xi. **API Type:** Information about API type.
    xii. **Mode:** Information about Mode.
    xiii. **Date:** Information about time stamp.



Error Log:

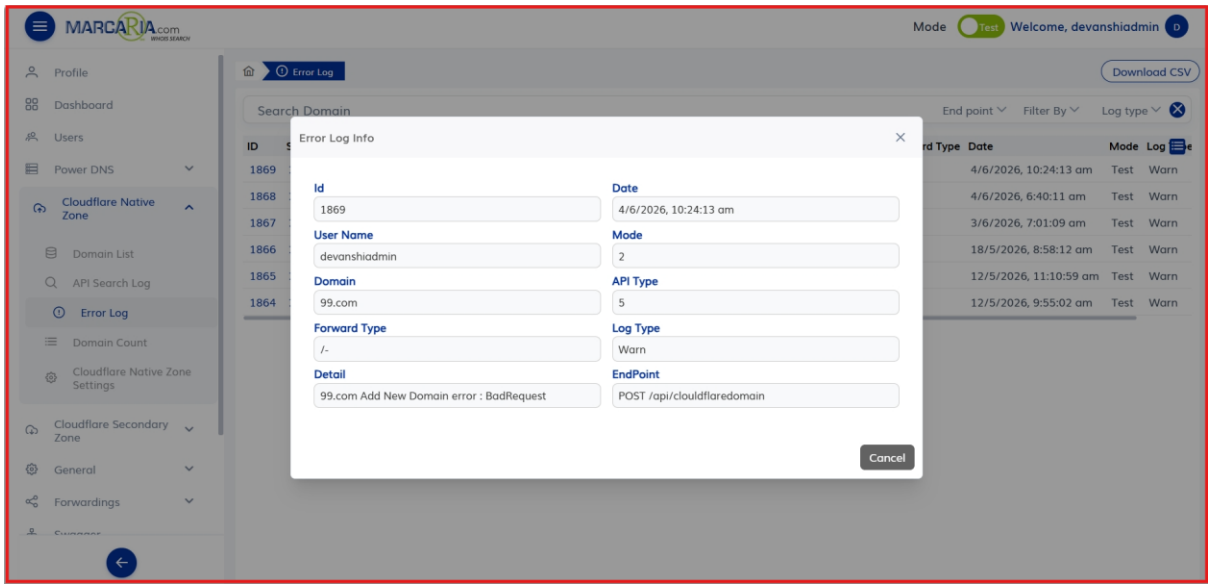
a. Here users are able to view the error log if any errors occurred.
b. Users can have the option to apply filters based on..
    i. **Search By Domain:** Enter the domain name which you want to search.
    ii. **Search By Endpoint Dropdown:** Select the endpoint from the list.
    iii. **Filter By Dropdown:** Can filter according to the beginning with, end with and contains.
    iv. **Log Type Dropdown:** Filter records according to the log type.

c. Users are able to view the following information..
    i. **ID:** Unique Identification number.
    ii. **SearchLog ID:** It is display the unique ID of the Search Log i.e. API Search Log.
    iii. **User Name:** Display user name.
    iv. **Domain:** Name of the domain in which error occurred.
    v. **EndPoint:** Name of the End Point.
    vi. **Detail:** It will display the errors in details
    vii. **Forward Type:** Display the forward type.
    viii. **Date:** Timestamp is displayed. (Display Date & Time)
    ix. **Mode:** Display information about Mode. (Live :1 , Test : 2)
    x. **Log Type:** Display information about the log type.
    xi. **API Type:** Display information about API type.
d. Here users are able to download CSV files of the error log.



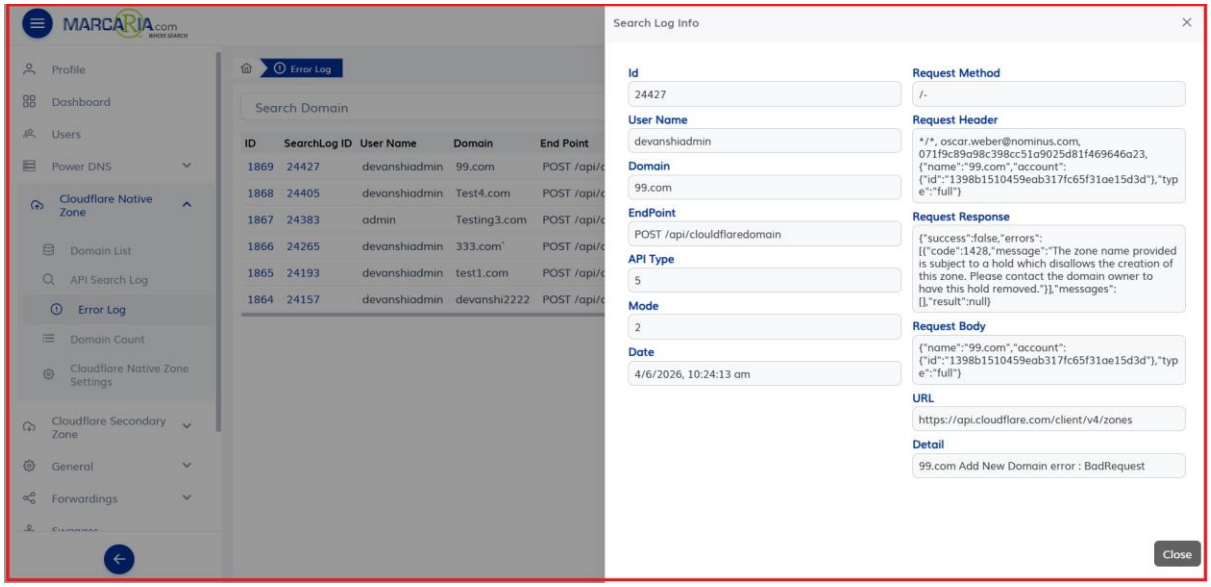
e. Click on the ID user will be able to view the Error Log Info popup with following details.
   i. **ID:** Unique identification number.
   ii. **User Name:** Display user name.
   iii. **Domain:** Display domain name.
   iv. **End Point:** Display End Point name.
   v. **Detail:** Display details about error.
   vi. **Forward Type:** Display the forward type.
   vii. **Date:** Display timestamp.
   viii. **Mode:** Display mode information.
   ix. **API Type:** Details of API type.
   x. **Log Type:** Display log type information.

f. All error log information will be displayed as disabled mode.



g. Click on the **Search Log ID** user will be able to view Search Log Info popup with following information…
   i. **Id:** Unique identification number.
   ii. **User Name:** Information about User Name.
   iii. **Domain:** Information about domain.
   iv. **End Point:** Information about End Point.
   v. **URL:** Information about URL.
   vi. **Detail:** Provides Information.
   vii. **Request Methods:** Information about method name.
   viii. **Request Header:** Information about requested header.
   ix. **Requested Response:** Information about requested response.
   x. **Request Body:** Information about request body.
   xi. **API Type:** Information about API type.
   xii. **Mode:** Information about Mode.
   xiii. **Date:** Information about time stamp.

h. All search log information will be displayed as disabled mode.



Note Error Log :

- **Error Log :**

1. **Message :** NULL

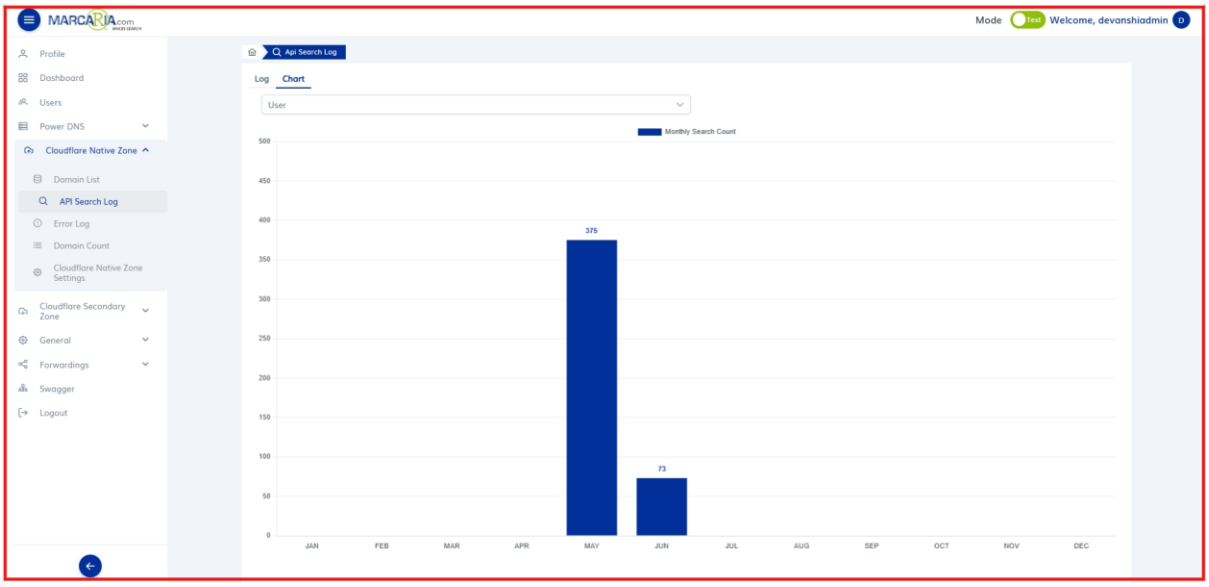
- If there is an error in the coding, the message will be displayed in the Message Column. (Message **:** Decoded String is not a valid)
- Id will not be displayed in Search Log Id. (Blank Display)

2. **Detail :**

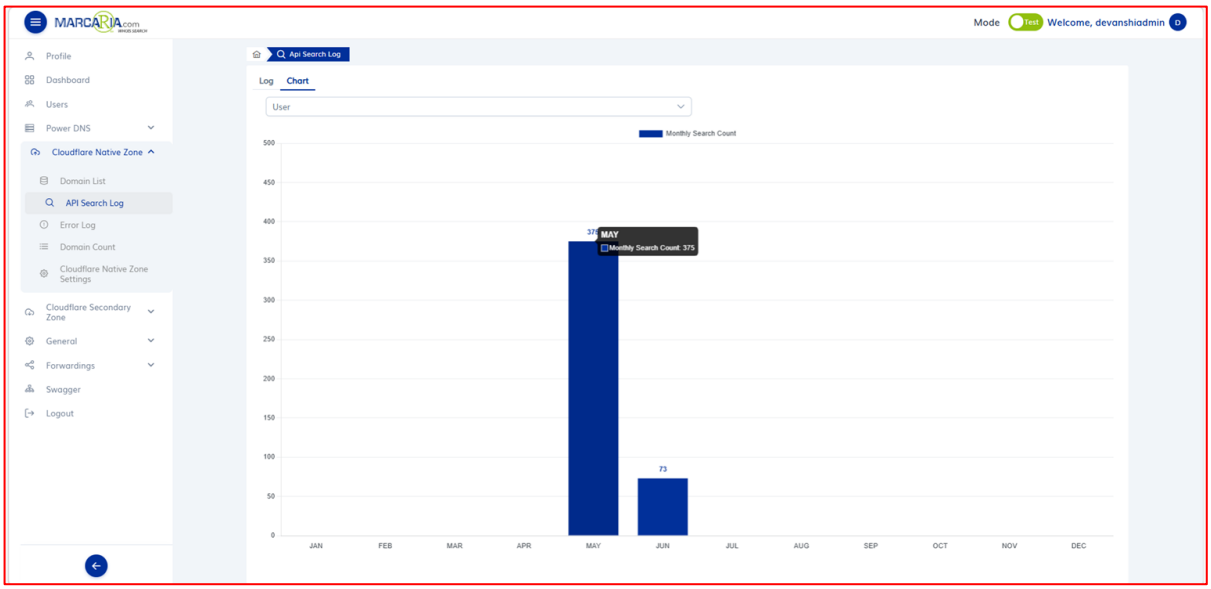
- If there is any error in the Api then the message will be displayed in the Detail column. (Message **:** System.ArgumentException)
- The ID will be displayed in the search log ID.

4. API Search Chart.

a. It will display the total of API searches in the chart form.
b. It will display months on X-axis and API search count on Y-axis.
c. Here users also can see the data according to the user by selecting the user available from the drop down.

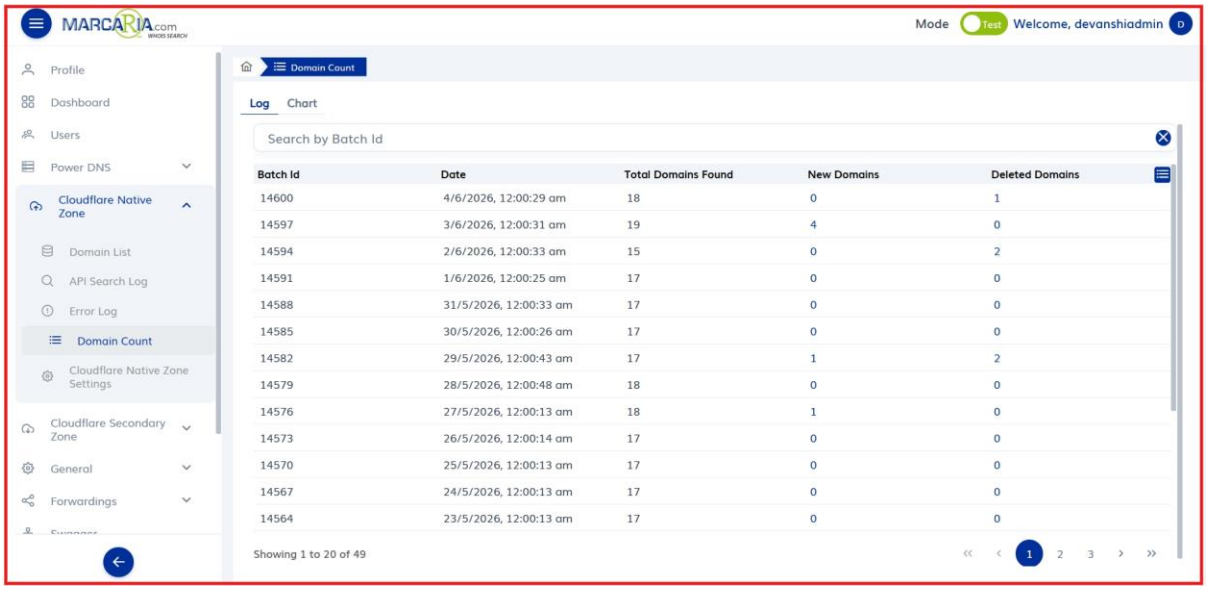


d. When the user hovers over the graph, the month name and monthly API search count will appear.

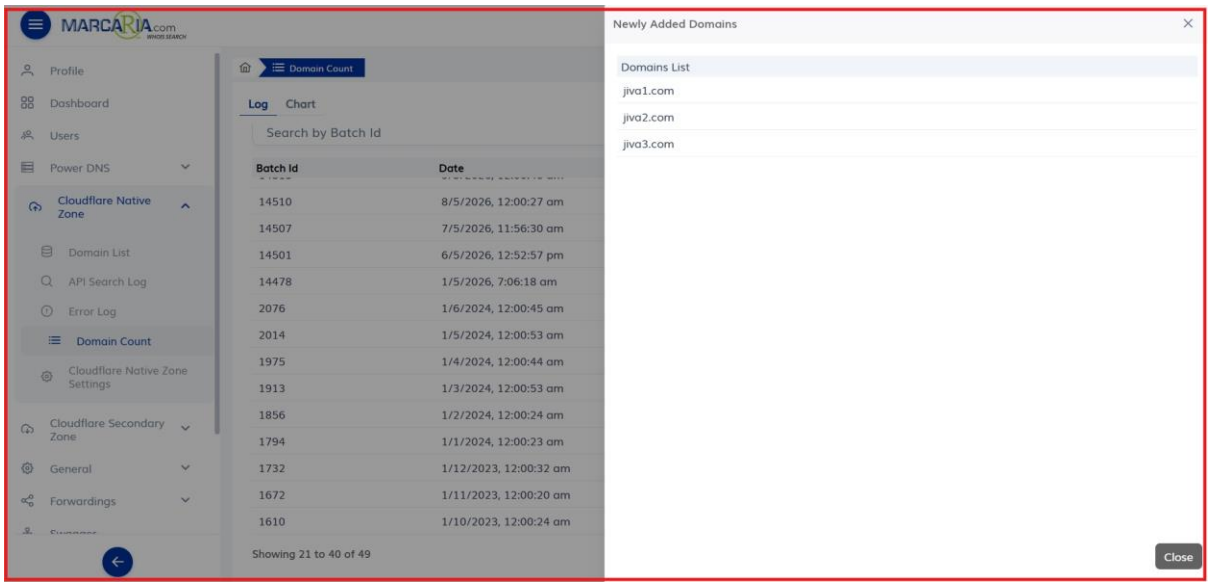


Domain Count:

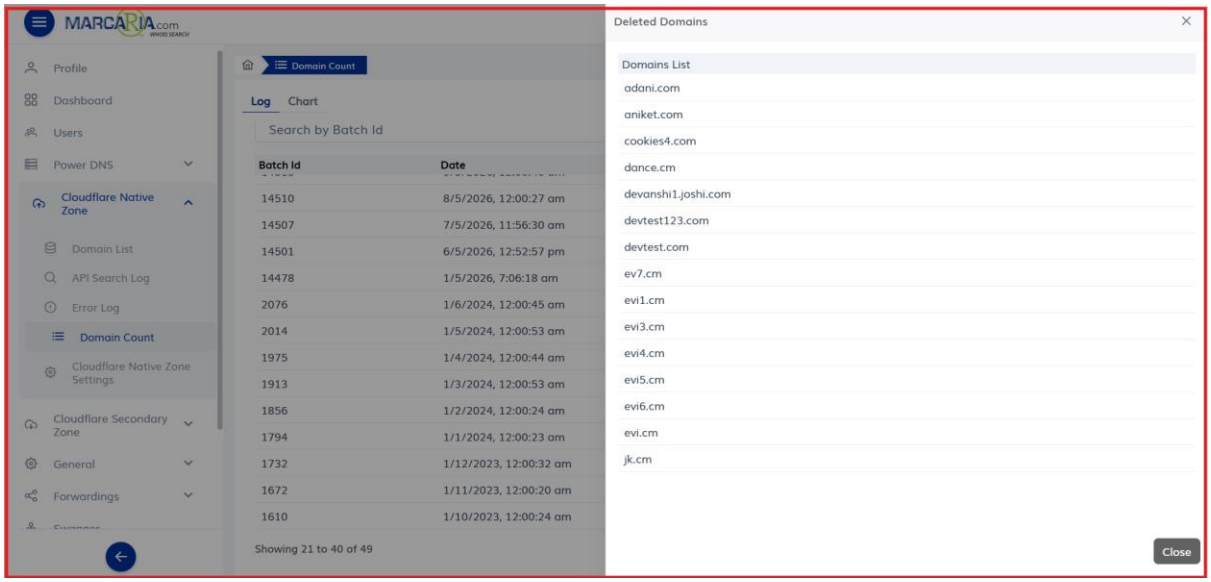
a. Here users can check the log of the created domain according to the date.
b. The Latest created domain will be displayed **Tomorrow**. (Domain Count)
c. Users are able to find the table with the following fields..
    i. **Batch ID:** It is a unique identification number.
    ii. **Date:** The timestamp of the domain creation.
    iii. **Total Domain Found:** It will provide information about the total domain creation till date.
    iv. **New Domain:** It will provide information about the newly created domain of the specific date.
    v. **Deleted Domain:** It will provide information about the deleted domain for the specific date.
d. Using a search box, users are able to search the records.
e. Using sorting functionality users are able to sort the data.



f. When the user clicks on the **new domain count**, the list of newly created domains will be shown.

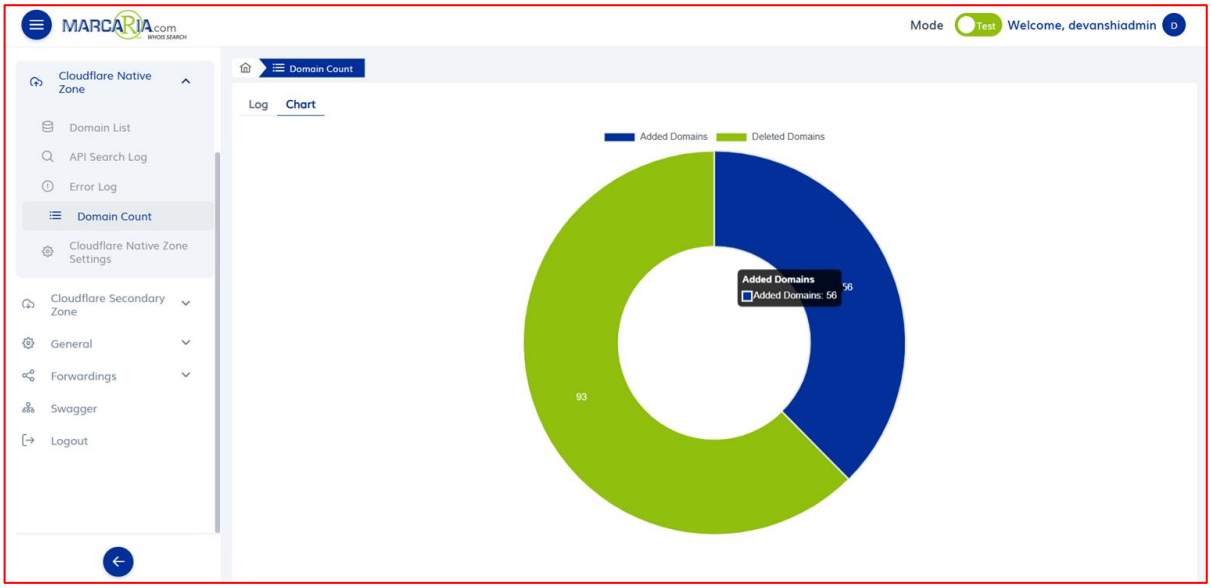


g. When the user clicks on the **deleted domain count**, the list of deleted domains will be shown.



Domain Count Chart.

a. Here users can view the data in the chart form.
b. Here users can see Added Domains and Deleted Domains.
c. Here Blue represents Added Domains and Green represents Deleted Domains.
d. When the user hovers over the chart, the Added Domain and Deleted Domain count will appear.



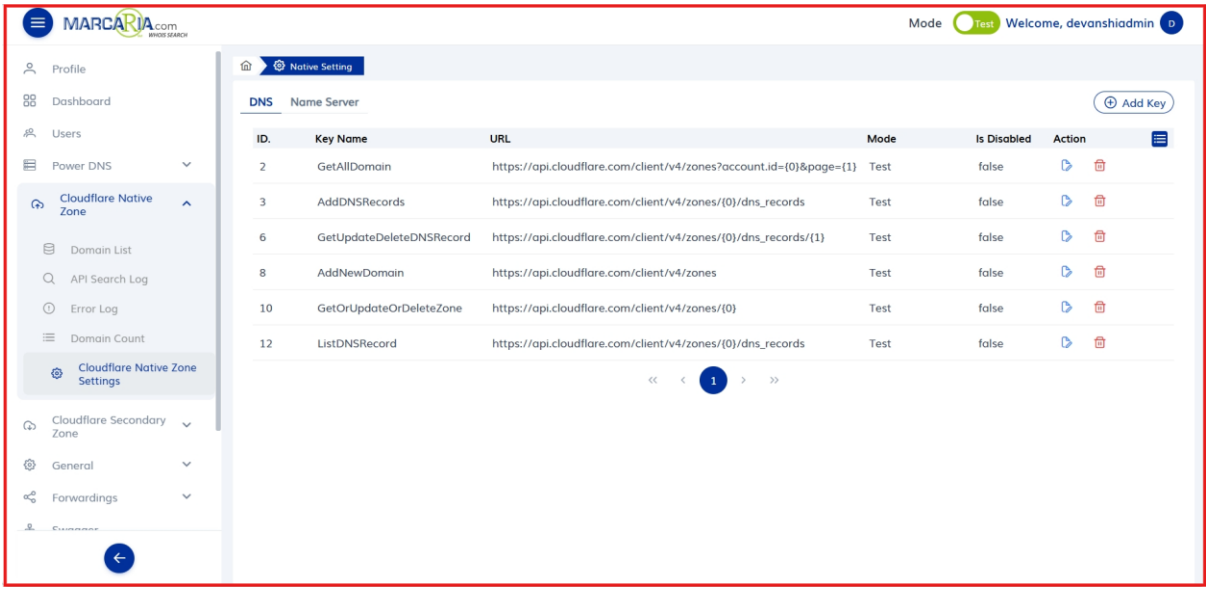
Cloudflare Settings:

Cloudflare settings details have two modules.

1. **DNS.**
2. **Name Server.**

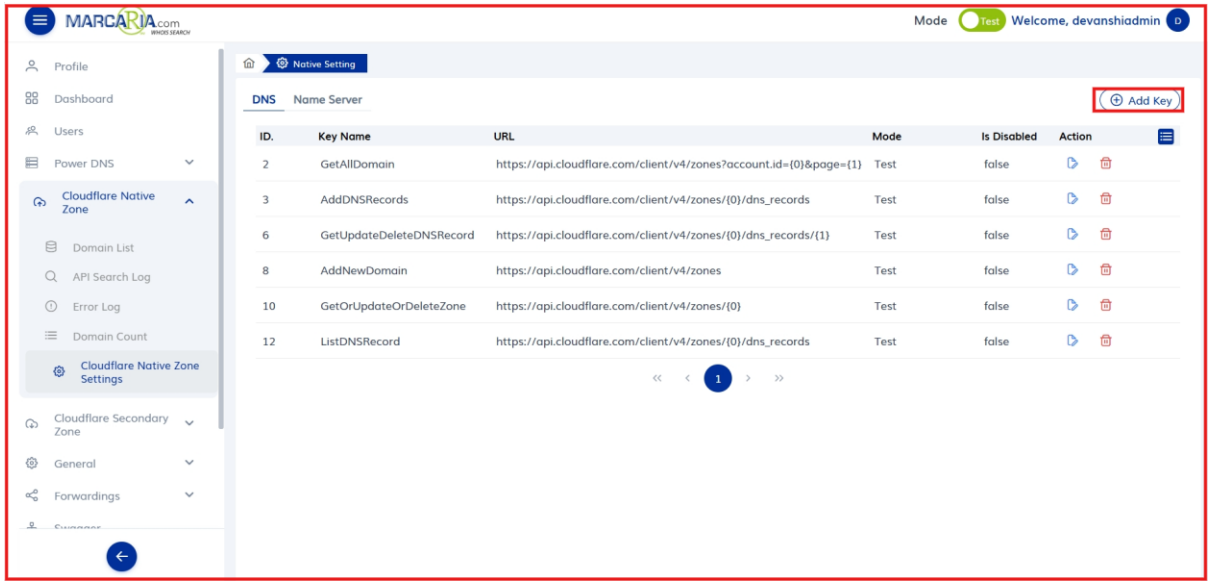
1. DNS:

- Users can map the Cloudflare Native API with the key name and this key name is used in coding to identify the Cloudflare API.
- Users can switch between **Test** and **Live** mode using the Mode toggle available in the header section.
- Here users can add Cloudflare DNS details using the **Add Key** button.
- Here users are able to view details like:
  - **ID:** Unique Identification number.
  - **Key Name:** It is the name which is used for creating reference of the API in the coding part.
  - **URL:** Display the URL which is assigned to the key. When a key is used in code it will indirectly call the URL.
  - **Mode:** Display the Mode information.
  - **Is Disabled:** Indicates whether the API configuration is disabled or active.
  - **Action:** User is able to Edit the information or Delete it.
- Pagination controls are available at the bottom of the grid.

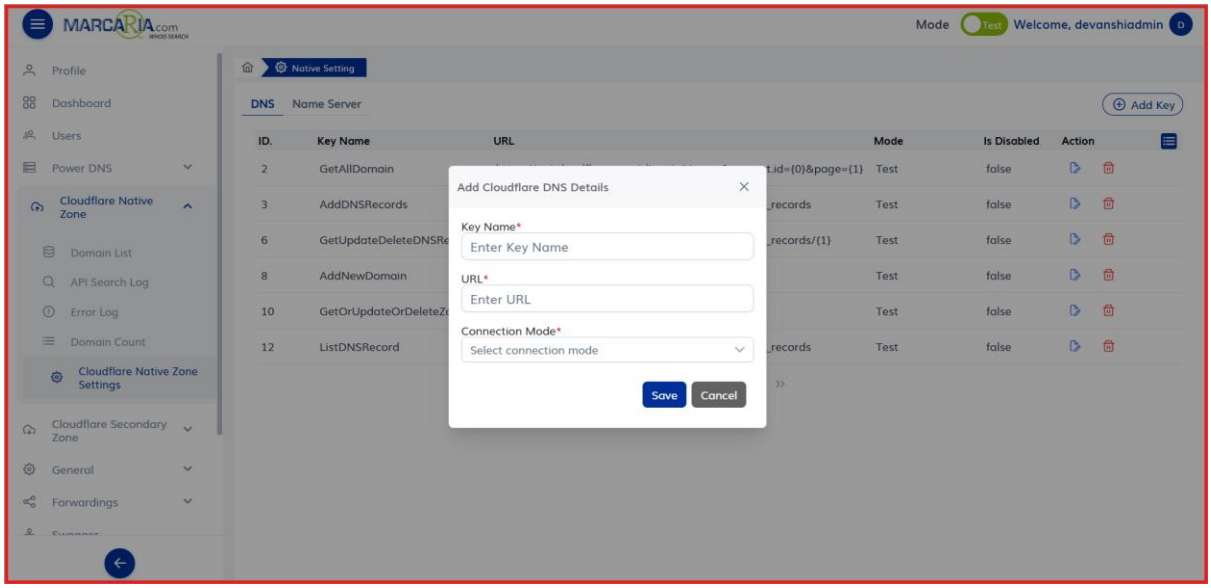


❖ Add Key Button:

- Once the user clicks on the Add Key button, the Add Cloudflare DNS Details popup will open.



- Users can enter the following details:
  - **Key Name** : Name used as a reference for the API.
  - **URL** : API endpoint URL.
  - **Connection Mode** : Select the required mode (Test/Live).
- **Submit Button:** Click on Submit button it will add DNS detail
- **Cancel Button:** Click on Cancel button pop will closed

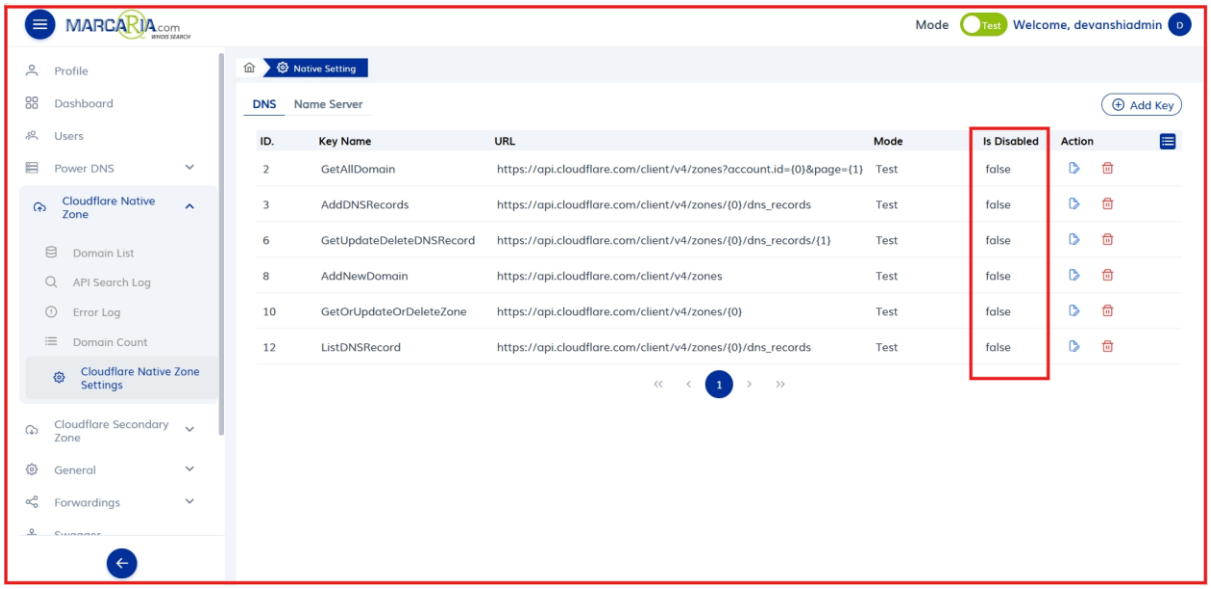


❖ Edit Button:

- Click on the Edit icon it will open the Edit Cloudflare DNS Details popup which allows users to edit the data.
  - **Key Name** : Name used as a reference for the API.
  - **URL** : API endpoint URL.

- **Connection Mode** : Select the required mode (Test/Live).
- **Is Disabled:** Users can enable or disable the DNS setting.

- If the user checks the **Is Disabled** checkbox and saves the data, the DNS setting will be marked as disabled.
- If the user unchecks the **Is Disabled** checkbox and saves the data, the DNS setting will remain active.

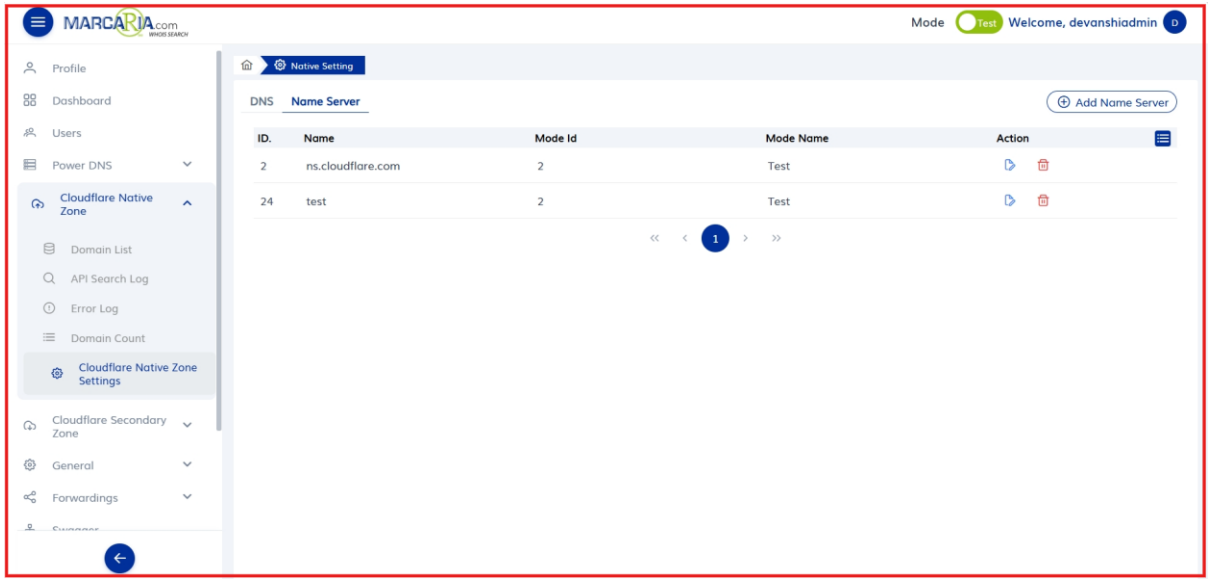


❖ Delete Button:

- Click on the Delete Button user is able to delete the DNS Detail and get the message like - **“Cloudflare Record Successfully Deleted”.**

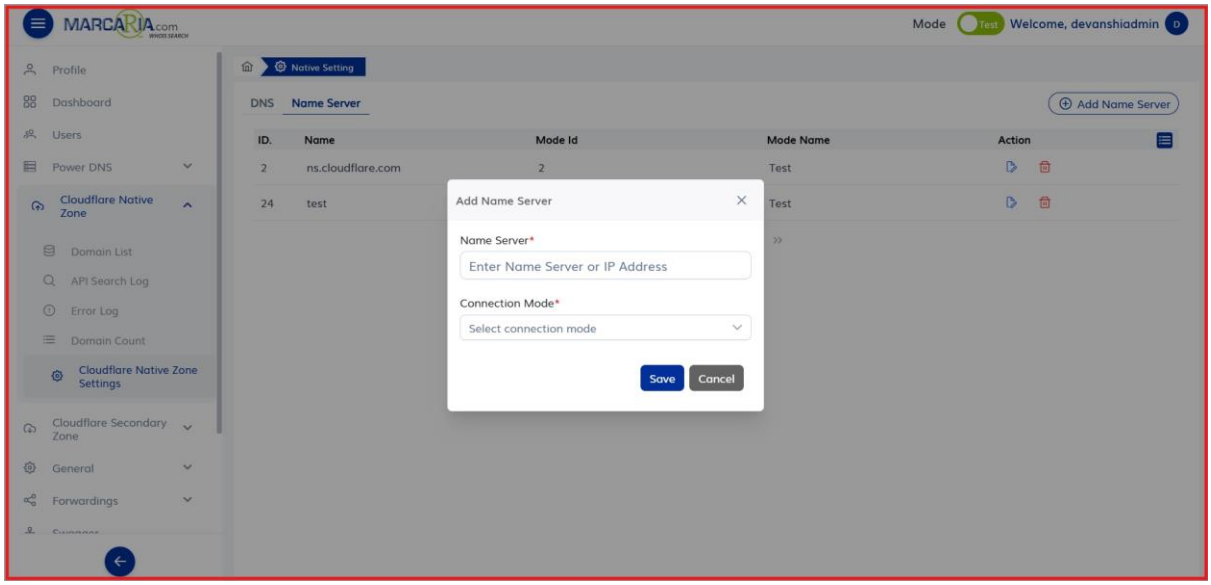
2. Name Server :

- When a user creates a new domain at that time the records which are added by default will be managed from the Cloudflare settings -> Name Server (Tab).
- Here, users are able to view information like..
  - **ID:** Unique IDentification number.
  - **Name Server:** Information about Name Server.
  - **Mode ID:** Information about mode id.
  - **Mode Name:** Information about mode name.
  - **Action field:** Provide Edit and Delete option.



❖ Add Name Server Icon:

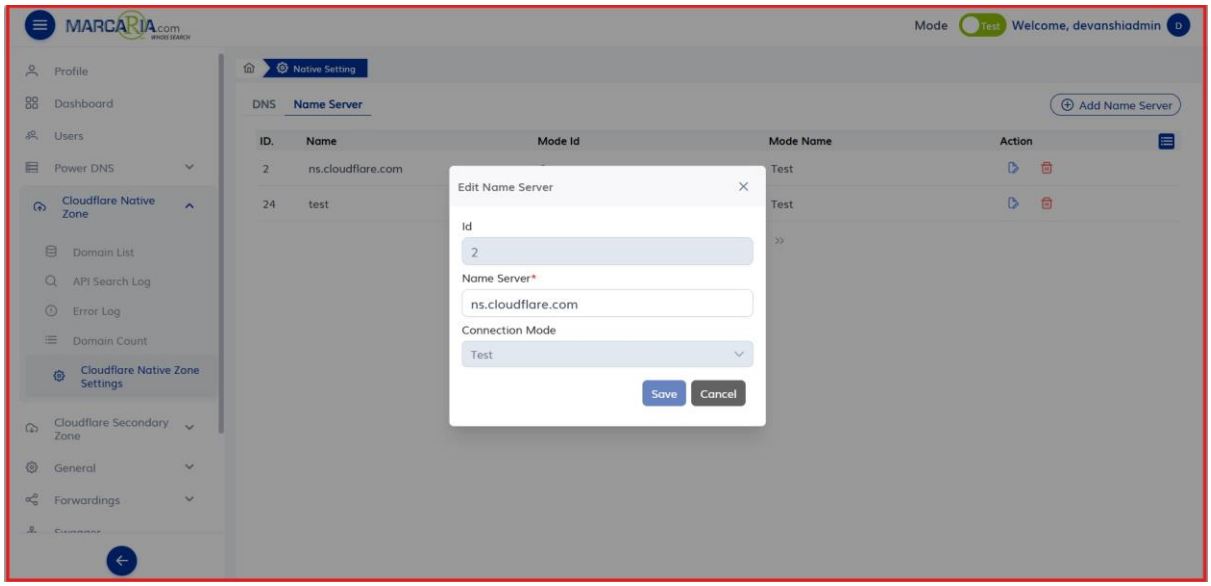
- Click on the Add Name Server button it will open the popup and the user can see the following fields for adding data.
- **Name Server:** Used to enter the Name Server.
- **Connection Mode:** Used to select the mode (Test/Live).



- **Save Button:** Click on save button it will add DNS server
- **Cancel Button:** Click on Cancel button pop will closed

❖ Edit Button:

- Click on the Edit icon pop will open like..
    - **ID:** Unique Identification number.
    - **Name Server:** Information about Name Server.
    - **Connection Mode:** Information about connection mode.
- Here users can edit the Name Server and Connection Mode.
- Users can select Test or Live from the Connection Mode dropdown.



❖ <u>Delete Button:</u>

- Click on the Delete Button user is able to delete the name server and get the message like - **“Name Server Successfully Deleted”.**

Cloudflare Secondary Zone

- Cloudflare Secondary Zone has 5 submodules.
    1. Domain List.
    2. API Search Log.
    3. Error Log.
    4. Domain Count.
    5. Cloudflare Secondary Zone Settings.

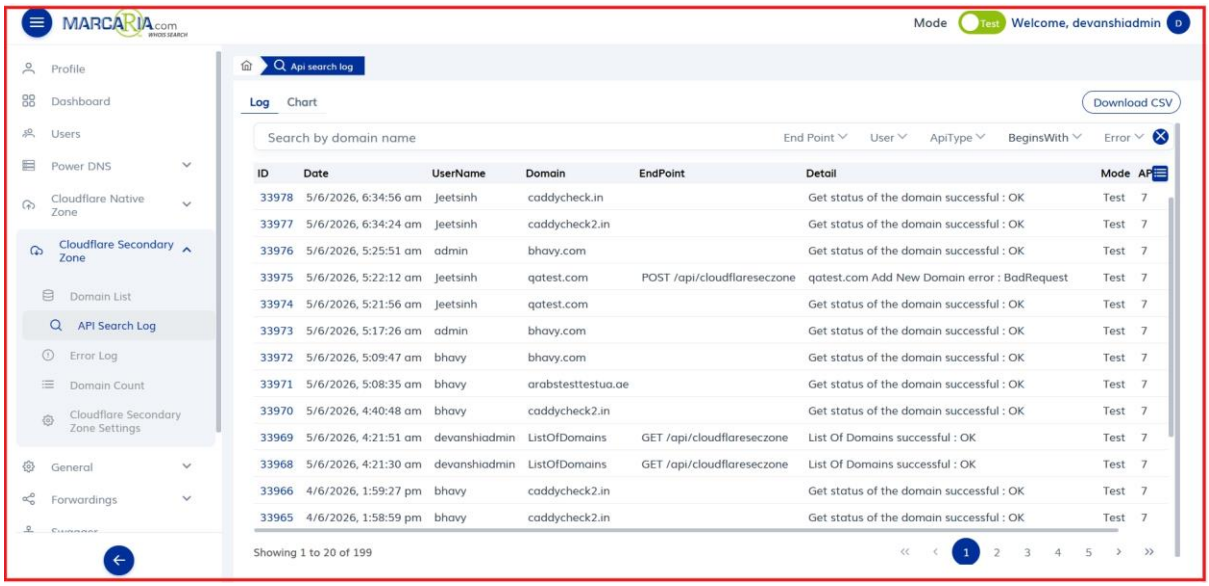
Domain List :

- When the user clicks on Cloudflare Secondary Zone, they can view options similar to those in PowerDNS.

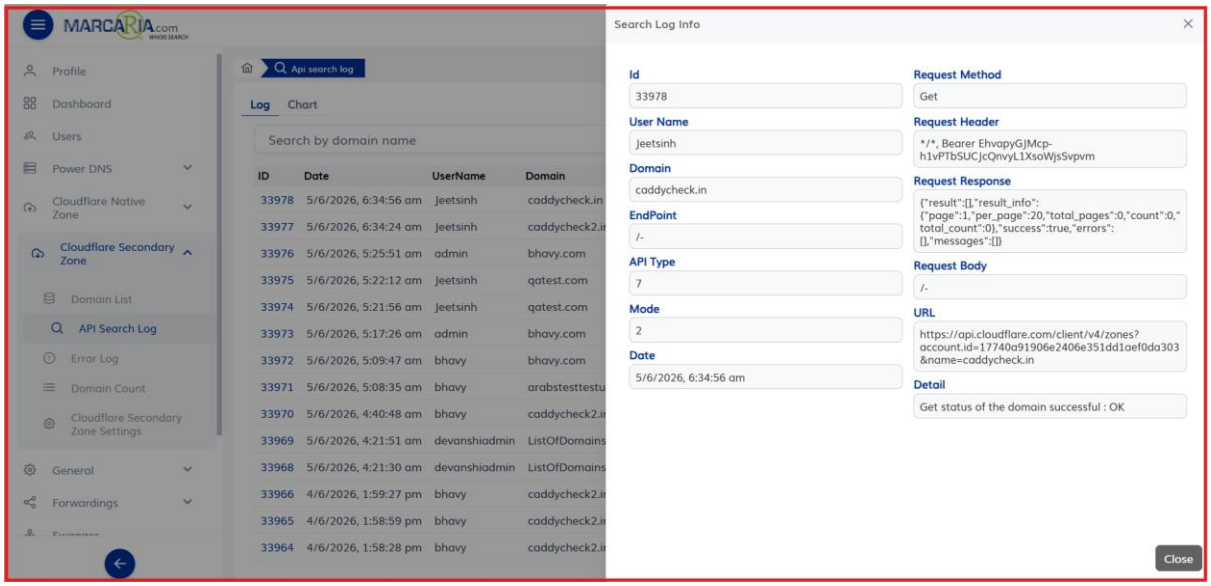
- On the domain list page, users can search for domains that are created as Cloudflare Secondary Zones.
- Here users are able to view details like.
  - No.: Unique record number.
  - Name: Display the domain name.
  - Status: Display the current status of the domain.
  - Created: Display the domain creation date and time.
  - Modified: Display the last modified date and time of the domain.
- Pagination controls are available for navigating records.

API Search Log:

- Users can view API search log details for Cloudflare Secondary Zones.
- Users are able to search logs by domain name.
- Users are able to see different filter options such as **Search Domain, Endpoint, User, API User, Filter By, and Error** for filtering API search log records.
- Here users are able to view details like **ID, Date, User Name, Domain, End Point, Detail, Mode, API Type, and Error** information for each API search log record.
- Users can download API Search Log data using the **Download CSV** button.



- When clicking on an API log ID, users can view the API search log info.



❖ API Search Chart:-

- Users have the ability to view API log charts.
- It will display the monthly API search count in chart form.
- Here users can see the API search count according to the selected user by choosing a user from the dropdown.
- The chart displays months on the X-axis and API search count on the Y-axis.

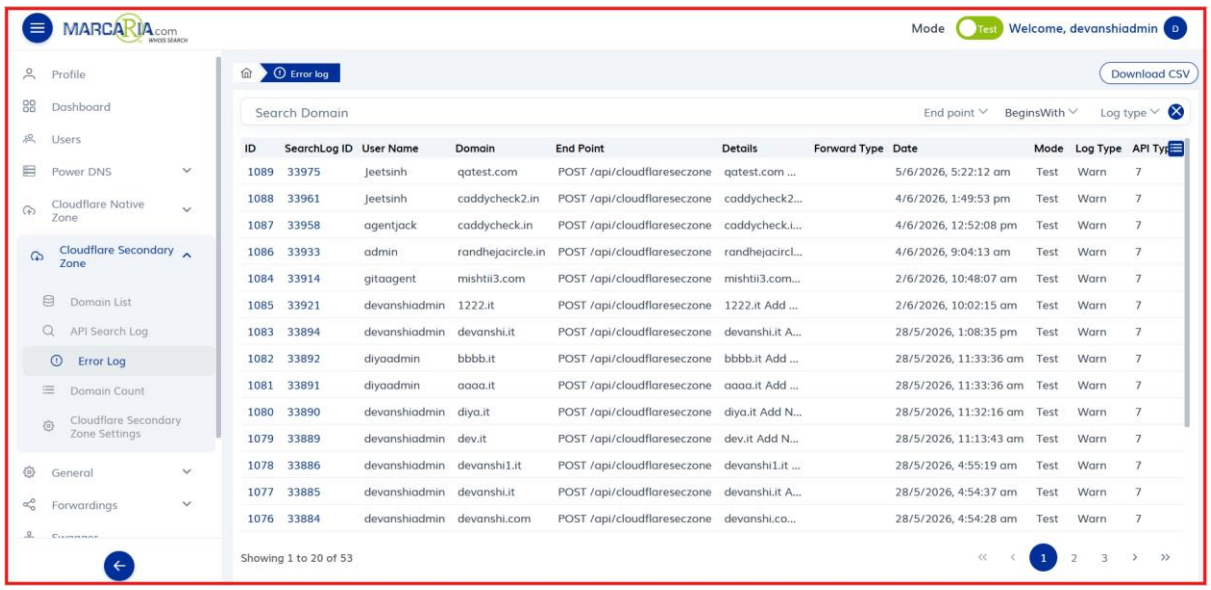


- When the user hovers over a chart bar, the month and corresponding API search count will appear.

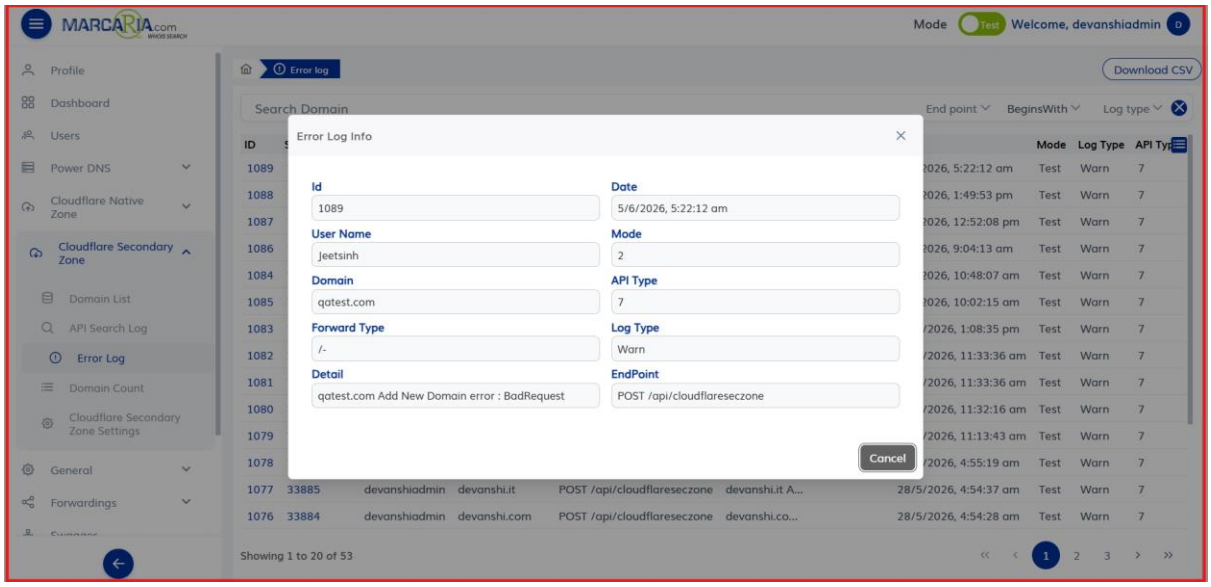


Error Log:

- Users can view API error log details for Cloudflare Secondary Zones.
- Users are able to filter error logs using different filter options such as **Search Domain, End Point, Filter By, and Log Type**.
- Here users are able to view details like **ID, Search Log ID, User Name, Domain, End Point, Details, Date, Mode, Log Type, and API Type** information for each error log record.
- Users can download Error Log data using the **Download CSV** button.



- When the user clicks on an ID, they can view the specific error log details.



Domain count :

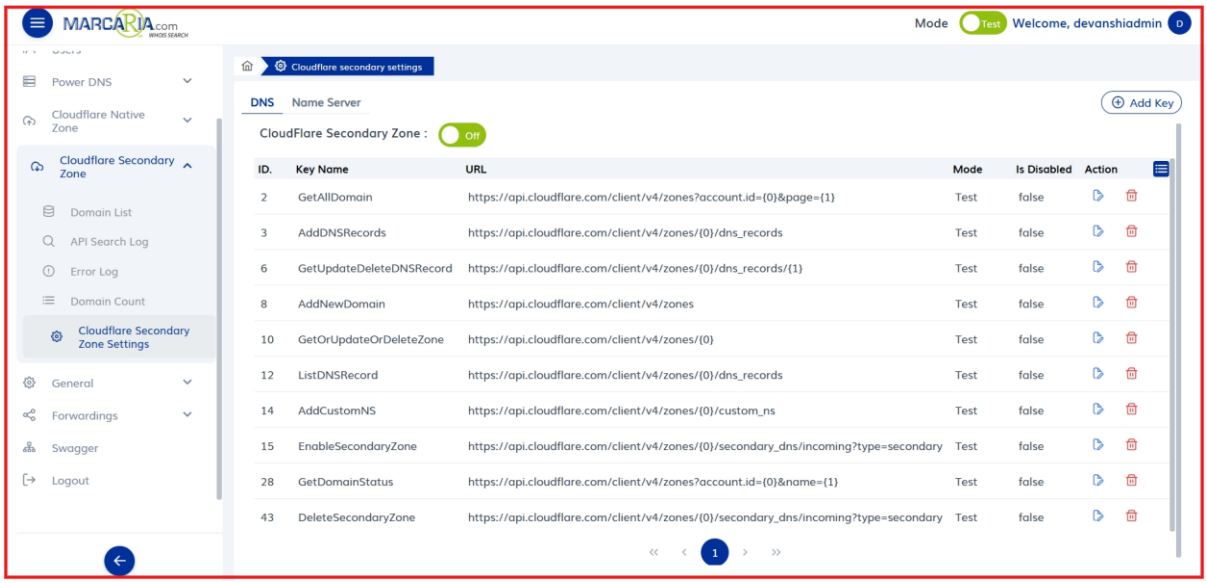
will not be visible to users, as per the client's requirements.

Domain Count chart:

will not be visible to users, as per the client's requirements.

Cloudflare Secondary Zone Settings:-

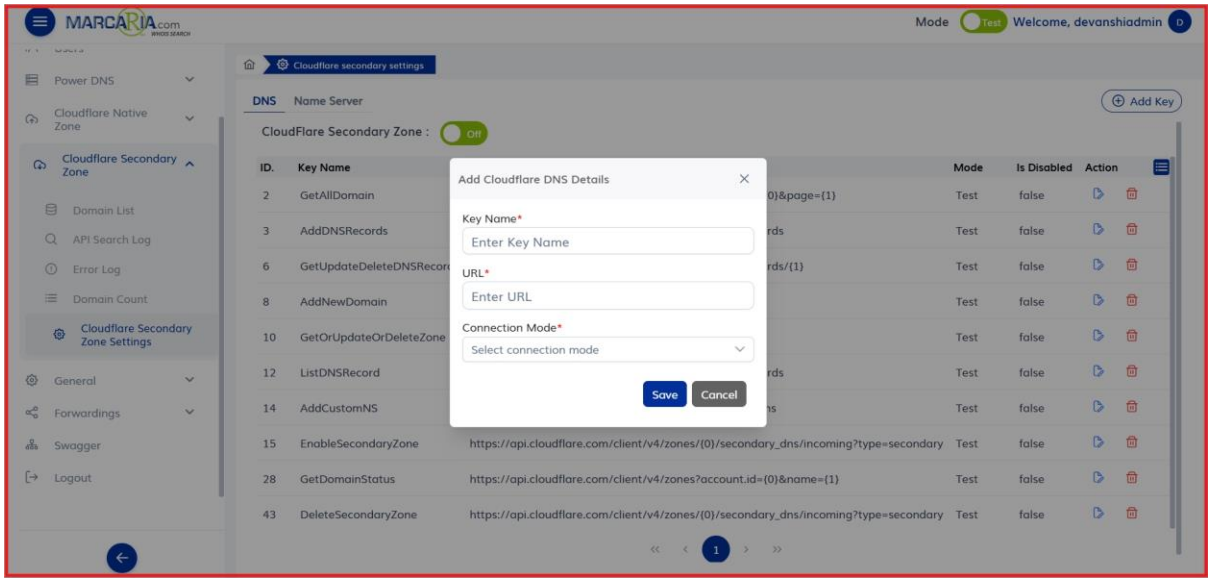
- To enable Cloudflare Secondary Zone, users must configure required authentication keys in the Cloudflare Secondary Zone Settings page as shown in the new UI.
- Users are required to add the following details:
  - **AccountID :-** 17740a91906e2406e351dd1aef0da303
  - **PeersID :-** b913eb944e8347cc981fbe391d02199b
  - **BearerToken :-** Bearer EhvapyGJMcp-h1vPTbSUCJcQnvyL1XsoWjsSvpvm



- Users can map the base (Cloudflare Secondary Zone) API with the key name and this key name is used in coding to identify the Base (Cloudflare Secondary Zone) API.
- Here users are able to enable or disable Cloudflare Secondary Zone using the ON/OFF toggle switch.
- Here users are able to view details like.
  - **ID:** Unique Identification number.
  - **Key Name:** It is the name which is used for creating reference of the API in the coding part.
  - **URL:** Display the URL which is assigned to the key. I.e. when a key is used in code it will indirectly call the URL.
  - **Mode:** Display the Mode information.
  - **Is Disabled:** Here if the user does not check the checkbox its value is N otherwise its value is Y. Here if users don't want to use the API longer at that time they can do Is Disabled checkbox checked instead of hard delete.

❖ Add Button :-

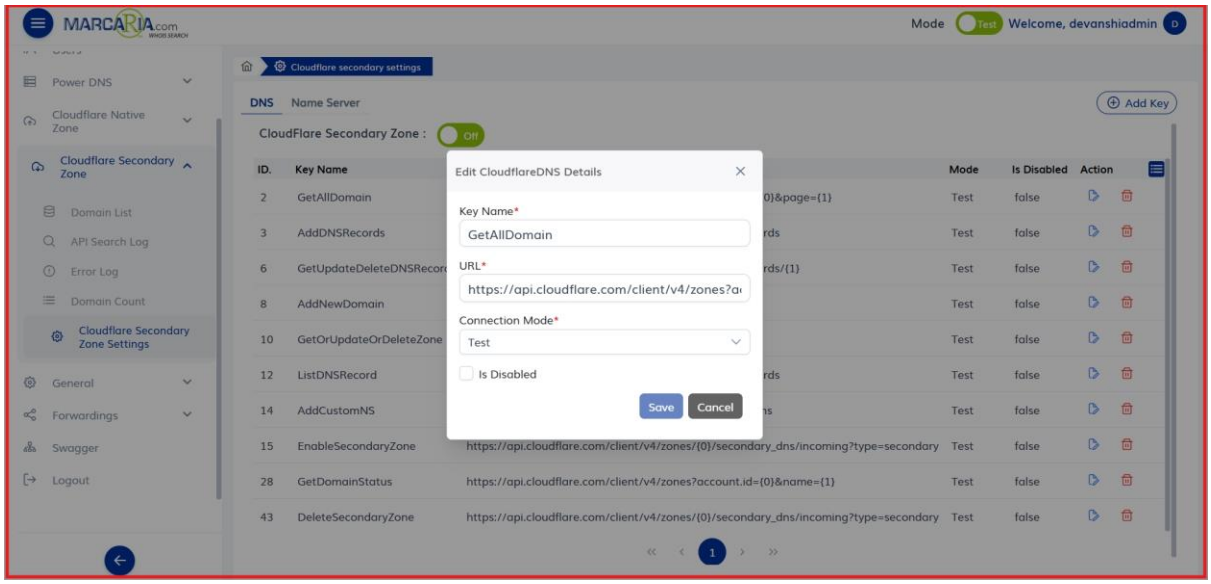
- Once the user clicks on the add key button it will show key name and URL text area and connection mode.
- **Submit Button:** Click on Submit button it will add DNS detail
- **Cancel Button:** Click on Cancel button pop will closed



- **Action:** The user is able to edit the information or delete it.

Edit Button:

- Click on the Edit icon it will open the key name , URL, connection mode and IsDisabled checkbox which allows us to edit the data.
- If the user checks the **Is Disabled checkbox** and saves the data, **Y** (Yes) will be displayed in the **Is Disabled** column of the Cloudflare DNS details page.
- If the user unchecks the **Is Disabled checkbox** and saves the data, **N** (No) will be displayed in the **Is Disabled** column of the Cloudflare DNS details page.



Delete Button:

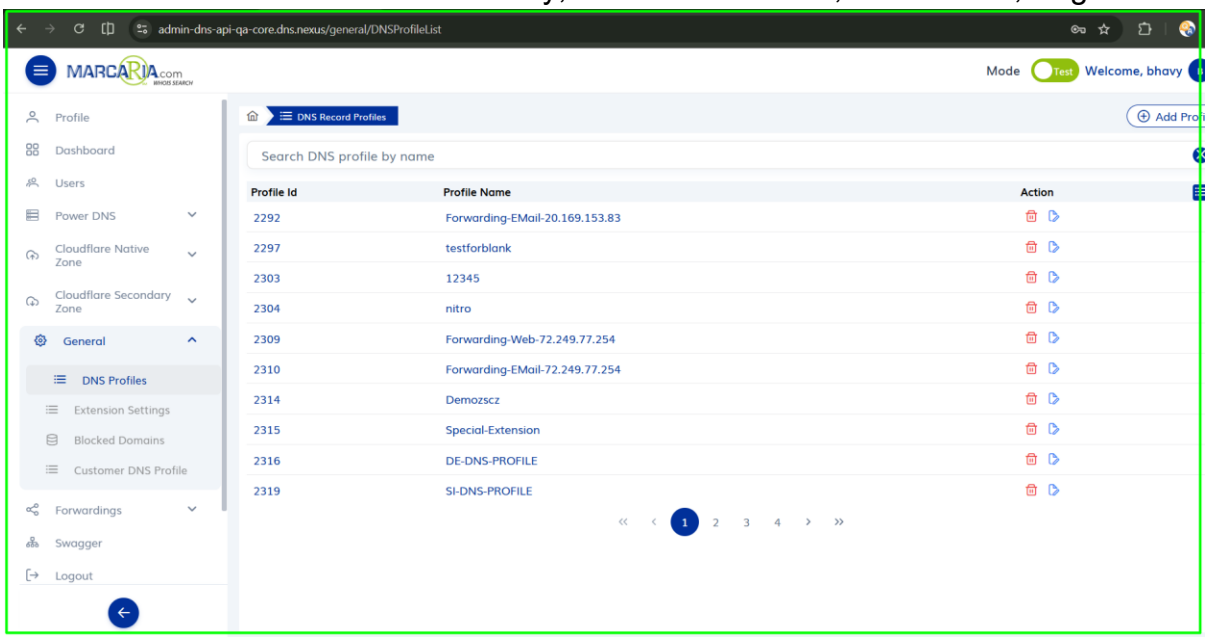
- Click on the Delete Button user is able to delete the Power DNS Detail and get the message like - **“Cloudflare DNS Record Deleted Successfully”.**

GENERAL SETTING

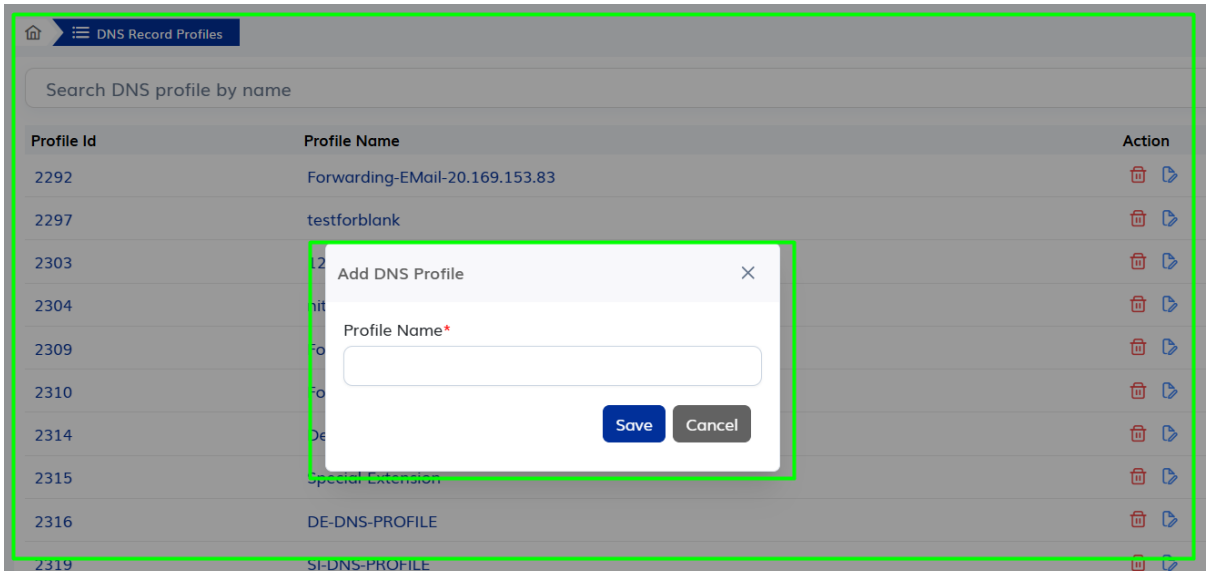
- General has 4 submodules.
    1. DNS Profile.
    2. Extension Settings.
    3. Blocked Domains.
    4. Customer DNS Profile.

DNS Profiles:

- Users can find out the DNS Profile menu option from the side menu panel.
- Click on it and the user will redirect to the DNS Record Profile page.
- On this page users can view tables with Profile ID, Profile Name, Action field with edit functionality, Add Profile Button, Search box, Pagination.

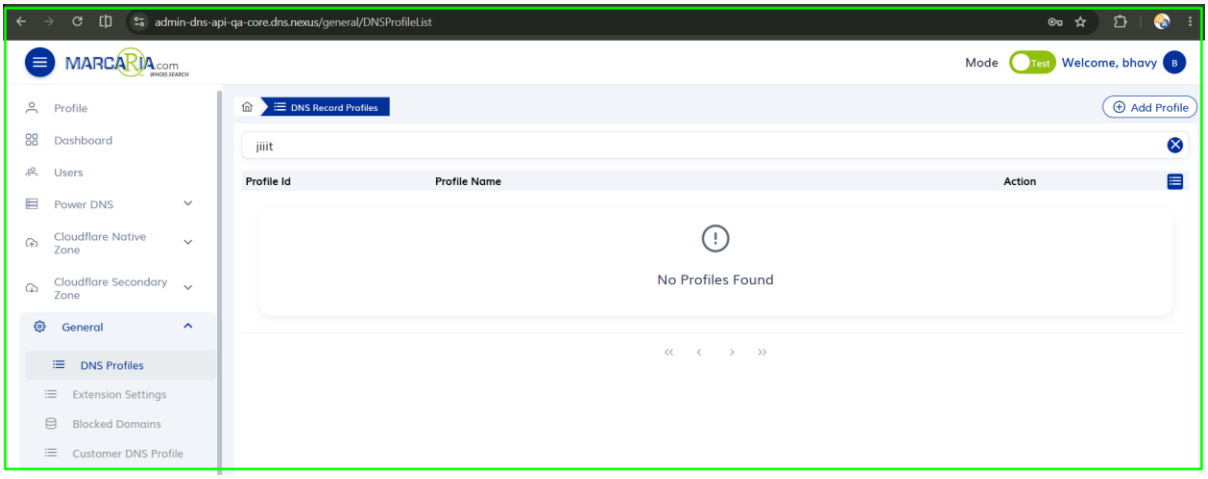


- **Add Profile Button:**
    - Users can click on the Add Profile button and it will open the popup of Add DNS Record Profile Name.
    - On this popup user is able to view Textbox for entering the profile name, Save & Cancel button.
    - If the entered profile name is already available in the list it allow to add it there is not any restrictions.

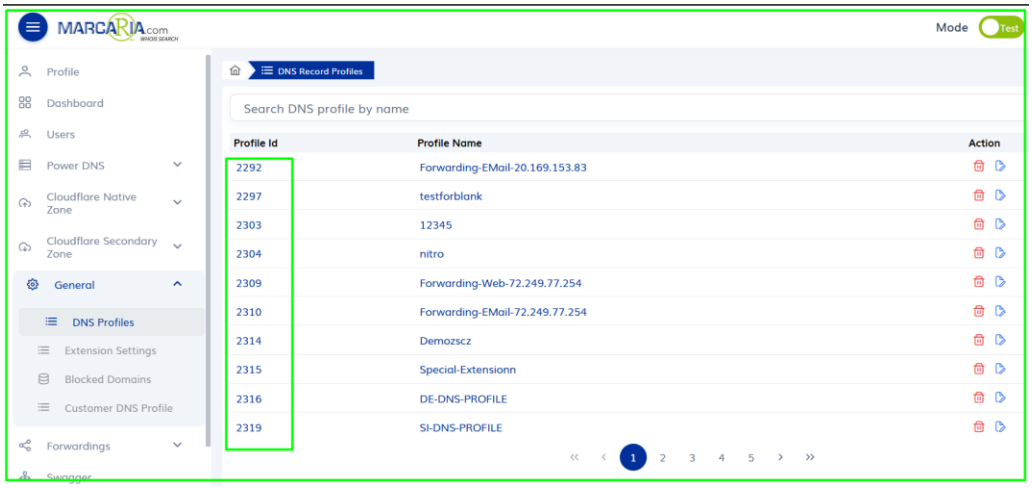


- **Save Button:**
  - Click on the Save button after entering it will create a profile with the same name which is entered and get a message like **"Profile Successfully Added".**
- **Cancel Button:**
  - Click on the cancel button and it will close the popup

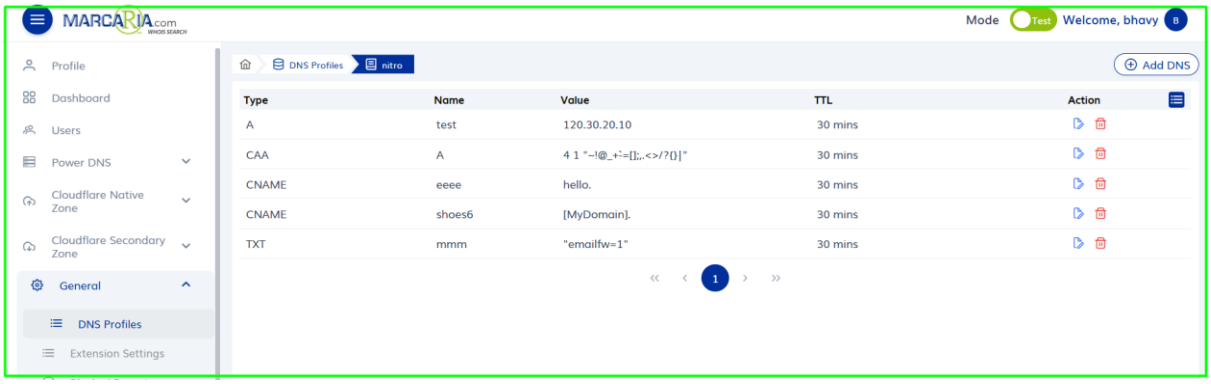
→ If a search record is not found, get a message like "No Profiles Found".



- **<u>Add Records in the created profile:</u>**
  - When user click on the profile ID OR profile name it will redirect to the DNS Record Page

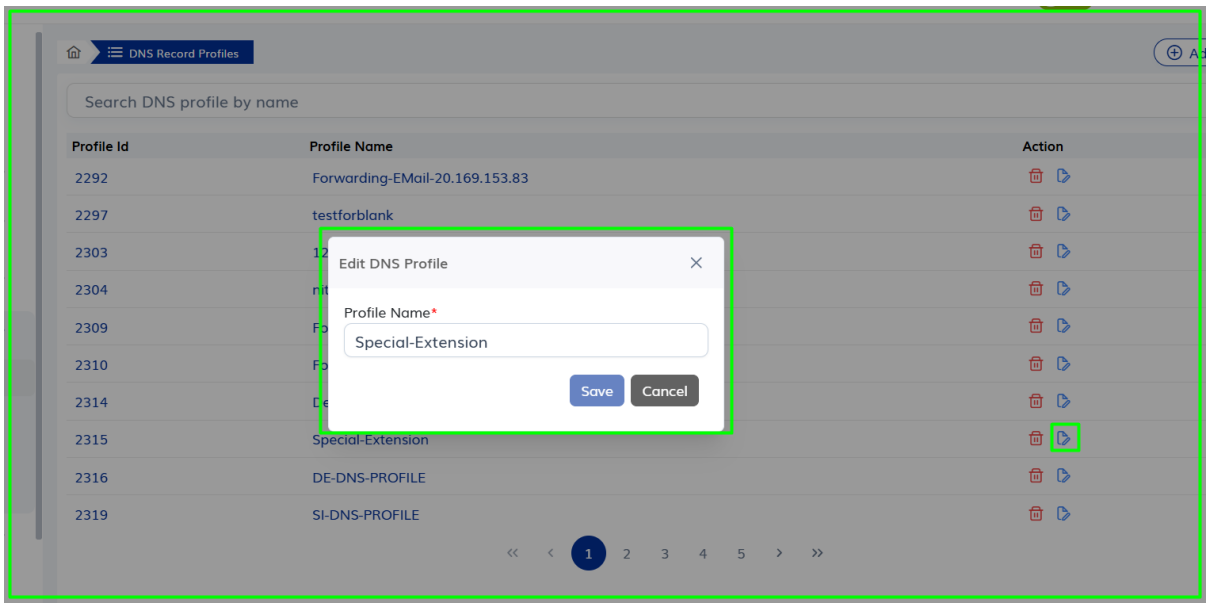


- From here, users are able to add the record for the profile.

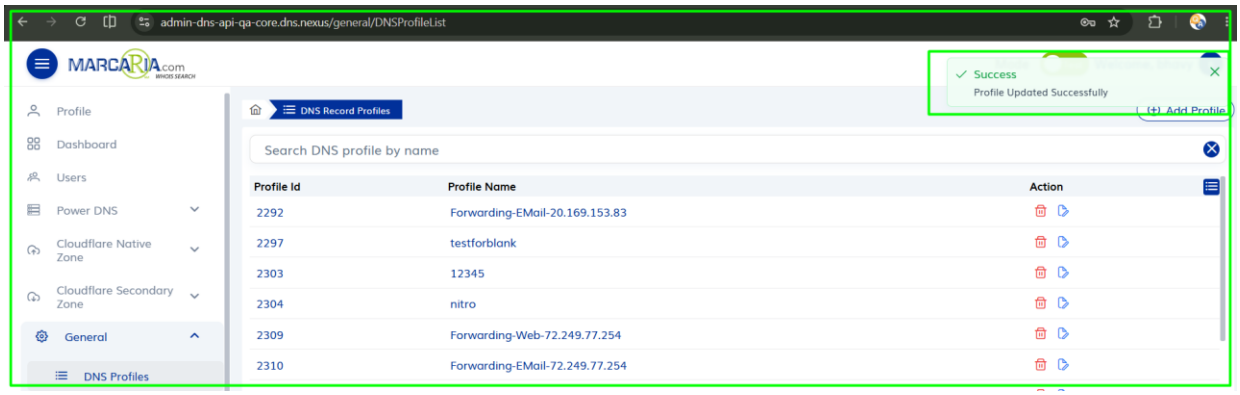


Action Column (Edit) :

- Click on Edit Icon. It will open the popup of Edit DNS Record Profile Name.



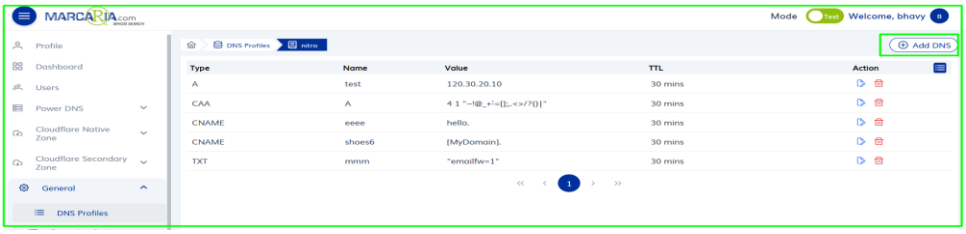
- Here after modifying the name click on **Save** button user can able to view new name in place of old name and also get the message like - **“Profile successfully updated”.**



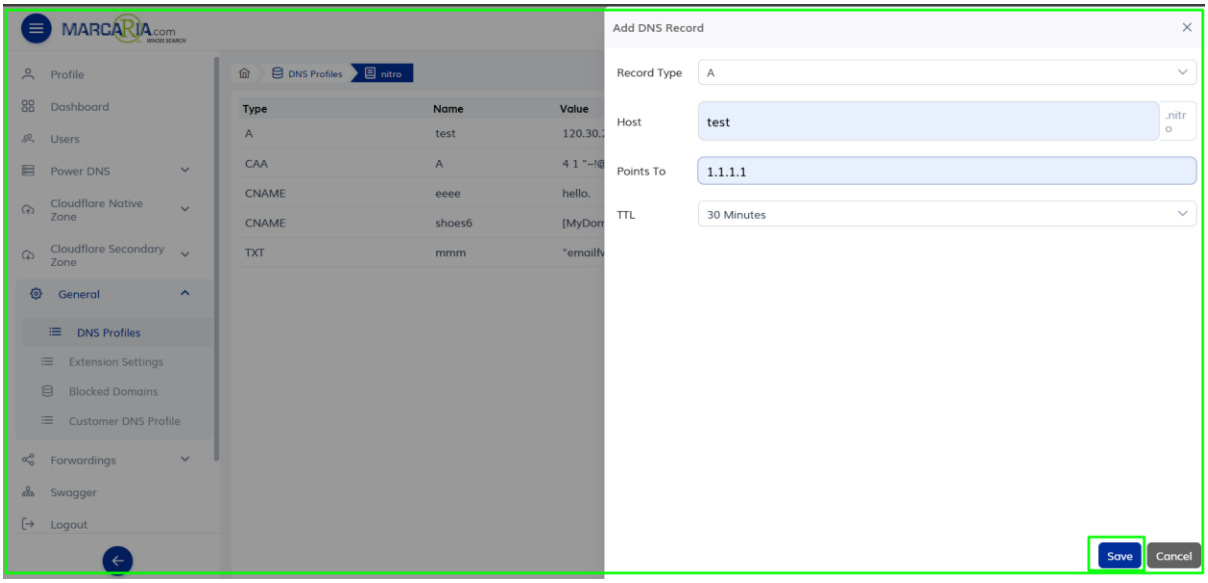
- **Cancel Button :** If you Click on cancel button, then the edit popup will close.

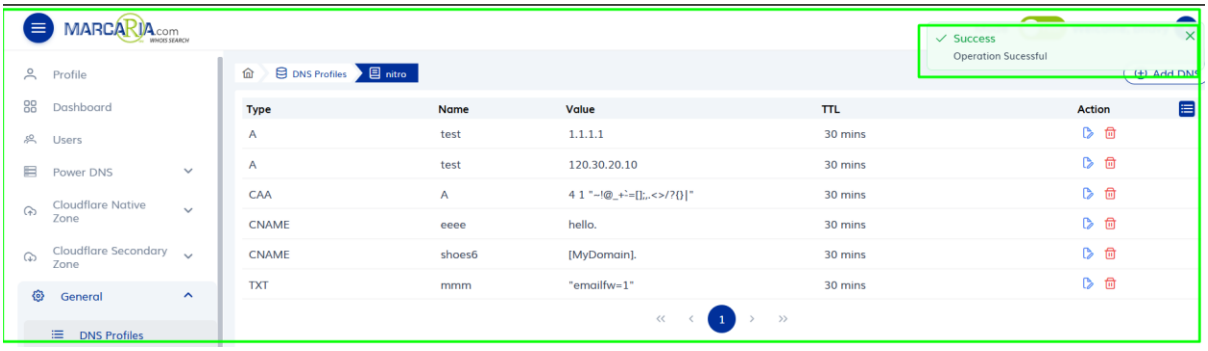
Add DNS Record Button:

- Click on the Add DNS Record button it allows the user to add records just similar to the DNS Record available in the Domain List.



- Click on Save button Record is added to the profile and get a validation message like - **“Record Successfully Added “**

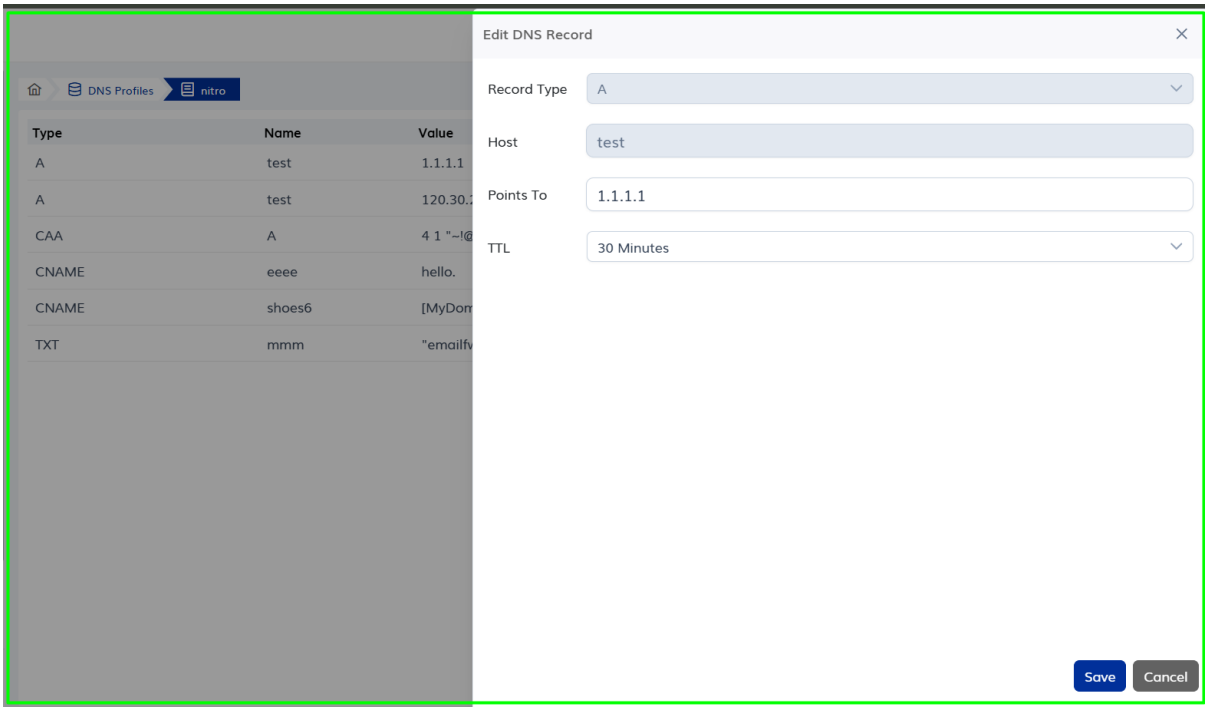




- **Edit Icon :**
  - Click on the edit icon from the Action field of the DNS records table.

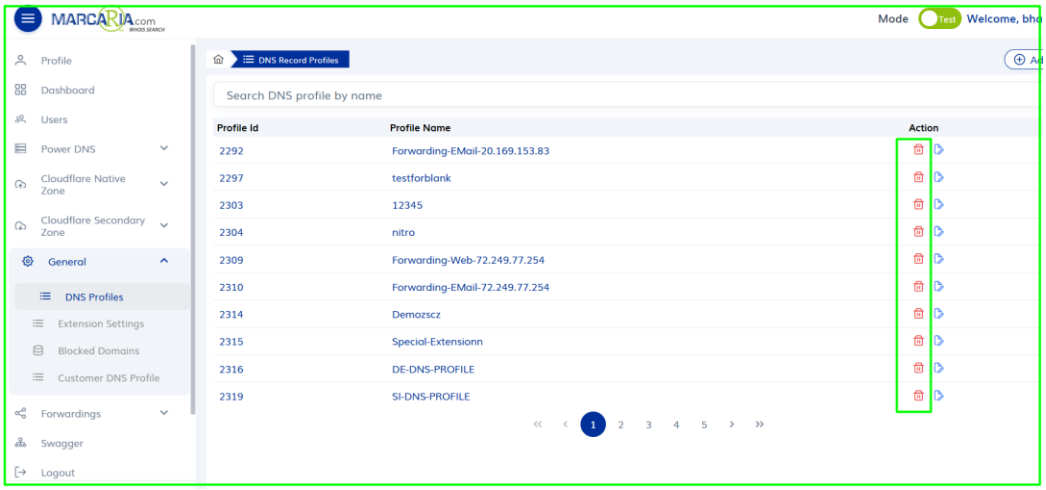


  - Users can get the popup with following details.
    - Fields of the record generated by the user.
    - Save button

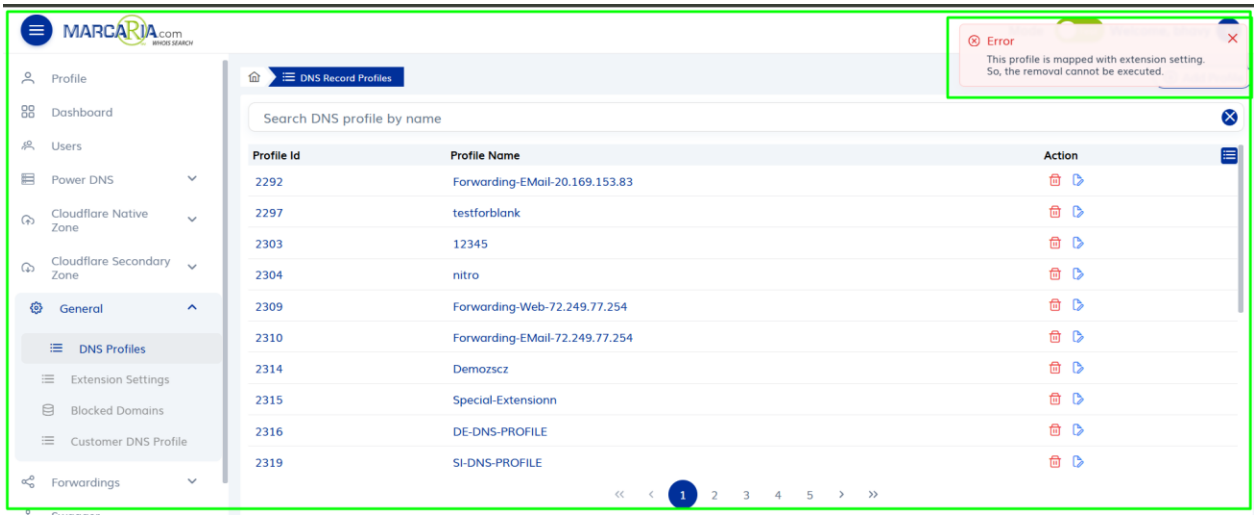


- **Save Button:**
  - Click on the Save button after editing it will edit the record and get a message like - **“ Profile Successfully updated”.**
- **Delete Button (For delete record):**
  - When the user clicks on the delete button, the record will be deleted
- **Cancel Button:**
  - Click on the cancel button and it will close the popup.

- **Delete Button (For delete DNS Profile) :**
  - Once a user opens the profile it is able to see the delete button besides the DNS record button.

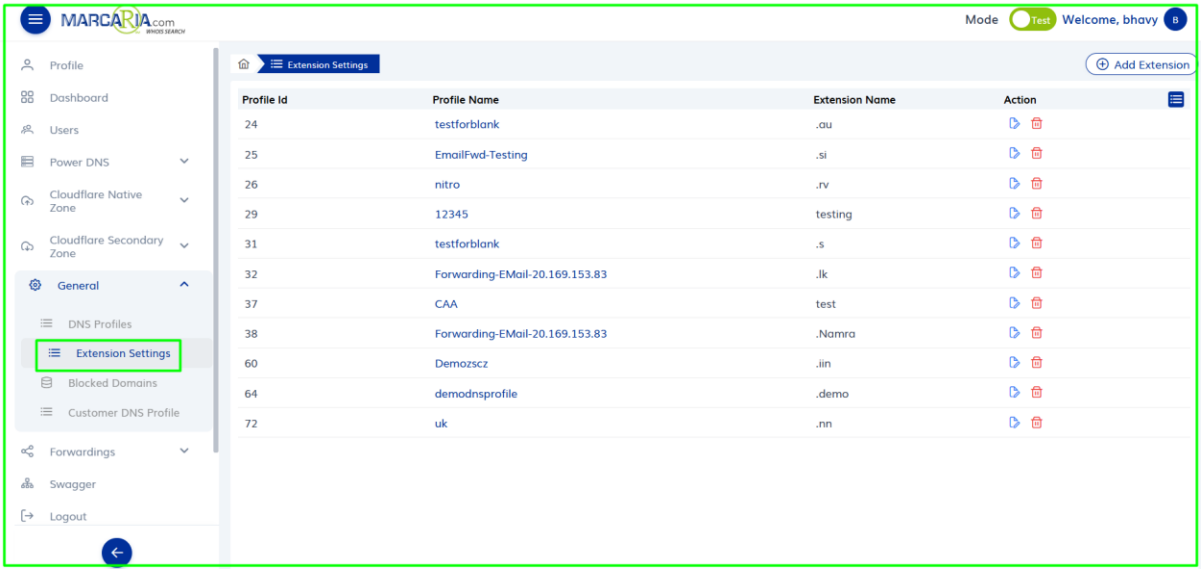


NOTE: A DNS profile cannot be deleted unless the extension record is deleted in the extension settings Because it is mapped to extension settings.

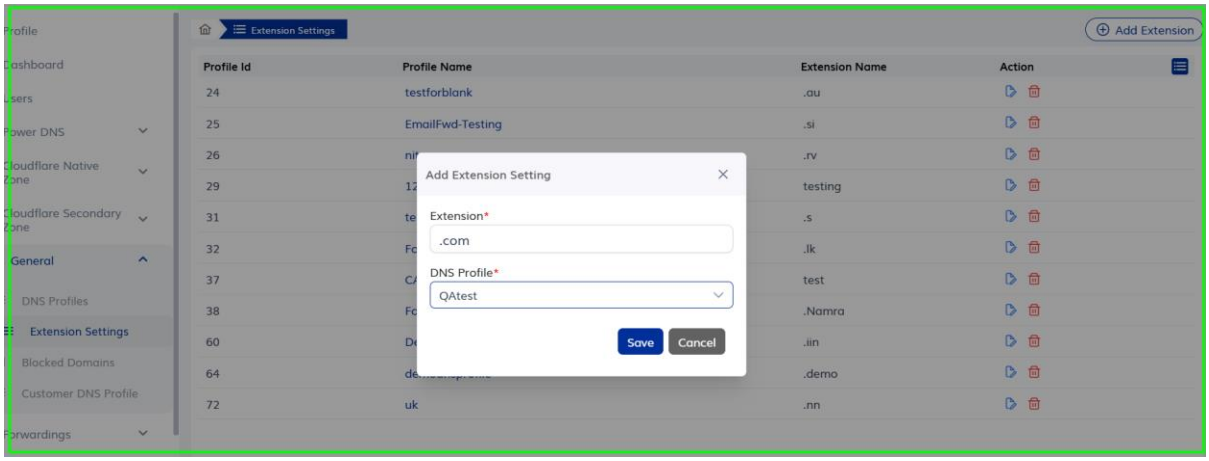


Extension Settings:

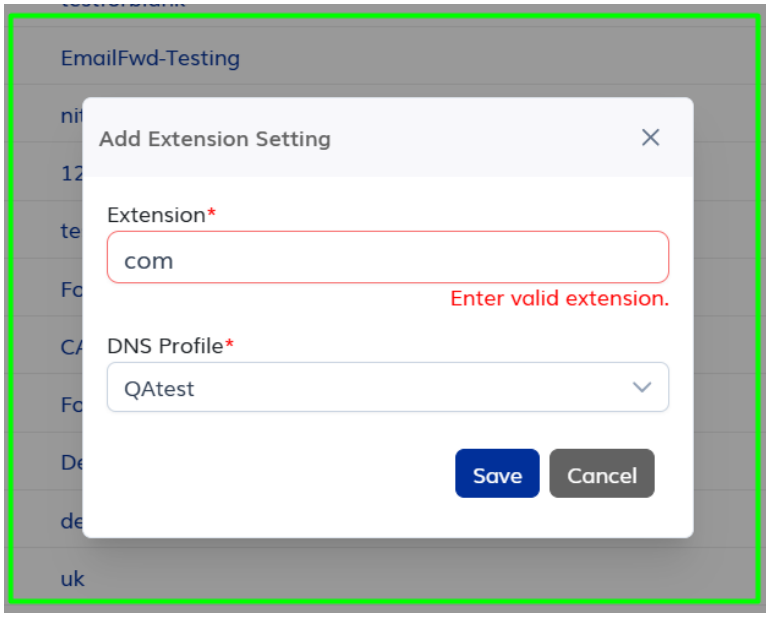
- Users can find out the extension settings menu option from the side menu panel.
- When users click on the DNS profile it will redirect to the DNS record page.
- On this page users are able to view tables with extension, DNS profile and action with edit functionality and add button.



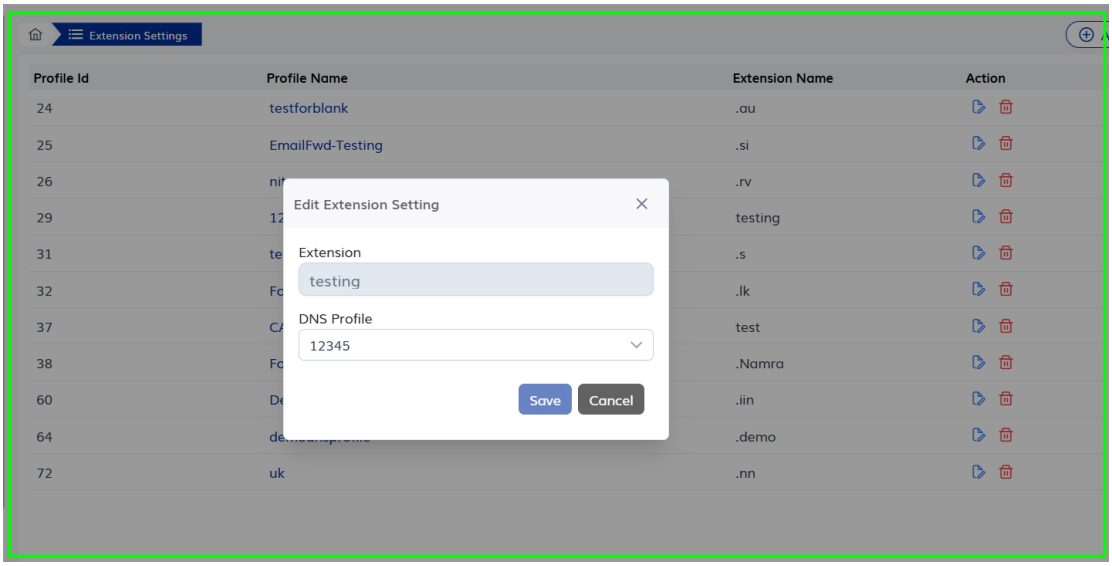
- **Add Button:**
  - Once the user clicks on the add button it will show extension text area and DNS profile dropdown.
  - Users can add any extension by startingfrom. **(Dot).** For example: **.com**



- If a user adds extension without entering dot then validation message will be shown: "Enter **value in valid extension format”**.



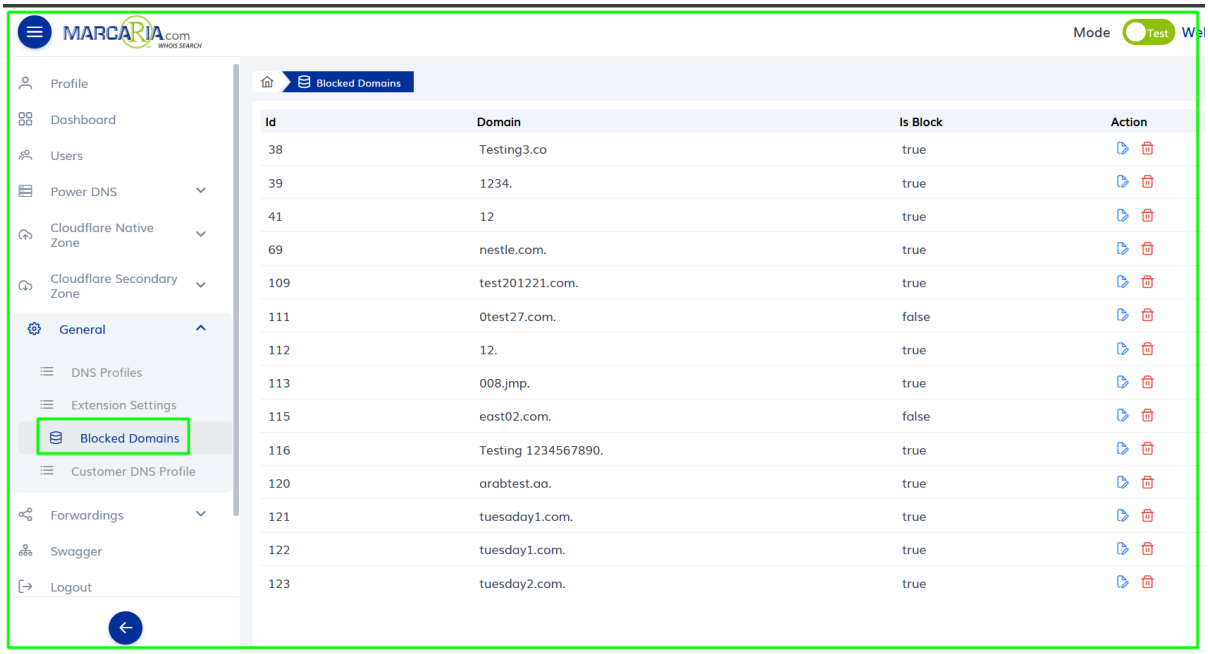
  - **Save Button:** Click on save button it will add DNS record
  - **Cancel Button:** Click on Cancel button pop will be closed
- **<u>Edit Icon :</u>**
  - Click on the edit icon from the Action field of the extension setting page.
  - Users can get the popup with following details..
    - Fields of the record which are generated by the user.
    - Save button
    - Delete button
    - Cancel button



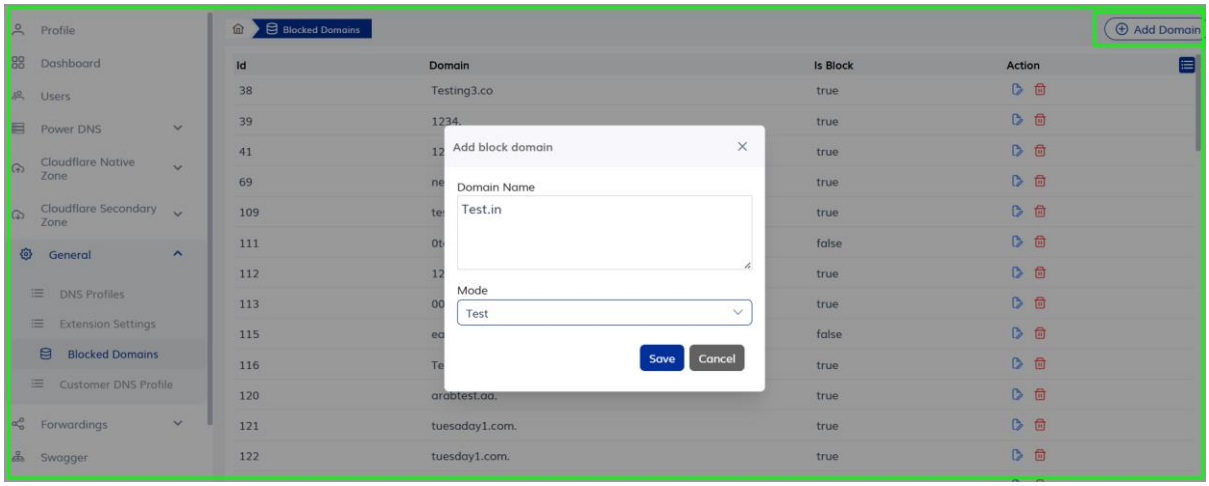
- **Extension** is disabled at the time of edit and DNS profile users can modify.
  - **Save Button:** Click on save button record will update in DNS record
  - **Cancel Button:** Click on Cancel button pop will closed
  - **Delete Button:** Click on the Delete Button user is able to delete the profile and get the message like - **"Extension Successfully Deleted".**
- **NOTE: An extension settings record in Extension Settings cannot be deleted until the DNS profile is deleted. Because it is mapped to a DNS profile.**

Blocked Domains:

- On this page users are able to view tables with ID, domain, Is blocked and action with edit functionality and add button.
- Blocked domains Record will be displayed in the domain list.
- Users can not add new records in the domain which are blocked.
- Using sorting functionality users are able to sort the data



- **Add Button:**
  - Once the user clicks on the add button it will show domain name text area and mode dropdown with live and test option.
  - Here, users are able to block multiple domains in the text area separated each by entering a domain name. i.e. every domain must be in a new line.



  - **Save Button:** Click on save button it will block the domain.
  - **Cancel Button:** Click on Cancel button pop will be closed.

- **Edit Icon :**
  - Users are able to click the edit icon from the Action field of the blocked domain page.
  - Users can get the popup with following details..
    - Domain name text area field
    - Mode dropdown
    - Is block dropdown

- Save button
- Delete button
- Cancel button

- The domain is **unblocked** when the user selects the **No option** from the **Is Blocked dropdown menu.**
  - **Users can add records to the domain only when the domain is unlocked**
- Email and Web forwarding is allowed in blocked domains.
  - **NOTE:** Users need to add (.) dot after the domain name when they are adding it in the blocked domain list.

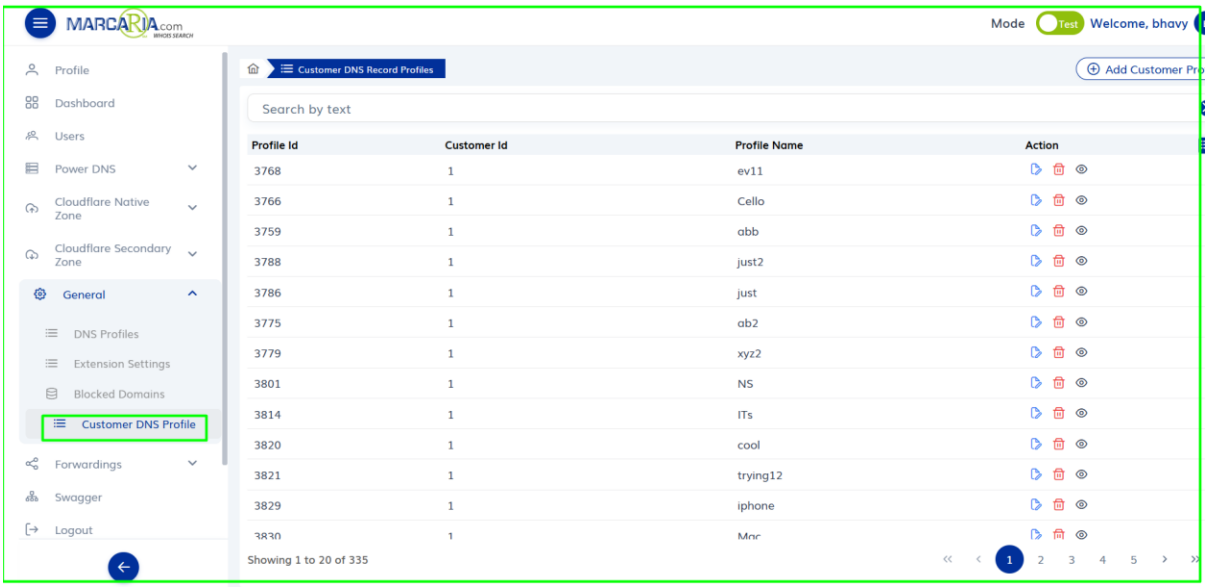
Client Confirmation regarding the who domain works:

1. Blocked domain behavior (your question from May 1 — apologies for the delay). Proposed behavior when a domain is marked as blocked in the DNS-Admin Panel:

- **Restricted:** all mutating operations — add/edit/delete DNS records, web/email forwarding changes, applying DNS profiles, Setup Secondary Zone, and domain deletion. This applies to NormalAgent users and to API calls (the API should return an explicit "domain is blocked" error, not a generic failure).
- **Still allowed:** read-only operations — viewing the domain and its records, and zone export.
- **Unblocking:** only Admin/Superuser can unblock a domain; the block/unblock action should be logged.

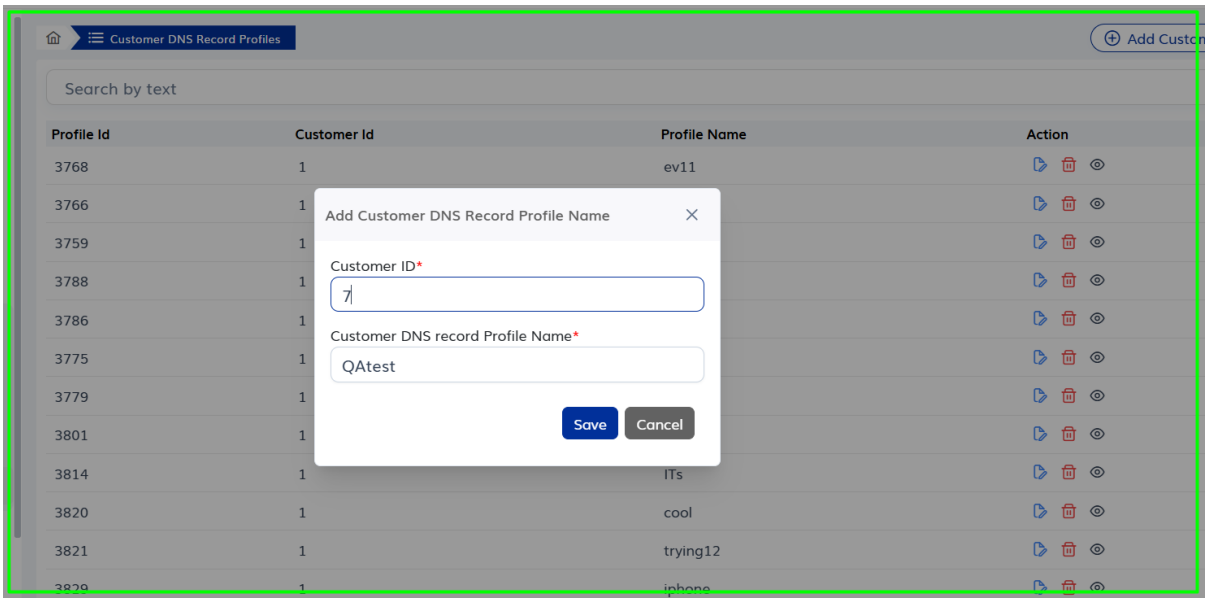
Customer DNS Profile:

- Users can find out the customer DNS Profile menu option from the side menu panel.
- Click on it and the user will redirect to the customer DNS Record Profile page.
- On this page users are able to view tables with Customer ID, Profile ID, Profile Name, Action field with edit functionality, Add Customer Profile Button, Search box, Pagination.



Add Customer Profile Button:

- Users can click on the Add customer Profile button and it will open the popup of Add customer DNS record profile name.
- On this popup user is able to view Textbox for entering the customer ID,customer DNS record profile name, Save & Cancel button.



Save Button:

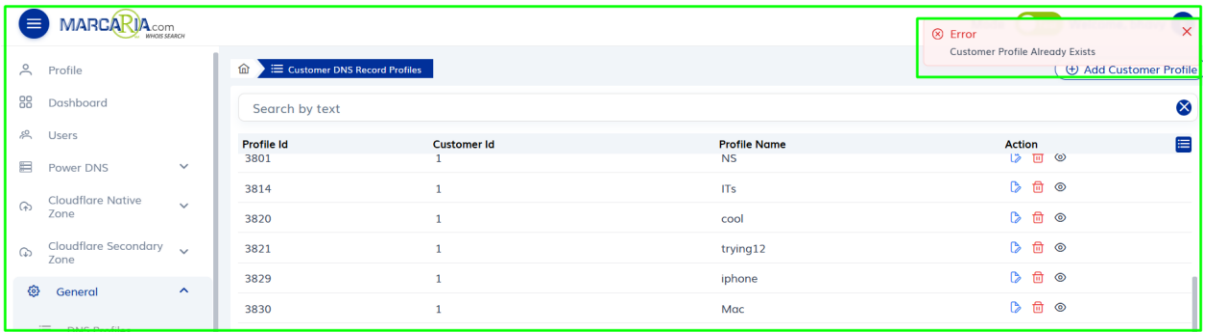
- Click on the Save button after entering it will create a profile with the same name which is entered and get a message like **“Customer Profile Successfully Added”.**

Cancel Button:

- Click on the cancel button and it will close the popup.

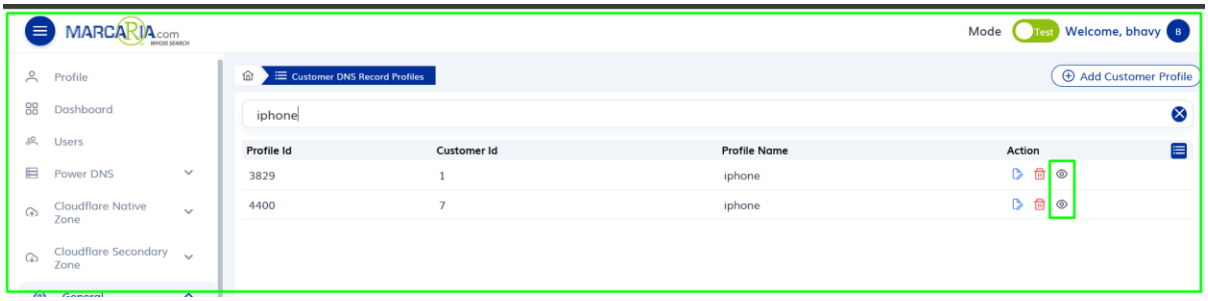
Here, users will be able to create multiple DNS profiles with the same customer ID.

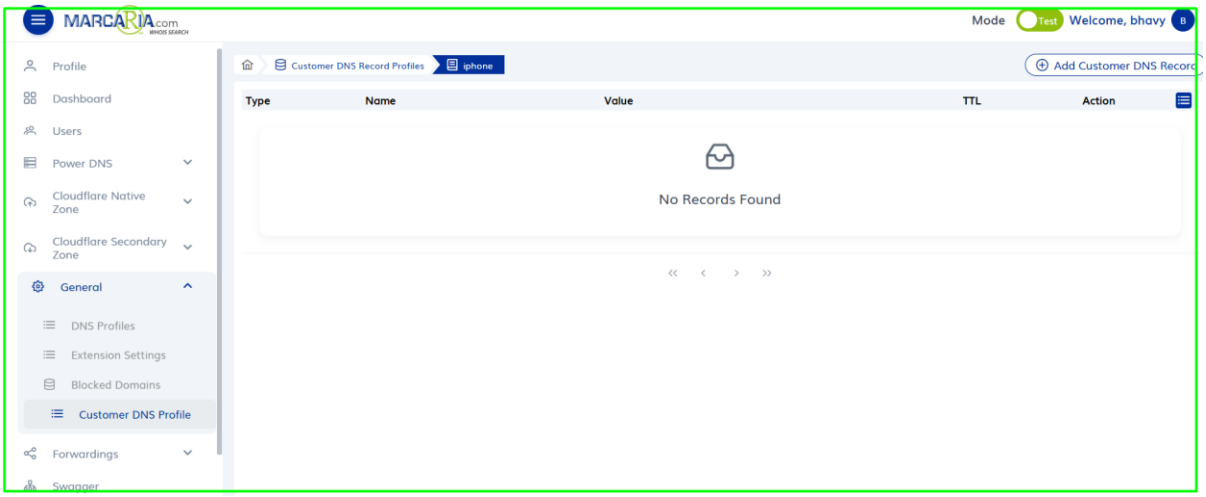
- If the user enters the customer ID and profile name that is available in the list, it will give a validation message like "**Customer Profile already exists!**"



- **Search Box:**
  - Users can enter the name in the search box and get the relevant result.
  - If a search record is not found user will get message like - **No matching records found".**

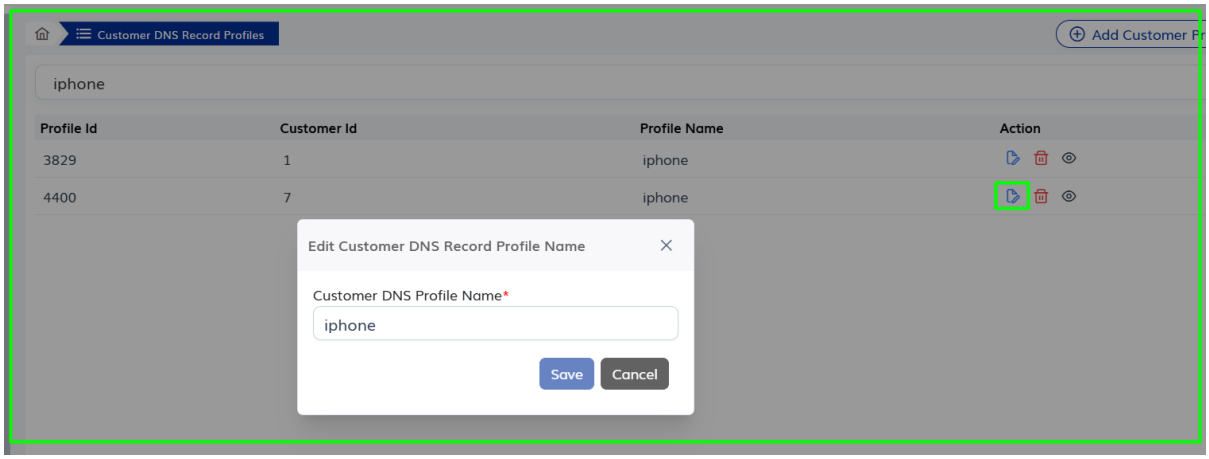
- **<u>Add Records in the Created Customer Profile:</u>**
  - When a user clicks on the **customer ID** OR **profile ID** OR **profile name** it will redirect to the customer DNS record page.
  - From here users are able to add the record for the profile.





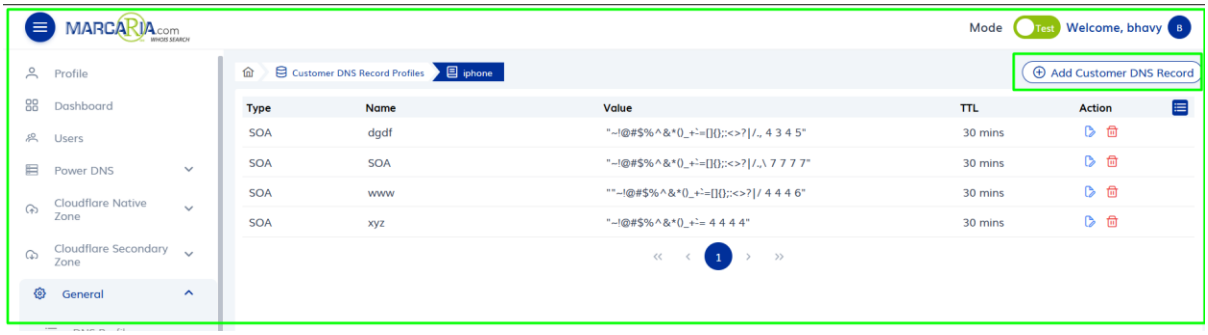
- **<u>Action Column (Edit) :</u>**

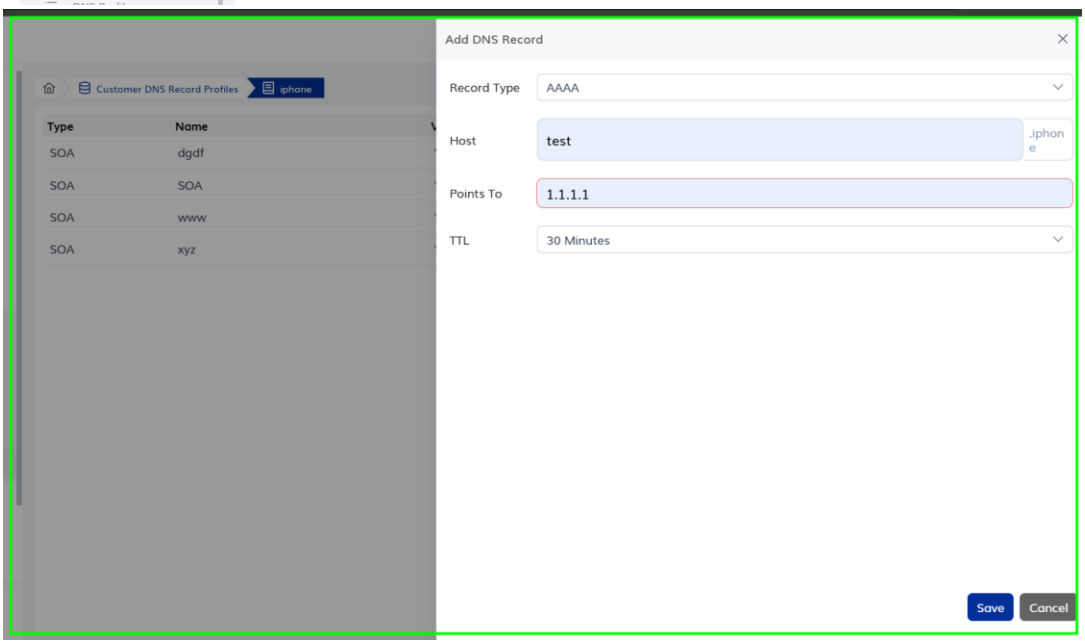
- Click on Edit Icon it will open the popup of edit customer DNS Record Profile Name with save and cancel button.



  - Save Button : Here after modifying the name click on Save button user is able to view the new name in place of the old name and also get the message like “Profile successfully updated”.
  - **Cancel Button :** If you Click on cancel button, then the edit popup will close.

- **<u>Customer DNS Record Button:</u>**
  - Click on the customer DNS Record button it allows the user to add records just similar to the DNS Record available in the Domain List

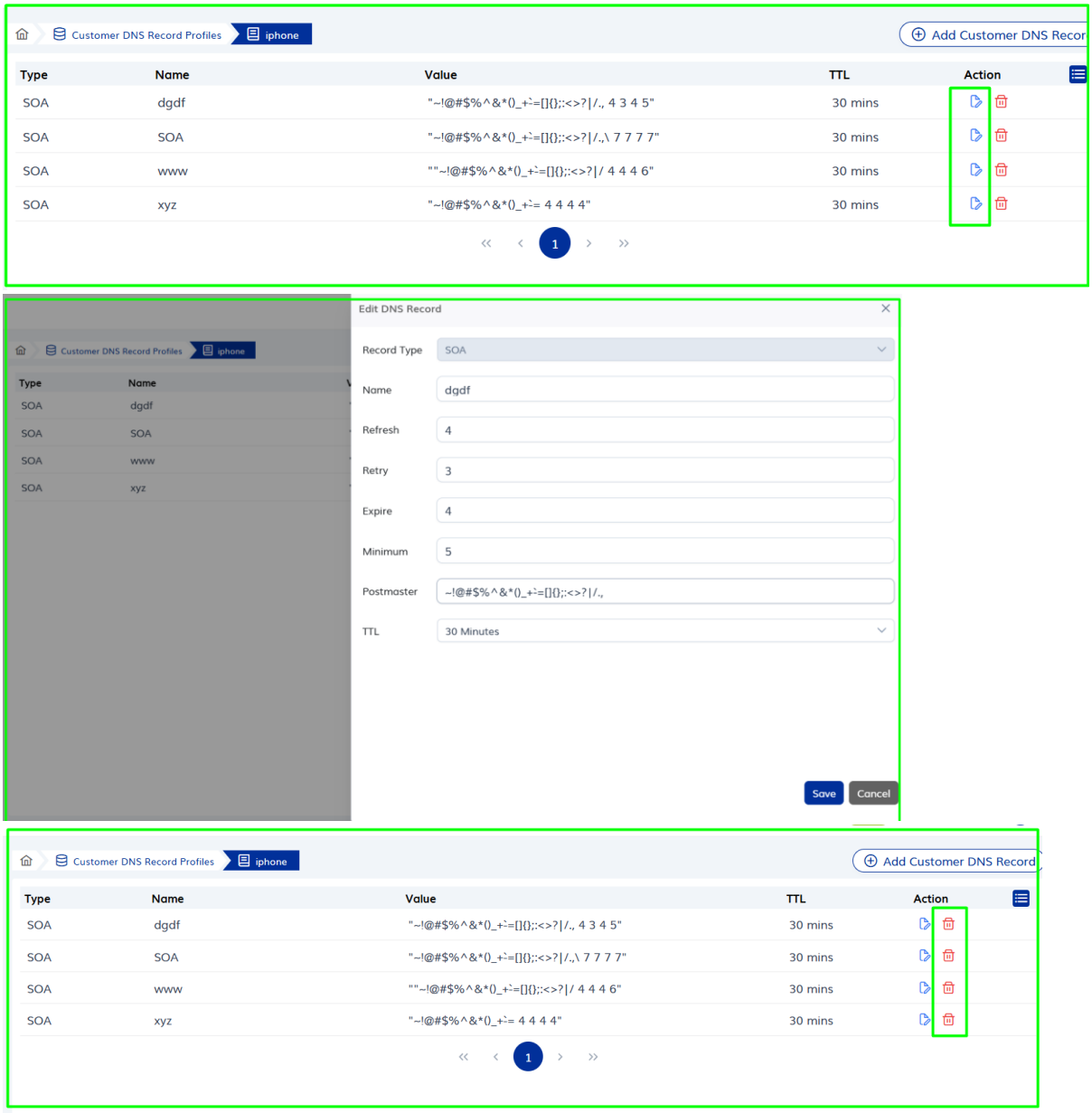




- Click on Save button Record is added to the profile and get a validation message like - **“Record Successfully Added”**

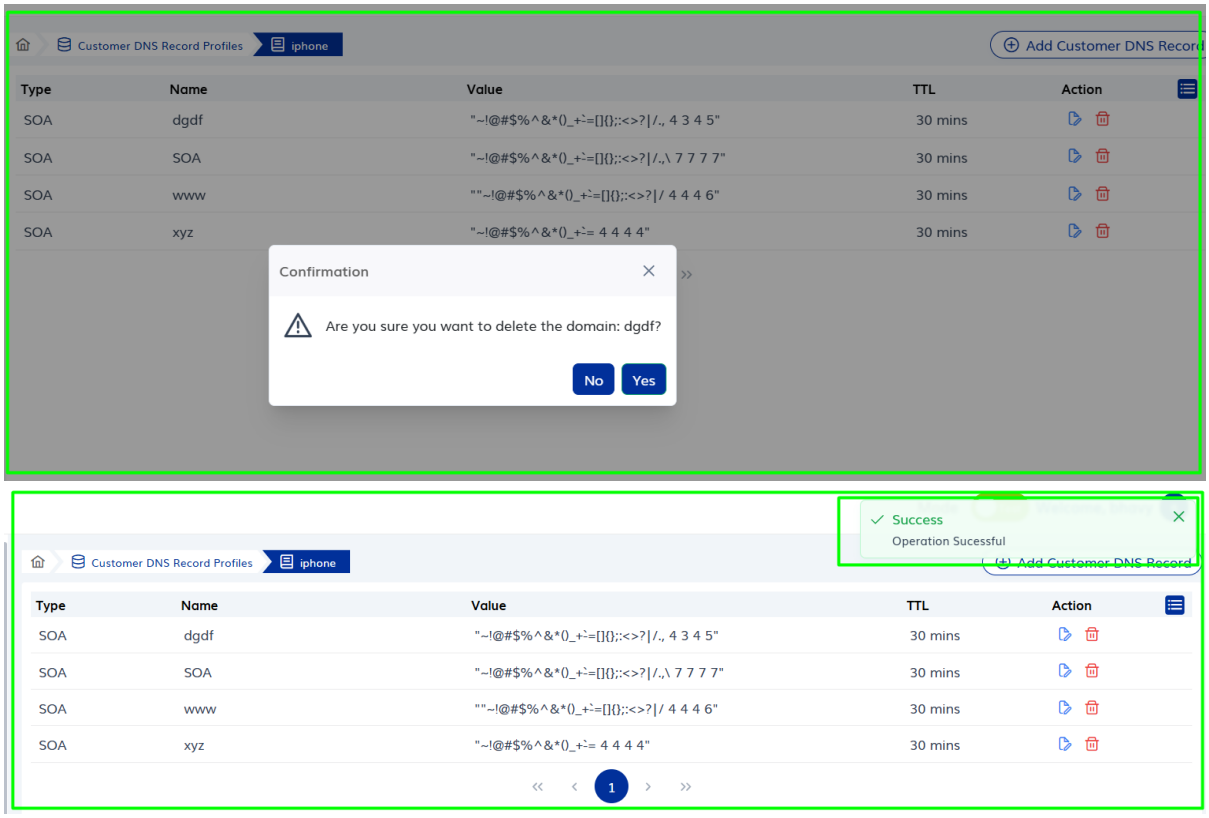
Edit Record Icon :

- Click on the edit icon from the Action field of the customer DNS records table.
- Users can get the popup with following details..
  - Fields of the record which are generated by the user.
  - Save button
  - Delete button
  - Cancel button
- **Type** and **hostname** are disabled at the time of edit and all other fields users can modify



- **Delete Button (For delete record):**
  - When the user clicks on the delete button, the record will be deleted.
- **Cancel Button:**
  - Click on the cancel button and it will close the popup.

- **Delete Button (For delete Customer DNS Profile) :**[Text Wrapping Break]
  - Once a user opens the profile it is able to see the delete button besides the customer DNS record button.
  - Click on the Delete Button user is able to delete the profile and get the message like - **“ Customer Profile Successfully Deleted”.**



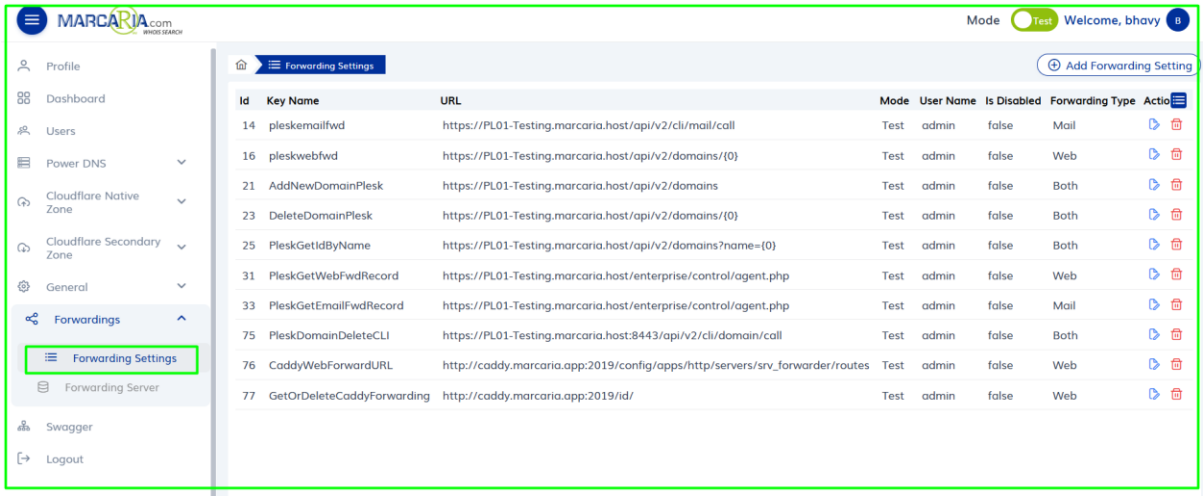
FORWARDING.

- Forwarding has 2 submodules.
  - Forwarding Settings.
  - Forwarding Server.

Forwarding Settings:

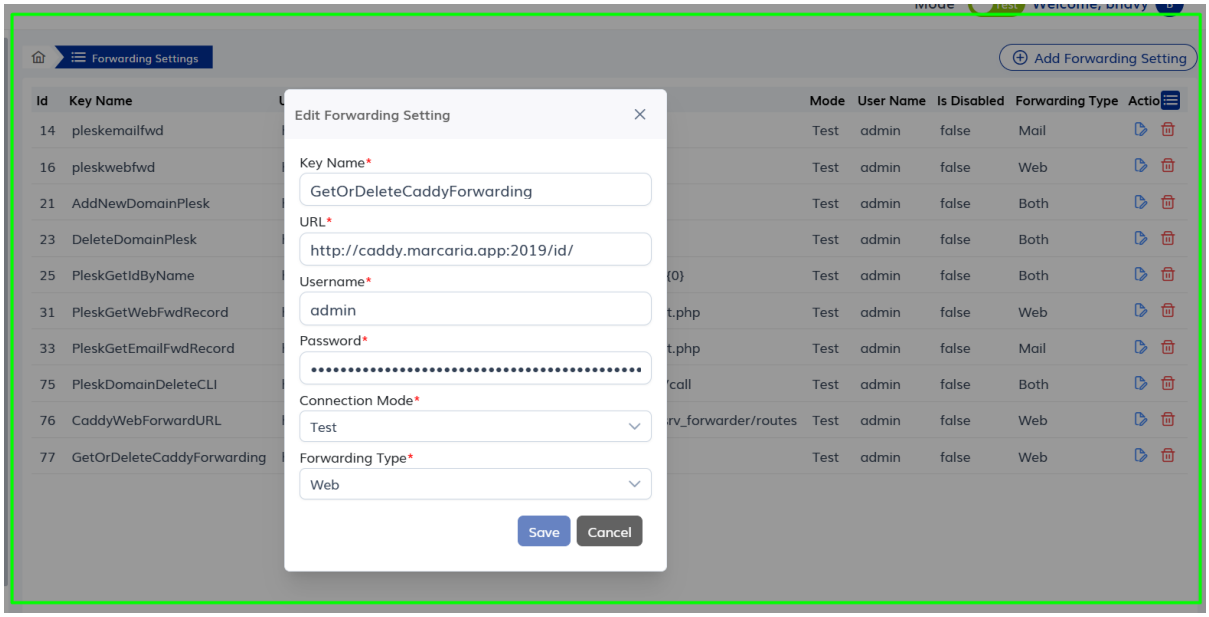
- It is used for mapping the plesk API.
- Here users are able to view the following details with search by mode dropdown and add button.
- Using sorting functionality users are able to sort the data.
- **ID:** Unique Identification number.
- **Key Name:** Key name which is used to map the plesk URL.
- **URL:** A plesk URL
- **Mode:** Users can set the mode either Live or Test.
- **User:** User name
- **Is Disabled:** Here if the user does not check the checkbox its value is N otherwise its value is Y. Here if users don't want to use the API longer at that time they can do Is Disabled checkbox checked instead of hard delete.
- **Type:** Select the type of forwarding i.e. Email Forwarding, Web Forwarding or both.

- **Action:** Users can edit or delete the records.[Text Wrapping Break]

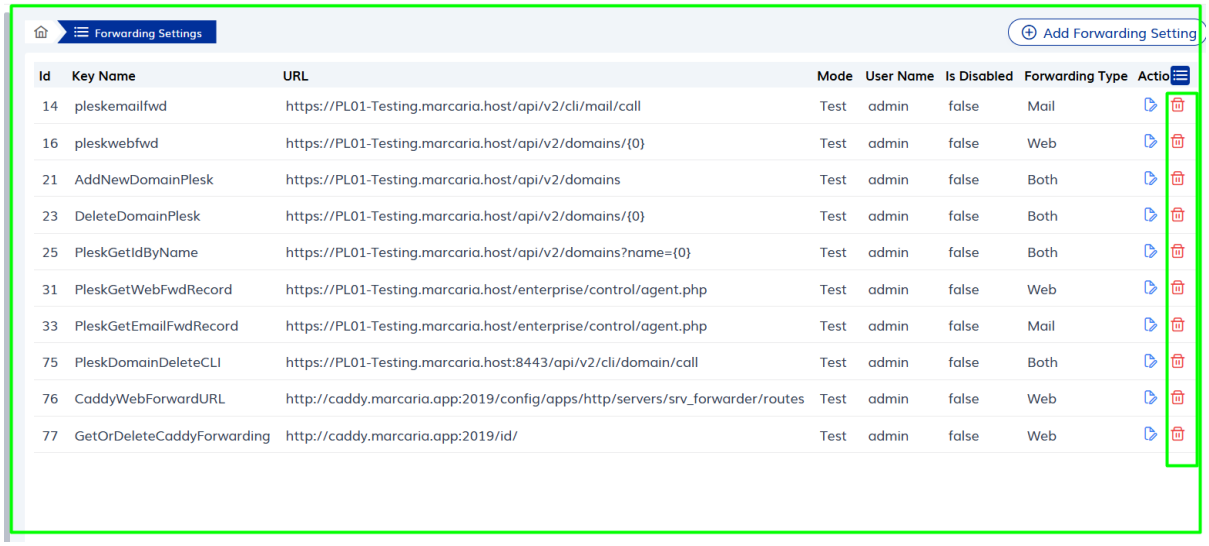


Edit Icon:

- **Key Name:** User can edit the Key name which is used to map the plesk URL.
- **URL:** User can edit the plesk URL
- **Mode:** Users can edit the mode either Live or Test.
- **User:** User can edit the User name
- **Password:** password for the user.
- **Is Disabled:** User can edit the status i.e. Here if the user does not check the checkbox its value is N otherwise its value is Y. Here if users don't want to use the API longer at that time they can do Is Disabled checkbox checked instead of hard delete.
- **Forward Type:** User can edit the type of forwarding i.e. Email Forwarding, Web Forwarding or both.
- **Save Button:** Click on the Save button user can save the modification.
- **Cancel Button:** Click on the Cancel button and the user can cancel the process.



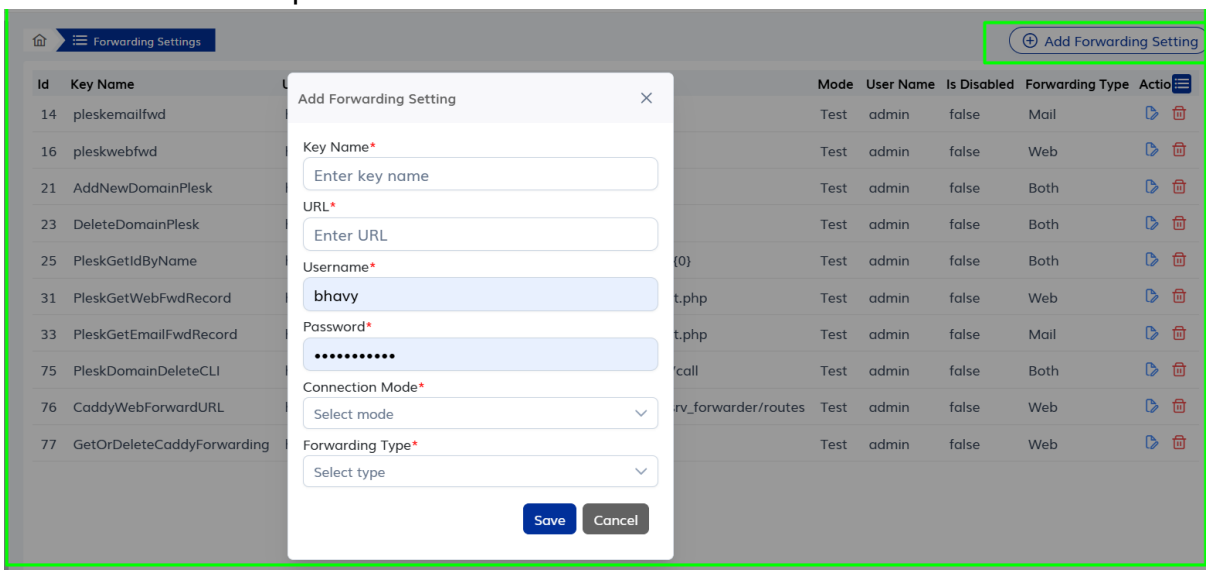
- **Delete Icon:**
  - By clicking on the delete user is able to delete the record.



Id	Key Name	URL	Mode	User Name	Is Disabled	Forwarding Type	Actio
14	pleskemailfwd	https://PL01-Testing.marcaria.host/api/v2/cli/mail/call	Test	admin	false	Mail	
16	pleskwebfwd	https://PL01-Testing.marcaria.host/api/v2/domains/{0}	Test	admin	false	Web	
21	AddNewDomainPlesk	https://PL01-Testing.marcaria.host/api/v2/domains	Test	admin	false	Both	
23	DeleteDomainPlesk	https://PL01-Testing.marcaria.host/api/v2/domains/{0}	Test	admin	false	Both	
25	PleskGetIdByName	https://PL01-Testing.marcaria.host/api/v2/domains?name={0}	Test	admin	false	Both	
31	PleskGetWebFwdRecord	https://PL01-Testing.marcaria.host/enterprise/control/agent.php	Test	admin	false	Web	
33	PleskGetEmailFwdRecord	https://PL01-Testing.marcaria.host/enterprise/control/agent.php	Test	admin	false	Mail	
75	PleskDomainDeleteCLI	https://PL01-Testing.marcaria.host:8443/api/v2/cli/domain/call	Test	admin	false	Both	
76	CaddyWebForwardURL	http://caddy.marcaria.app:2019/config/apps/http/servers/srv_forwarder/routes	Test	admin	false	Web	
77	GetOrDeleteCaddyForwarding	http://caddy.marcaria.app:2019/id/	Test	admin	false	Web	

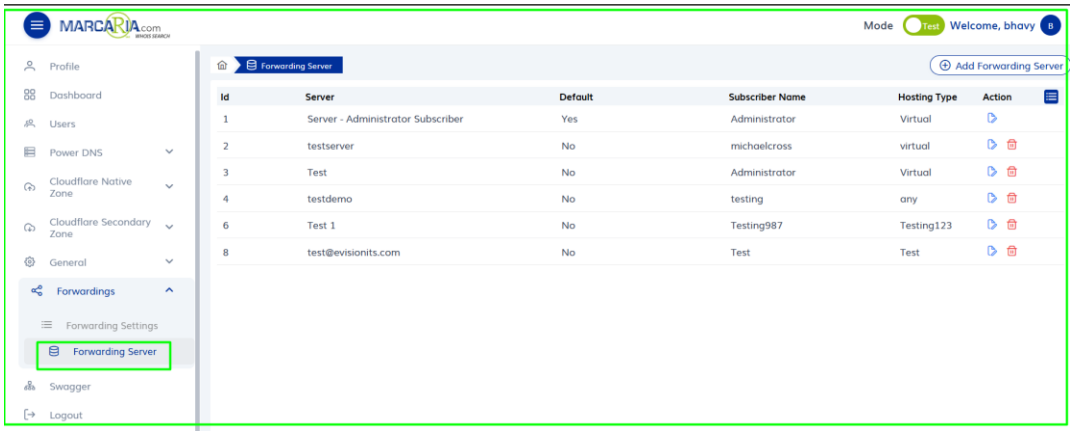
- **Add Button:**
  - Click on the Add button user will redirect to the Add Forwarding Details Page.
  - Here users are able to view/add following information..
    - **Key Name:** Key Name user for the plesk URL.
    - **URL:** A plesk URL.
    - **User Name:** Display the User Name.
    - **Password :** Password for the User.
    - **Connection Mode:** Select the mode i.e. Live : 1 or Test : 2
    - **Forward Type:** Select the forwarding type i.e.
      - Mail Forwarding,
      - Web Forwarding,
      - Web & Mail Forwarding.
    - **Submit Button:** Click on Submit button it will submit the data.

- **Cancel Button:** Click on Cancel button it will cancel the process.



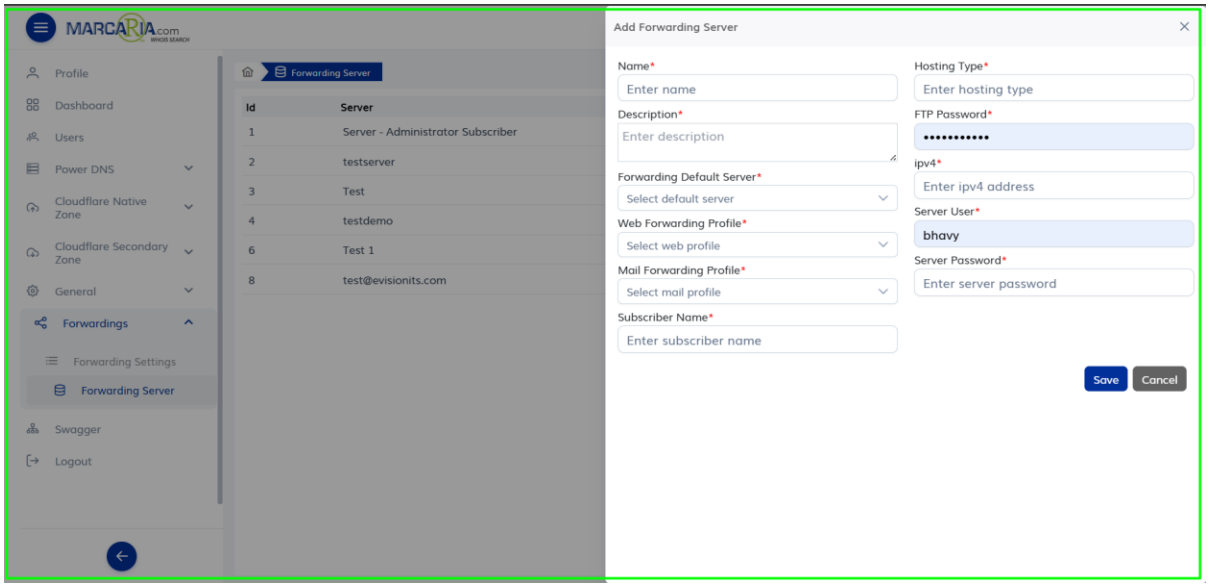
Forwarding Server:

- Click on the Forwarding Server menu and the user will be able to view List of Forwarding Servers.
- On this page users are able to view tables with fields Server ID, Server, Default,show entries tab,search tab and Add Forwarding Button.
- Here users see **Default = Yes** in front of the profile which is "**Active**" and **Default = No** in front of the profile which is **"InActive"**.
- When creating a new profile at that time the newly added profile will have **Default = Yes**. It means the newly added profile will be considered as "**Active".**
- Using sorting functionality users are able to sort the data.



- **Add Forwarding Server:**[Text Wrapping Break]
  - Click on this button and the user will get a popup with following information..
    - **Name:** Provide name to the forwarding server.[Text Wrapping Break]

- **Description:** Add description for the forwarding server.[Text Wrapping Break]
- **Forwarding Default Server:** Here checkbox is available if user checks the checkbox it will make profile as Default = Yes and make the profile **"Active".** [Text Wrapping Break]**NOTE:** Once a user makes the profile Active it will not be able to **"Deactivate the profile"**. Users can change the status by creating the new profile.
- **Web Forwarding Profile:** Select the profile which is used to add the records. Here users are able to view all the profiles in drop down which are added from the DNS **Profiles menu**.[Text Wrapping Break]
- **Mail forwarding Profile:** Here user can able to view all the profile in drop down which are added from **DNS Profile menu**[Text Wrapping Break]
- **Subscriber Name:** Add the name of the Subscriber.(Plesk)[Text Wrapping Break]
- **Hosting Type:** Add Hosting type.(Plesk)[Text Wrapping Break]
- **FTP Password:** Add password for the FTP(Plesk)[Text Wrapping Break]
- **IPV4:** Add the IPV4 address.(Plesk)[Text Wrapping Break]
- **Server User:** Add name of the server user(Plesk)[Text Wrapping Break]
- **Server Password:** Add server password.(Plesk)



Master Domain Creation Flow with Secondary DNS enables:

- When a domain is created with Secondary DNS **enabled**, the system establishes the domain as a Master zone in PowerDNS.

- The system validates that the PowerDNS zone has been created with the expected Master configuration; the corresponding Cloudflare zone is available, the required Incoming Secondary configuration and Peer information are present.
- The system also validates the SOA serial and ensures that a valid **`soa_serial > 0`** is available before considering the Master/Secondary relationship successfully established
- Any temporary synchronization or propagation delay between PowerDNS and Cloudflare is handled through the defined retry and verification mechanism.
- The domain should be reported as successfully created only after the required PowerDNS, Cloudflare, zone-transfer, SOA, and database validations have been completed successfully.
- Below is the Expected Log for Domain creation in Cloudflare Secondary zone API Search log:

- ## Race Condition Handling
- The Secondary Zone workflow handles race conditions that can occur when a domain is deleted and immediately recreated. External systems such as Cloudflare and PowerDNS may require additional time to complete deletion or release existing resources. To accommodate this behavior, the system retries operations at the defined intervals of **0, 2, 4, 8, and 10 seconds**. This prevents temporary resource availability issues from being incorrectly treated as permanent failures.
- ## Missing Resource Handling
- The system differentiates between an expected missing resource and an actual operational failure. For example, during cleanup, a `404` response may indicate that the requested resource has already been deleted. Such cases can be treated as successful/idempotent cleanup where appropriate. This prevents repeated cleanup operations from producing misleading failures.
- ## Incorrect Configuration Handling
- The system validates configuration received from Cloudflare and other external services before using it as the basis for a successful result. If an expected configuration is missing, malformed, or inconsistent, the system does not blindly continue as though the setup were complete. Depending on the scenario, the configuration may be retried, repaired through the WebJob, logged as a warning, or reported as a hard failure.
- ## API Validation and Error Handling
- The Secondary Zone APIs validate the result of each required operation and return an overall result based on the complete workflow rather than an individual successful API response. The system handles successful responses, missing resources, invalid configurations, external-service failures, and retry scenarios according to the defined business rules. A successful API response from Cloudflare or PowerDNS alone does not guarantee successful Secondary Zone setup if subsequent mandatory validation or database persistence fails.
- ## Database Synchronization

- The application database maintains the state and relevant information associated with Secondary Zone domains. After successful processing, the corresponding database record is created or updated with the appropriate status and configuration information. Database persistence is therefore treated as an important part of the overall transaction and must remain consistent with the actual state of Cloudflare and PowerDNS

- ## Database Failure Handling

- The system also handles scenarios where the database becomes unavailable while the external APIs continue to return successful responses. For example, if all required Cloudflare and PowerDNS APIs return `200` responses, but the database connection is disrupted, the system must not incorrectly report the entire operation as successful. The database persistence failure must be detected, logged, and reflected in the final operation result according to the implemented error-handling behavior.

- Kindly refer to the Test Cases file for a detailed steps above all related scenarios From TC – 10 to 62

Deferred verification implementation:

- A deferred verification process is performed after domain creation, based on the configured verification interval (60 or 120 seconds). The process verifies whether the domain has been successfully created and whether the expected peer is correctly linked.
- If the expected peer is not linked, the deferred verification process automatically attempts to repair the configuration and logs the details of the repair operation.
- The logs below show the deferred verification process and the corresponding verification/repair activities:

  sokhda180.com Cloudflare Secondary deferred verification. Classification: DeferredVerificationVerified. OperationId: 6b08a9fc-1bdd-4270-afa8-f8b2f0518755. Status: Verified. Attempt: 1. RepairPerformed: False. Message: The expected peers were present, force AXFR completed and the fresh verification succeeded.
- If you want to check the auto repair related scenario performed this Test Case steps: TC: 69

Automated Web Job Verification

- The automated WebJob periodically checks Secondary Zones that have recently been created or are awaiting verification.
- The WebJob validates the required Cloudflare and PowerDNS configurations and identifies zones that have not reached the expected state.

- This provides an additional layer of verification beyond the synchronous domain creation process and ensures that transient external-system delays or configuration issues are detected after the initial operation.
- All the logs related to the Web Job & Safety Sweep will be logged in **tblWebJobCloudflareSecondaryVerification** [Note: `All Log records older than 90 days are deleted by webjob`]

48-hour SafetySweep

- The 48-hour SafetySweep is a backup check. It is not redundant with the every-minute pending verification.
- The difference is:
- **Deferred verification:** Processes a `PendingVerification` operation after a create/recreate. Once it becomes `Verified`, the minute-by-minute processor will not normally check that operation again.
- **48-hour SafetySweep:** Once daily, it creates new `SafetySweep` operations for recently created zones and rechecks them—even if their original Provision operation already became `Verified`.
- Your test currently proves the first path:
    - Create domain → `PendingVerification`.
    - Delete peer before verification completes.
    - Every-minute WebJob claims that pending operation.
    - WebJob discovers the missing peer and repairs it.
    - Operation becomes `Verified`.
- The SafetySweep protects against this different timeline:
    - Create domain.
    - Deferred WebJob verifies it successfully → `Verified`.
    - Cloudflare asynchronously removes the peer later.
    - There is now no pending operation for the minute WebJob to claim.
    - Daily SafetySweep selects the recently created zone.
    - It creates a new `SafetySweep` operation.
    - The normal deferred processor claims that operation and repairs the peer.
- This is important because the original production race can occur after the initial request and potentially after the first successful verification. The sweep provides a second independent opportunity to discover delayed peer loss.

- The limitation is intentional: it covers zones created during the configured lookback—currently 48 hours. It is designed to catch delayed damage associated with provisioning/recreation, not arbitrary peer removal on every historical Cloudflare zone.
- If you want to check the auto repair by Safety Sweep related scenario performed this Test Case steps: TC: 70

Master Domain Creation with Secondary DNS Disabled

- The system also supports creation of a domain without Secondary DNS functionality. In this scenario, the domain is created as a **Native zone** in PowerDNS and no Cloudflare Secondary DNS configuration is provisioned. The system must ensure that Secondary DNS-specific operations, such as Cloudflare zone creation, Incoming Secondary configuration, peer creation, or AXFR validation, are not triggered when the Secondary option is disabled. This provides a clear separation between standard Native domain creation and Master/Secondary DNS domain creation. If Secondary DNS is enabled later for the existing domain, the system can initiate the required Secondary DNS configuration as a separate operation.

SWAGGER.

- By clicking on the swagger button it will be redirected too the DNS Swagger site.

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